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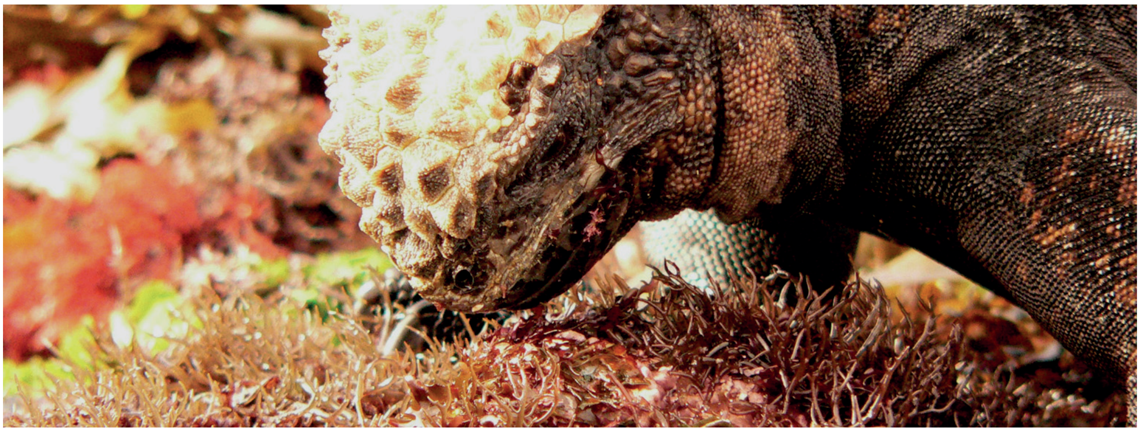

Energy flow and trophic structure of Galápagos shallow rocky reef systems along a gradient of productivity and artisanal fisheries

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Erklärung

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Zusammenfassung

Das Galápagos Archipel befindet sich ungefähr 1000 km vor der Festlandküste Ecuadors. Diese charismatische Zusammenstellung von verschiedenen Inseln hat besondere geologische und ozeanographische Eigenschaften, die die Entwicklung eines einzigartigen marinen Ökosystems mit hoher Artenvielfalt und Endemismus beeinflusst haben.

Das Marine Reservat Galápagos (MRG) wurde von der ecuadorianischen Regierung 1998 nach Maßgabe eines speziellen Managementplans deklariert, um Fischereiaktivitäten zu regulieren und gefährdete Arten zu schützen. Die langjährige Entwicklung der einfachen Kleinfischerei und Naturereignisse wie die El-Niño-Südliche-Oszillation (ENSO) hatten jedoch große Veränderungen in der trophischen Struktur von marinen Lebensgemeinschaften zur Folge.

Das Ziel dieser Dissertation ist ein besseres Verständnis, wie Veränderungen der Primärproduktion, der Temperatur und der Fischereiintensität innerhalb einer biogeographischen Region die trophische Struktur und die Diversität sowie die Verbreitung von Schlüsselarten beeinflusst. Veränderungen im marinen Ökosystem von Galápagos aufgrund von El Niño und Fischerei wurden ausgiebig beschrieben. Allerdings wurden Veränderungen im Nahrungsnetz, wie zum Beispiel Veränderungen von Energieflüssen, trophischen Interaktionen und der Struktur des Systems, die von diesen beiden Faktoren beeinflusst werden, unzureichend dargelegt.

Die trophischen Modelle im Zeitraum 2004-2008 wurden mit Hilfe der Anwendung des „Ecopath with Ecosim" (EwE) Ansatzes erstellt, um die Unterschiede zwischen den drei großen biogeographischen Regionen im MRG zu beschreiben: das westliche, kalte Auftriebssystem – Bolivar Channel, das süd-zentrale, gemischte System – Floreana und das hohe nördliche, tropische System – Darwin und Wolf.

Die Informationen, die für die Erstellung dieser trophischen Modelle gefordert werden, basieren hauptsächlich auf fünf Jahren ökologischen Unterwasser-Surveys, ozeanographischen Monitoring und Populationszählungen der Charles Darwin Foundation und des Nationalparks Galápagos. Zusätzliche Informationen wurden aus

anderen Quellen, wie aus der Literatur, Berichten und anderen trophischen Modellen, entnommen. Die Beschreibung der Unterschiede der Systemstrukturen, Energieflüsse und ökologischen Charakteristika von jeder biogeographischen Region basierte auf quantitative Ökosystemparameter, die mit Hilfe von Ecopath geschätzt worden sind. Zudem wurden die Auswirkungen der Kleinfischeri und El Niño auf die marinen Ökosysteme Galápagos‘ mit Ecosim untersucht, um den Mechanismus hinter den beobachteten Veränderungen in den Dynamiken des Ökosystems zu verstehen.

Die wichtigen Ergebnisse dieser Studie suggerieren Folgendes:

1) Das Bolivar Channel-System gehört zu den produktivsten Zonen im MRG und ist das größte System hinsichtlich der Energieflüsse, die mit dem vom Humboldtstrom beeinflussten peruanischen Auftriebsküstensystem verglichen werden kann.

2) Die entlegensten Inseln Galápagos’, Darwin und Wolf, zeigten eine komplexe trophische Struktur mit großen Prädatoren, welche 55 % der Gesamtsystembiomasse ausmachten.

3) Die Größe der Unterschiede der festgestellten Räuberbiomasse zwischen dem Darwin und Wolf-System und dem Floreana-System ist konsistent mit anderen Untersuchungen, die entlegene und bevölkerte Inseln im Pazifischen und Indischen Ozean miteinander vergleichen.

4) Die Gesamtbiomasse ist ähnlich hoch im Bolivar Channel- und Floreana-System, aber signifikant geringer im Darwin und Wolf-System. Die Verteilung von Biomasse in Funktion der trophischen Stufe unterscheidet sich zudem sehr zwischen dem tropischen und dem kalten System.

5) In Übereinstimmung mit verschiedenen Ökosystem-Indizes, die mit Hilfe von Ecopath berechnet worden sind, ist die Reife und Entwicklung im entlegenen Darwin und Wolff-System, welches am wenigsten von anthropogenen und natürlichen Störungen beeinflusst ist, höher.

6) Arten, die als Schlüsselarten identifiziert worden sind, unterschieden sich zwischen den biogeographischen Regionen. Haie im Darwin und Wolf-System kontrollieren mittlere Prädatoren, welche abundanter in diesem entlegenen Gebiet sind. Wogegen Seelöwen das kalte Auftriebssystem Bolivar Channel dominieren.

7) Aufgrund einer starken Abnahme der Nährstoffe und des Phytoplanktons während eines heftigen El Niño Ereignisses kontrollieren bottom-up-Effekte weitgehend das System. Insgesamt zeigten die Systemeigenschaften eine

ähnliche Reaktion wie andere Auftriebssysteme an der peruanischen Küste, indem sie Störungen der trophischen Flüsse verursachten und das System auf einen niedrigen (aber höchst effizienten) Entwicklungsstand hielten.

8) Die Modellsimulationen zeigten, dass starker Fischereidruck im entlegenen Darwin und Wolf-System die trophische Struktur des Systems in Richtung eines alternativen, von Seeigeln beherrschten Systemzustands sehr beeinträchtigen würde. Damit würden biogeographische Unterschiede mit anderen Regionen nicht mehr evident sein.

Zusammenfassend zeigten alle drei verglichenen Systeme klare Unterschiede in ihren ozeanographischen Eigenschaften sowie in den Auswirkungen der Fischerei. Die Selektivität der Zielarten der lokalen Kleinfischerei spielt eine wichtige Rolle bei der Strukturierung der benthischen Lebensgemeinschaft. Allerdings sind direkte Effekte räumlich variabel und hängen zum Teil von der Nähe zu Fischereihäfen ab. Strategien zum Schutz müssen die biogeographische Funktionalität und die strukturellen Unterschiede von jedem System berücksichtigen, um die Effektivität des aktuellen räumlichen Managementplans des MRG zu verbessern. Das aktuelle Programm zum Fischerei-Monitoring muss evaluiert und angepasst werden, damit bessere geographisch kodierte Daten erzeugt werden, die einen robusten Beitrag zur Evaluierung der Fischerei-Effekte bereitstellen würden.

Abstract

The Galapagos Archipelago is located in the Eastern Tropical Pacific at about 1000 km off the coast of mainland Ecuador. This charismatic set of islands has particular geological and oceanographic characteristics that have influenced the development of a unique marine ecosystem with high species diversity and endemism.

The Ecuadorian government in 1998 created the Galápagos Marine Reserve, under a special management plan to regulate fishing activities and to protect the vulnerable species. However, a long history of small-scale artisanal fishery and natural events such as the El Niño Southern Oscillation (ENSO) have resulted in major changes in the trophic structure of marine communities.

This thesis aims to better understand how changes in primary production, temperature and fishing intensity within a biogeographic regional influence the trophic structure and the diversity and distribution of key species. Changes in the marine ecosystem influenced by El Niño and fisheries have been extensively evident in Galápagos. However, alterations resulting in the food chain such as changes in energy flows, trophic interactions and system structure influenced by these factors have been poorly described.

Trophic models for the period 2004-2008 were constructed using the "Ecopath with Ecosim" (EwE) software to describe differences between the three major biogeographic regions in the Galapagos Marine Reserve (GMR): the west cold upwelling system – Bolivar Channel, the south-central mixed system – Floreana and the far-north tropical system – Darwin and Wolf.

Information required for the construction of these trophic models were mainly based on five years of underwater ecological surveys, oceanographic monitoring and population censuses carried out by the Charles Darwin Foundation (CDF) and the Galapagos National Park. Additional information was obtained from other sources such as literature, reports and other trophic models. The description of differences in the system structures, energy flows and ecological characteristics of each biogeographic region were based on quantitative ecosystem parameters estimated by Ecopath.

Additionally, the effects of artisanal fisheries and El Niño on the Galápagos ecosystems were explored with Ecosim to understand the mechanisms behind the observed changes in the ecosystem dynamics.

The important results of this study suggest that:

1) The Bolivar Channel system is the most productive zones in the Galápagos Marine Reserve and is the largest system in terms of energy flows, which can be comparable to Peruvian upwelling coastal systems influenced by the Humboldt Current.

2) Darwin and Wolf, the most remote islands in Galápagos, showed a complex trophic structure dominated by large-predators, which accounted for 55% of the total system biomass.

3) The magnitude of observed predatory biomass differences between the Darwin and Wolf and Floreana systems is consistent with other studies comparing remote and populated islands in the Pacific and Indian Ocean.

4) Total biomass is similarly high for the Bolivar Channel and Floreana system, but significantly lower for the Darwin and Wolf system. In addition, biomass distribution by trophic level differs greatly between tropical and cold systems.

5) According to several ecosystem indices calculated by Ecopath, maturity and development is higher in the remote Darwin and Wolf system, where the anthropogenic and natural-oceanographic perturbations are lowest.

6) Species identified as keystone differed between biogeographic regions. Sharks are keystone species in the Darwin and Wolf system where they control intermediate predators, which are far more abundant in this remote area, while sea lions dominate the cold-upwelling Bolivar Channel system.

7) During a strong "El Niño" event, bottom-up effects largely control the system, due to the strong decreases of nutrients and phytoplankton. Overall the system characteristics showed a similar response to other coastal upwelling systems of the Peruvian coast, by causing disruptions to trophic flows and by keeping the system at a low (but highly efficient) development state.

8) Model simulations showed that heavy fishing pressure in the remote Darwin and Wolf system would greatly affect the system's trophic structure, thus showing an alternative system state, dominated by sea urchins and where biogeographic differences with other regions will not be anymore evident.

In conclusion, the three systems compared in this study showed clear differences in their oceanographic settings as well as in their fishery impacts. The selectivity of target species by the local artisanal fishery plays an important role in structuring the benthic community. However, direct effects are spatially variable, depending in part on the proximity to fishing ports. Conservation strategies need to take into account the biogeographic functionality and structural differences of each system to improve the effectiveness of the actual spatial management plan in the Galápagos Marine Reserve. In addition, further studies are recommended to improve the models for future exploration of management strategies. The current fishing-monitoring program must be evaluated and adapted to produce better geo-referenced data, which will provide a robust input to evaluate the fishing effect in the Galápagos Marine Reserve.

Resumen

El archipiélago de Galápagos, ubicado en el Pacifico Este Tropical a 1000 km de la costa de Ecuador continental, presenta características oceanográfica y geológicas particulares, que han influenciado en un ecosistema marino único con alta diversidad de especies y endemismo.

Galápagos fue declarado como Reserva Marina por el gobierno ecuatoriano en 1998 bajo un plan de manejo especial para la regulación de actividades pesqueras y la protección de especies vulnerables. Sin embargo, una larga historia de pesca artesanal a pequeña escala y eventos naturales como el fenómeno de El Niño han resultado en importantes cambios en la estructura trófica de las comunidades marinas.

Esta tesis pretende incrementar el conocimiento de los procesos tróficos considerando un gradiente de producción primaria, temperatura e intensidad pesquera, dentro de un esquema biogeográfico regional que influye en la diversidad y distribución de especies claves. Cambios ocurridos en el ecosistema marino influenciados por El Niño y la pesca artesanal han sido ampliamente descritos en Galápagos. Sin embargo, alteraciones resultantes en la cadena trófica como cambios en los flujos de energía, interacciones tróficas y estructura del ecosistema influenciados por estos factores han sido pobremente descritos.

Modelos tróficos, para el período 2004-2008, fueron construidos utilizando el software "Ecopath with Ecosim" (EwE) para describir diferencias entre las tres mayores regiones biogeográficas del la Reserva Marina de Galápagos (GMR): el sistema frio de afloramiento oeste – Canal Bolivar, el sistema de mezcla sur-central – Floreana y el sistema tropical lejano-norte – Darwin y Wolf.

Información requerida para la construcción de estos modelos tróficos se basó principalmente en cinco años de monitoreo ecológico, monitoreo oceanográfico y censos poblacionales realizados por la Fundación Charles Darwin (CDF) y La Dirección del Parque Nacional Galápagos. Información adicional fue obtenida de otras fuentes como literatura, reportes y otros modelos tróficos. La descripción de las diferencias estructurales de los sistemas, flujos de energía y características ecológicas de cada

región biogeográfica se basan en datos cuantitativos e índices ecosistémicos estimados por Ecopath. Adicionalmente, efectos de la pesca artesanal y El Niño sobre el ecosistema de Galápagos fueron explorados con Ecosim, para observar posibles cambios en la dinámica del ecosistema.

Los resultados relevantes de este estudio sugieren que:

1) El sistema del Canal Bolívar es una de las zonas más productivas de la Reserva Marina de Galápagos y el más grande en términos de flujos de energía, pudiendo ser comparable con los sistemas de afloramiento costeros peruanos influenciados por la corriente de Humboldt.

2) Darwin y Wolf, las islas más remotas en Galápagos, muestran una estructura trófica compleja dominada por una gran agregación de peces predadores que representan el 55% de la biomasa total del sistema.

3) La magnitud de las diferencias entre la biomasa de predadores observada entre Darwin y Wolf y el sistema de Floreana es consistente con otros estudios que comparan islas remotas y pobladas en el Océano Pacífico e Índico.

4) La gran cantidad de biomasa en el sistema del Canal Bolívar y Floreana es similar, pero significativamente menor en el sistema de Darwin y Wolf. Adicionalmente, la distribución de la acumulación de biomasa por nivel trófico difiere mucho entre los sistemas tropicales y fríos.

5) De acuerdo con varios índices ecosistémicos estimados por Ecopath, el estado de desarrollo y madurez en el sistema de Darwin y Wolf es mayor, mientras que las perturbaciones antropogénicas y oceanográfica son menores.

6) Especies identificadas como clave diferían entre las regiones biogeográficas. Los tiburones son especies clave en el sistema de Darwin y Wolf controlando depredadores intermedios, que son mucho más abundantes en esta zona, mientras que los leones marinos dominan en el sistema de afloramiento del Canal Bolivar.

7) Durante un evento fuerte de El Niño, los efectos de *bottom-up* controlan gran parte del sistema, debido a la gran reducción de nutrientes y fitoplancton. Las características generales del sistema muestran una respuesta similar a otros sistemas de afloramiento en la costa peruana, causando interrupciones en los flujos tróficos y manteniendo un nivel bajo en el desarrollo (pero muy eficiente).

8) Las simulaciones muestran que una fuerte presión pesquera en el sistema de Darwin y Wolf cambiaría enormemente la estructura trófica del sistema,

mostrando un estado alternativo, dominado por los erizos de mar, donde las diferencias biogeográficas ya no son evidentes.

En conclusión, los tres sistemas comparados en este estudio han mostrado diferencias en su entorno oceanográfico e impacto de la pesca. La selectividad de la pesca artesanal local por especies objetivo juega un papel importante en la estructuración de la comunidad béntica. Sin embargo, los efectos directos son variables espacialmente, dependiendo en parte de la proximidad a los puertos pesqueros. Las estrategias de conservación deben considerar las diferencias biogeográficas de cada sistema en términos funcionales y estructurales para mejorar la eficacia del plan espacial de manejo en la Reserva Marina de Galápagos. Además, se recomiendan estudios adicionales que permitan el mejoramiento de los modelos para futuras exploraciones enfocados a estrategias de manejo. El programa de monitoreo de pesca actual debe ser evaluado y adaptado para producir mejor georreferenciación de la información para proporcionar una entrada sólida para evaluar el efecto de la pesquería en la Reserva Marina de Galápagos.

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General Introduction

There is a strong global need to conserve and manage marine recourses (Costanza et al. 1997, Hourigan et al. 1998). One way to inform decision makers and marine park managers it to provide ecological interaction data about the dynamic nature of multifaceted systems. Modelling food webs (e.g. Ecopath with Ecosim) is considered a key element of an “ecosystem-based” approach to management (EPAP 1998). By providing a holistic approach, these models identify key shifts in marine populations arising from changes in prey availability and predation mortality and can simulate structural changes in marine systems occurring after natural disturbances (Aydin et al. 2007). In order for ecosystem-based models to actually reflect the system dynamics Aydin et al. (2007) suggests the need to improve 1) specific indicators to compare energy flow, 2) sensitivity of species to perturbations and the effects of predator-prey interactions, 3) additional data on ecological important but poorly understood species, and 4) alternative management strategies to consider long-term changes in food web structure and marine ecosystem services.

The mass-balance modelling approach (Ecopath) used for this study is a food web comparative analysis method developed by Laevastu and Favorite (1979) and posteriorly generalized and extended by Polovina (1985) and Christensen et al. (1992, 2000, 2004, 2008). Ecopath calculates the consistent estimation and comparison of mass-balance model between multiple marine ecosystems. The module Ecosim enables the temporal dimension to be considered and provides the theoretical means to project changes in the system’s component biomasses in respond to, harvest intensity, management policy implementation, and mortality induced by predation or natural events (e.g. El Niño South Oscillation events). Ecopath with Ecosim (EwE) is designed to use estimations of biomass, production, consumption, catches, and predator-prey interactions in a system, with either single species or functional group compartments. EwE models are ideally suited to the incorporation of data coming from single-species fishery surveys or underwater ecological monitoring. EwE models are also very useful for making summaries of the available data and trophic flows in a system model and also identifying gaps in one’s knowledge about an ecosystem (Christensen and Pauly 1993) to improve model outputs.

Ecosystem based models are widely used; for example, within the Galápagos Marine Reserve (GMR), EwE was employed to evaluate the artisanal fishery and conservation strategies for the sea cucumber (*Isostichopus fuscus*) in the shallow rocky reef of Floreana Island (Okey et al 2004). In this case, EwE accurately predicted the posterior collapse of the sea cucumber *I. fuscus* as a result of an unsustainably high artisanal fishery pressure. In this case, EwE was shown to resolve very specific conservation and management issues with a mass-balance modelling approach, which clearly demonstrated the importance of ecological data collection and analysis for improving our understanding of sustainable use of marine resources.

Oceanographic features of the Galápagos Marine Reserve

Located in the eastern equatorial Pacific, the Galápagos Marine Reserve (GMR) lies in an area of particular oceanographic conditions with a marked separation between warm waters in the north and cool waters in the south (Harris 1969, Palacios 2004, Fig. 1). Primary production is generally oligotrophic along the latitudinal sea surface temperature (SST) gradient except within the wind–induced equatorial upwelling condition localized on the western, which is topographically droved by the Equatorial under-current (EUC) (Gordon et al. 1998, Lindley and Barber 1998, Palacios 2004).

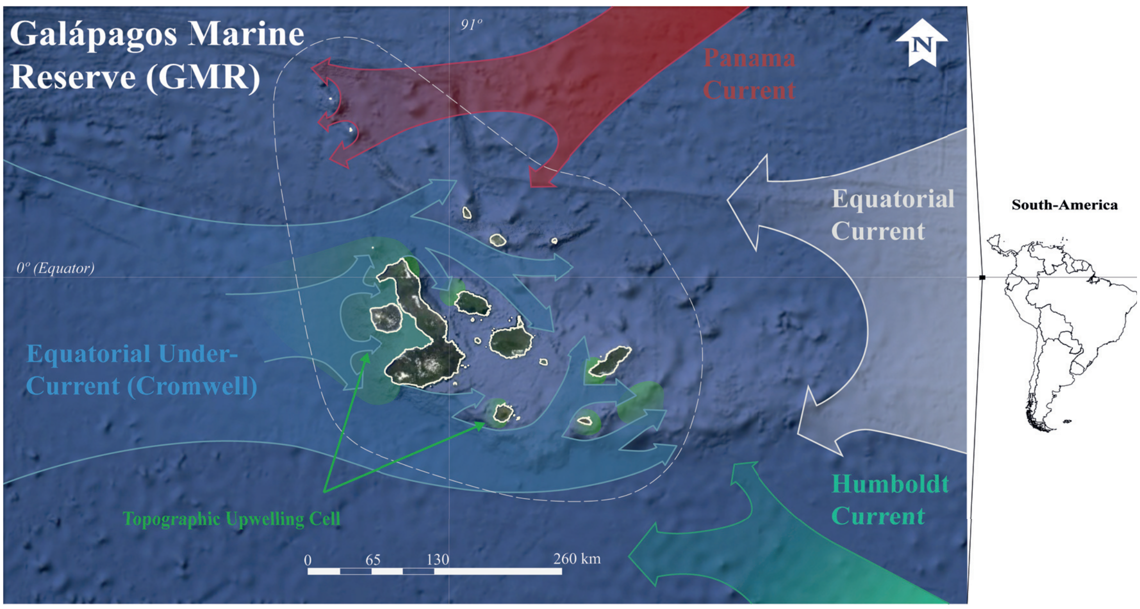

Fig. 1. Oceanographic characteristics in the Galápagos Archipelago (Source: base map – Google Earth and CDF).

Phytoplankton biomass, as estimated from chlorophyll a (Chl-*a*), shows a regional West-East gradient due to the excursions of the EUC at the southeast corner of the GMR, where small scale topographic upwelling occurred (Fig. 2). Enhanced Chl-*a* in the western side of the archipelago shows the greatest variability in terms of seasonal amplitude along the Equator (Palacios 2004). The inter-annual semi-permanent pool of Chl-*a* formed by the upwelling on the western side of the archipelago also responses to a localized source of iron from the western islands platform (Palacios 2004). The continuous influence of upwelling on the western region of the Galápagos Archipelago clearly contributes to the long-term establishment and evolution of distinctive biota (Abbott 1966, Glynn et al. 1983, Palacios 2004).

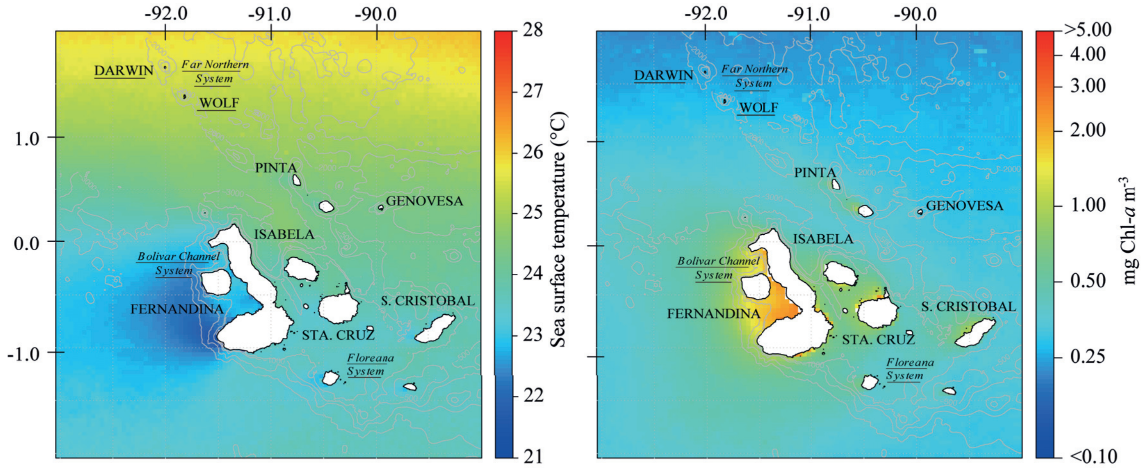

Fig 2. Sea surface temperature (SST) and chlorophyll-*a* (Chl-*a*) average for the period 2003-2014. (MODIS-Aqua daytime SST, 2003-2013, 4 km resolution) and Chlorophyll-*a* (Chl-*a*) (Globcolour merged product, 2003-2014, 4 km resolution).

Biogeography of the Galápagos Archipelago

The Galápagos Archipelago is one of the most interesting biogeographical settings on earth, primarily because it is the only tropical archipelago lying at the intersection of a major warm- and cold-water current system. Historically the region has been classified according to the naturally occurring temperature gradients dividing the archipelago into five regions (north, west, south, central and central mixing, Fig. 3; Harris 1969). Followed by biogeographical characteristics (Wellington 1975) and recently based on extensive biological and ecological characteristics (Edgar et al. 2004a). The work of Edgar et al. (2004a) proposes three major biogeography regions based on the community composition of the reef fishes, mobile macroinvertebrates, and benthic sessile organisms (Fig. 3). The three major bio-regions are: "*Western*" temperate–cold region, the "*Northern*" tropical-warm region, and "*South-central/eastern*" mixed temperate-subtropical region. These three biogeography regions are defined by their distinctive and unique biota, relating to tropical (e.g. manta rays, reef sharks, corals), temperate (sea lions, kelp) and subantarctic (fur seals, penguins, albatross) latitudinal regions in addition to components of the Indo-Pacific, Peruvian, Panamic-Caribbean, and local endemic species (Edgar et al. 2004a, James 1991, Kay 1991, McCosker and Rosenblatt 1984).

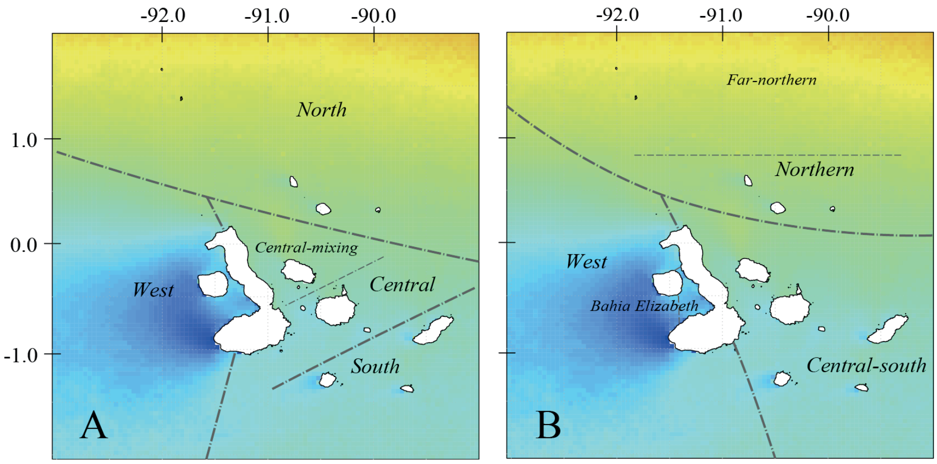

Fig. 3. A) Regionalization of the Galápagos Archipelago proposed by Harris (1969), and B) Actual biogeographic scheme proposed by Edgar et al (2004a).

El Niño – La Niña induced community changes in the Galápagos Marine Reserve

Historical information indicates the Galápagos marine ecosystems are not well adapted to extreme thermal impacts. Intertidal shores, shallow rocky, and coral reef habitats across Galápagos changed substantially during the 1982/83 and 1997/98 El Niño events (Glynn 1994, 2001, Robinson and del Pino1985, Fig. 4). These extreme thermal anomalies are characterised by elevated temperatures (persisting for over 12 months; Fig. 5), and major declines in dissolved nutrients causing a decline in phytoplankton productivity and increased UV radiation reaching the benthos. From an ecosystem-based perspective this also resulted in a prolonged reduction of biomass at the base of the marine food chain (Robinson and del Pino 1985).

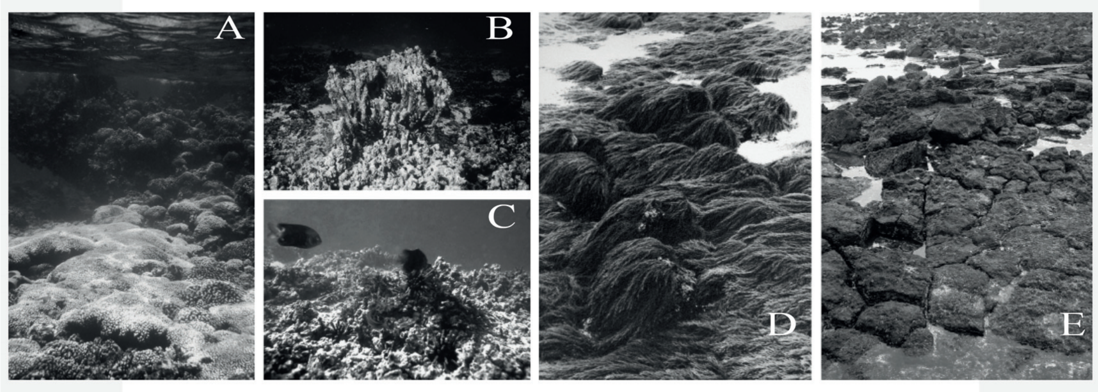

Fig. 4. Community changes induced by El Niño. Photo A, B and C, Devils Crown – Floreana Island, A) *Pocillopora damicornis* reef in 1982, B) remnants of coral framework with numerous urchins (*Eucidaris galapagensis*) in 1986 (B) – 1987 (C). Photo D and E show the rocky intertidal shore – front of the Charles Darwin Foundation in 1974 and 2003. A) Brown endemic alga (*Bifurcaria galapagensis*), distributed from the low intertidal to 6 m depth on moderately exposed coasts of southern and central islands (Wellington 1975). Probably extinct alter El Niño 1982/83 (Robinson and del Pino 1985) (Photo Archive, CDF - Marine Area Photographic Data Base, used as material for oral presentations, and posteriorly published in Edgar et al. 2010: A – B, photos P. Glynn; C – E, photos F. Rivera, D, photo G. Wellington)

Oceanographic warming associated with both El Niño and global climate change can directly influence species' populations (Helmuth et al. 2006, Thomas et al. 2004, 2006). Populations of endemic species, which occur mainly in the upwelling-western region, such as the Galápagos penguins (*Spheniscus mendiculus*), flightless cormorants (*Phalacrocorax harrisi*), Galápagos fur seals (*Arctocephalus galapagoensis*) and Galápagos sea lions (*Zalophus wollebaeki*) were the most affected (Robinson and del Pino 1985, Trillmich and Limberger 1985), through the bottom-up food deprivation and heat stress that restructured the entire benthic communities. Other habitat changes on shallow Galápagos reefs catalyzed by the 1982/1983 El Niño included loss of coral reefs through bleaching and subsequent sea urchin bioerosion (Glynn 1990, 1994, Glynn and Wellington 1983), and loss of most macroalgal beds (Robinson and del Pino 1985). A total of 95 – 99% of coral reef cover was lost from Galápagos between 1983 and 1985. All known coral reefs based on calcareous frameworks died and subsequently disintegrated to rubble and sand (Glynn 1994).

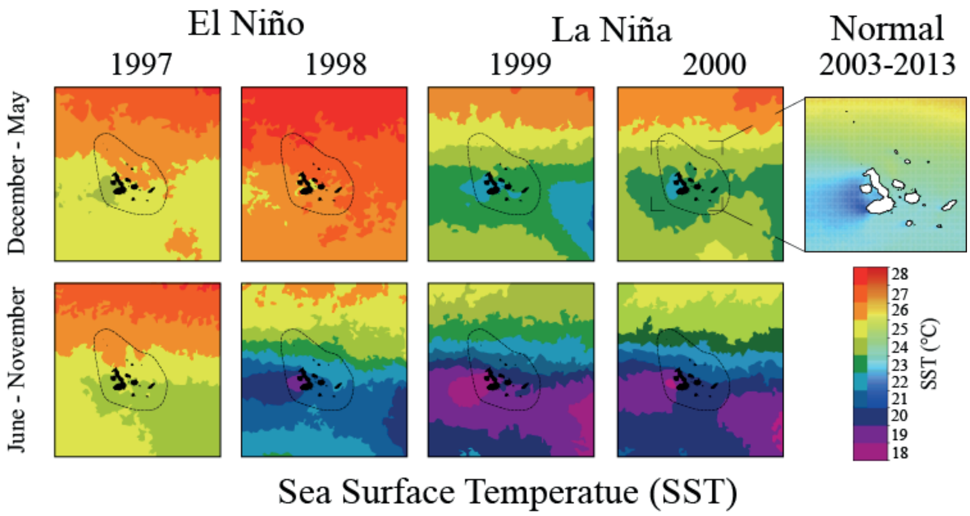

Fig. 5. Sea Surface Temperature (SST) anomalies shown as annual average value during El Niño (1997-1998) and La Niña (1999-2000) events in the Galápagos Marine Reserve (GMR). Warm season, December-May, and cold season, June-November.

Interactions between natural perturbation (e.g. climate change, El Niño) and fishing are suggested to have far-reaching consequences for marine species (Harley and Rogers-Bennett 2004, Hughes et al. 2003, Edgar et al. 2005, Pandolfi et al. 2005). In the Galápagos Archipelago, published studies and local observations spanning the last 40 years show a low degree of population recovery for many marine species since the catastrophic declines during El Niño events (e.g. Edgar et al. 2010, Robinson 1985) and excessive fishing pressure (Ruttenberg 2001, Sonnenholzner et al. 2009). Within this thesis is the first attempt to apply an ecotrophic approach to understanding the effects of El Niño on shallow rocky reef communities of the Galápagos Archipelago. An EwE spatial-temporal dynamic analysis is used to describe the changes in the ecosystem from an energy flow perspective, which leads to a greater understanding of the mechanisms shaping the community structure. Specifically, the Bolivar Channel in the Western region of the archipelago was modelled to investigate the influence of topographical upwelling during El Niño (see Chapter 2).

Effects of artisanal fisheries

The Galapagos archipelago has a long and rich history of artisanal fisheries, which include diverse target species ranging from large pelagic fish (e.g. tuna, wahoo) to reef associated holothurians. Reef communities' structure and habitats exhibit clear spatial changes according to the amount of artisanal fishing pressure in the Galápagos

Archipelago (Fig 6, Edgar et al 2010). This correlates to fishing effort and distance from fishing ports because relatively few fishers are prepared to undertake multiday trips (Born et al. 2003). However, fishing effort is not independent of the abundance and distribution of marine sources, high densities of resource species tend to attract fishers. Heavily fished sites could potentially possess low densities of resource species because fishing pressure has depleted resource species at those sites (Edgar et al 2010).

Densities of large predatory fishes (sharks, groupers, jacks, mackerel and snappers) increase with distance from fishing ports, as well as densities of spiny lobsters (*Panulirus penicillatus* and *P. gracilis*) and slipper lobsters (*Scyllarides astori*) in Galápagos. However, predatory fish populations also have been overexploited, including the endemic Galápagos grouper *Mycteroperca olfax*, the most heavily targeted reef species and biomass dominant amongst the benthic predatory fishes. This vulnerable species of grouper has been characterized as functionally extinct in the south-central region of the Galápagos Archipelago (Okey et al. 2004, Ruttenberg 2001).

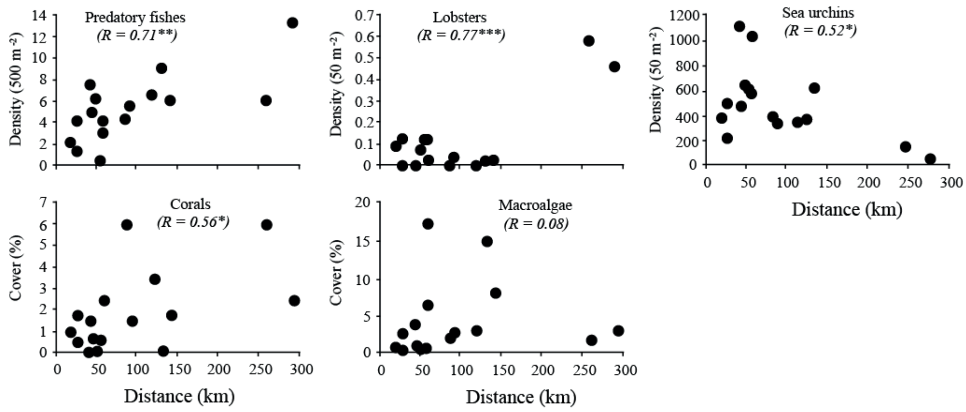

Fig. 6. Scatter plots shows relationships between total densities of large predatory fishes, spiny lobsters, sea urchins, corals and macroalgae and distance from fishing port. Pearson's correlation coefficients, with associated significance values based on two-tailed tests, are shown in parenthesis (for more details see: Edgar et al. 2010)

Overfishing in recent decades has reduced ecosystem resilience in Galápagos (Edgar et al. 2010). Specifically, overfishing has weakened predatory control exerted by lobsters and fishes on populations of urchins (primarily *Eucidaris galapagensis*, *Tripneustes depressus*, *Lytechinus semituberculatus*), resulting in greatly increased grazing pressure and loss of macroalgae as well as corals coverage (Edgar et al. 2010, Sonnenholzner et al. 2009). Significantly lower densities of urchins have been observed in well-protected marine areas where natural predators occur in high abundance, such as lobsters (*Panulirus* spp.) and hogfish (*Bodianus diplotaenia*), as compared to intensely fished sites in the Galápagos Archipelago and in the Eastern Tropical Pacific (Edgar et al. 2011).

Removal of higher predators by selective artisanal fisheries in the Galápagos Archipelago has demonstrated cascading effects on the community structure in heavily fished sites. In such a system state, the natural variability of the system structure and trophic interactions are reduced and in consequence fishing down the food web (Pauly et al. 1998). Predictable changes in certain species/components of the system, which is either directly or indirectly a result from over-fishing, was simulated by Esosim and showed varying scenarios of fishing efforts and their consequences in the system structure (see Chapter 3).

Aim of the thesis

The main goal of this thesis is to comparatively analyze the food web structure of three representative shallow rocky reef systems from each of the major biogeographical regions described by Edgar et al. (2004) The model outcomes are focused on variability of ecosystem functionality along a primary productivity, temperature and artisanal fishery gradient. The data used to validate and construct the steady-state models are based on the census data collected by the Charles Darwin Foundation (CDF) from 2004 to 2008. This data includes ecologically standardized underwater census, fishery landing reports, and marine mammals, seabirds, marine iguanas and sea turtles population censuses.

The specific objectives are to:

1) Integrate the information from the ecological surveys and the fishery data to construct a holistic quantitative energy flow model in order to identify and quantify main biological components and energy flow pathways.

2) Standardize a steady-state model with which to provide realistic comparisons and simulations scenarios of systems response to different fishing and climate pressures.

3) Identify the specific particularities of each biogeographical region of the Galápagos Archipelago through the identification of keystone species or

groups in the system, and explore the effects of biomass changes of these components to understand how they may impact other ecosystem components.

4) Elucidate the mechanisms behind the observed changes in the Galápagos Archipelago caused by the El Niño 1997/98 event by forcing model changes in primary producers.

5) Provide a holistic approach to improve conservation strategies for the management of the Galápagos Marine Reserve, by providing the first interpretation of the trophic differences between biogeography shallow rocky reef systems. This is important for effective resource based management and for setting conservation priorities.

Publication outline

This thesis consists of four chapters. Chapter 1 describes the characteristics of a topographic upwelling system in the western area of the Galápagos Archipelago. This study clearly demonstrates a bottom-up control driving the food web due to high primary productivity, which supports the larger marine organisms in a higher trophic level. Chapter 2 describes and compares observed biomass data and modelling simulation to understand the response of tropical upwelling marine communities to prolonged periods of low primary productivity and high temperatures that occurred during El Niño events. Subsequently, Chapter 3 describes a large-predator dominated system located in the remote northern area. Contrasting characteristics were evident when compared with the western upwelling system. Simulations of a potential increase in fishing efforts in the northern system reveals an alternative system state dominated by sea urchins, highlighting the importance of increase conservation strategies to maintain the functionality of this large-predator dominated system in a trophic balance state. Finally, in Chapter 4, a revisited steady-state model of the Floreana Island (Okey et al. 2004) was carried out for the same period considered by the model described in Chapters 1 and 3, to allow comparison between modelled systems and to understand differences along the natural gradient of primary productivity, temperature, and artisanal fishery intensity.

1) The Bolivar Channel Ecosystem of the Galápagos Marine Reserve: Energy flow structure and role of keystone groups

Authors: Diego J. Ruiz, Matthias Wolff

The idea was developed by DJ Ruiz with support from M Wolff. Data analyzing and writing were conducted by DJ Ruiz with improvements by M Wolff. (*Published in Journal of Sea Research*)

2) El Niño induced changes to the Bolivar Channel Ecosystem (Galápagos): comparing model simulations with historical biomass time series

Authors: Matthias Wolff, Diego J. Ruiz, Marc Taylor

The idea was developed by M Wolff and DJ Ruiz. Data analyzing was conducted by DJ Ruiz, writing was conducted by M Wolff and DJ Ruiz with improvements by M Taylor. (*Published in Journal of Marine Ecology Progress Series - MEPS*)

3) Elucidating fishing effects in a large-predator dominated system: the case of Darwin and Wolf Islands (Galápagos)

Authors: Diego J. Ruiz, Matthias Wolff

The idea was developed by DJ Ruiz with support from M Wolff. Data analysis and writing were conducted by DJ Ruiz with improvements by M Wolff. (*Re-submitted to Journal of Sea Research*)

4) Trophic flow structure of shallow rocky reef systems of the Galápagos Marine Reserve along a biogeographical gradient

Authors: Diego J Ruiz, Matthias Wolff, Marc Taylor

The idea was developed by DJ Ruiz with support from M Wolff. Data analysis and writing were conducted by DJ Ruiz with improvements by M Wolff and M Taylor. (*Submitted to Journal of Sea Research*)

1

The Bolivar Channel Ecosystem of the Galápagos Marine Reserve: Energy flow structure and role of keystone groups

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Abstract: The Bolivar Channel Ecosystem is among the most productive zones in the Galápagos Marine Reserve. It is exposed to relatively cool, nutrient-rich waters of the Cromwell current, which are brought to the photic zone through topographic upwelling. The BCE is characterized by a heterogeneous rocky reef habitat covered by dense algae beds and inhabited by numerous invertebrate and fish species, which represent the food for higher predators including seals and sharks and exploited fish species. In addition, plankton and detritus based food chains channel large amounts of energy through the complex food web. Important emblematic species of the Galápagos Archipelagos reside in this area such as the flightless cormorant, the Galápagos penguin and the marine iguanas. A trophic model of BCE was constructed for the habitats < 30 m depth that fringe the west coast of Isabela and east coast of Fernandina islands covering 14% of the total BCE area (44 km^2). The model integrates data sets from sub tidal ecological monitoring and marine vertebrate population monitoring (2004 to 2008) programs of the Charles Darwin Foundation and consists of 30 compartments, which are trophically linked through a diet matrix. Results reveal that the BCE is a large system in terms of flows (38695 t km^{-2} $year^{-1}$) comparable to Peruvian Bay systems of the Humboldt upwelling system. A very large proportion of energy flows from the primary producers (phytoplankton and macroalgae) to the second level and to the detritus pool. Catches are high (54.3 t km^{-2} $year^{-1}$) and are mainly derived from the second and third trophic levels (mean TL of catch = 2.45) making the fisheries gross efficiency high (0.3%). The system's degree of development seems rather low as indicated by a P/R ratio of 4.19, a low ascendency (37.4%) and a very low Finn's cycling index (1.29%). This is explained by the system's exposure to irregular changes in oceanographic conditions as related to the EL Niño Southern Oscillation. Most important keystone groups of large relative impact over other system compartments are sharks and marine mammals. In addition, the important role of macroalgae, sea stars and urchins, phytoplankton and barracudas should be emphasized for their great contribution to the trophic flows and biomass of the system.

Keywords: trophic modelling, mass balance, Bolivar Channel System, Galápagos, Ecopath with Ecosim.

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Introduction

The Galápagos Archipelago is located in the Eastern Tropical Pacific, approximately 1000 km off the coast of Ecuador, South America. The islands represent the summits of volcanoes that emerged from the sea approximately 1-3 million years ago (Christie et al. 1992). These islands sit on a relatively shallow platform (< 200 m) surrounded by deeper waters (> 1 000 m). They are under the confluence of three dominant oceanic currents (Chavez and Bursca 1991): the South-equatorial Current conformed by the confluence of the Panama Current from the North-east and the Peru or Humboldt Current from the south east and the Equatorial Undercurrent (EUC) or Cromwell Current from the west (Muromtsev 1963), which brings cold upwelling waters mainly to the western part of the Archipelago.

The shallow habitats of the Bolivar Channel (Fig. 1) (west of the Archipelago), are heavily influenced by these cold and nutrient-rich upwelling waters. They are conformed mainly by rocky reef areas of bedrock, boulder, cobbles and sand patches that spread along the coastline, interrupted by small sandy beaches and mangrove patches. The Bolivar Channel system supports high resource biomasses of small pelagic fish such as sardines, thread-herrings, anchovies, pompanos and mackerels that in turn are the prey for substantial populations of top predators such as sharks, tunas, wahoos, barracudas, dolphins, seabirds and marine mammals (Feldman 1985, 1986), many of which actively visit the rocky reefs to feed.

This productive pelagic system surrounds, and interfaces with the benthic rocky reef habitats of Fernandina and Isabela islands on the west and east sides of the channel. High production and accumulation of phytoplankton through the confluence of ocean currents as well as dense macro-algae beds provide elevated levels of primary production to these reefs. Many planktivorous fish that inhabit the rocky reef areas feed in the water column. The most prominent of this group is the medium sized gringo (*Paranthias colonus*), which frequently forms dense shoals of several hundred specimens but there are many other pelagic and benthic planktivorous species. The plankton also supports high biomasses of benthic filter-feeders that include gorgonians (*Muricea* sp. and *Pacifigorgia* sp.), zoanthids (*Parazoanthus* spp.), sponges (*Aplysilla* sp. and *Carmia* sp.), polychaets (*Spirobranchus giganteus*) and ahermatypic corals (*Tubastraea* sp.).

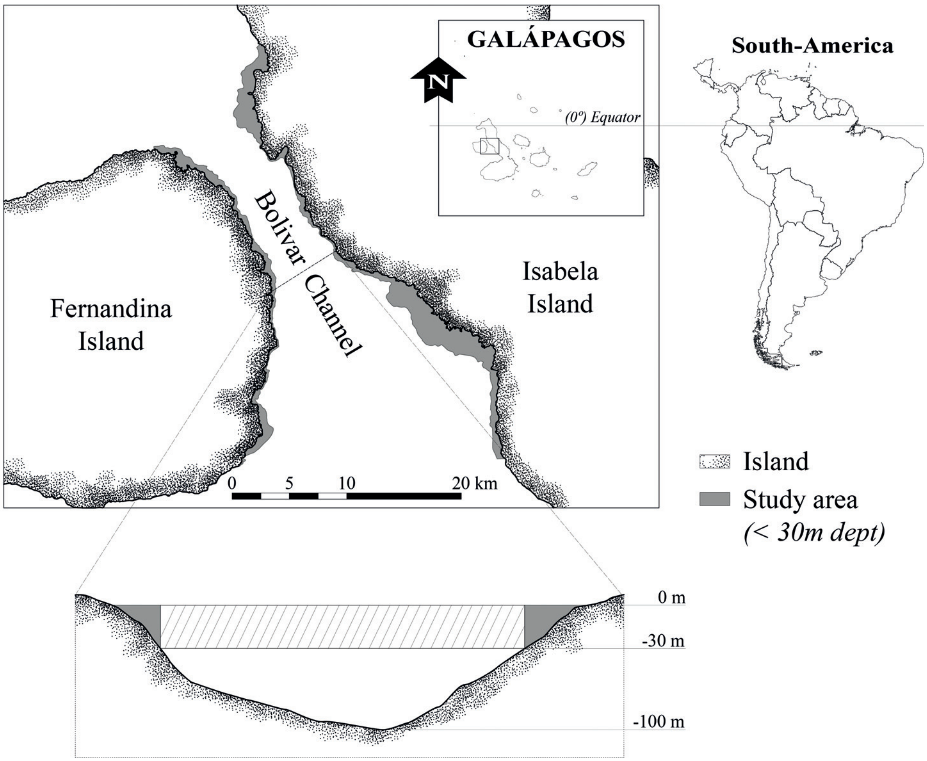

Fig. 1. Bolivar Channel System, located between Isabela and Fernadina islands, west of the Archipelago. The traverse cut show the bathymetric profile of the Bolivar Channel, the areas in grey are shown the zones considered for the model and the lined area represents the pelagic zone which connects the study areas.

In some parts of the Bolivar Channel macroalgae (*Ulva* sp., *Sargassum* sp., *Spatoglossum* sp. and *Heterosiphonia* sp.) form dense beds, while others are covered with filamentous algae, encrusting algae or/and benthic diatoms, providing large food sources for several invertebrates, fish and other vertebrates, including the marine iguanas (*Amblyrhynchus cristatus*).

Two abundant species of sea urchins (*Lytechinus semituberculatus*, *Eucidaris galapagensis*) cause intense levels of grazing on the benthic primary producers (Ayling 1981, Breen and Mann 1976, Himmelman and Lavergne 1985). Herbivorous fish include damsels, surgeon and parrotfishes. Marine turtles (*Chelonia mydas*) are also present, in some cases in high abundances. Several species of sea cucumbers (*Holothuria* spp. and *Isostichopus fuscus*) are also highly abundant in this area and make use of the rich detritus food supply in the area.

Omnivorous reef fishes are mainly conformed by butterflies, damsels and chopas, which consume both algae and small invertebrates. Grunts, wrasses and angel fishes are mainly invertebrate feeders, as are the carnivorous invertebrates, which include lobsters (*Panulirus* spp. and *Scyllarides astori*), crabs, sea stars (*Pentaceraster cumingi* and *Asteropsis carinifera*) and gastropods (*Pleuroploca princeps*).

In the higher trophic levels we find piscivorous fishes like the endemic grouper of Galápagos (*Mycteroperca olfax*), different species of snappers, and octopus, sea lions (*Zalophus wollebaeki*), the Galápagos penguin (*Spheniscus mendiculus*) and the flightless cormorant (*Nannopterum harrisi*) should also be mentioned here. Several sharks like the Galápagos shark (*Carcharhinus galapagensis*) and the white tip-reef shark (*Triaenodon obesus*), also feed in these areas and interact with the pelagic system.

The trophic structure of the BCE is subjected to direct and indirect changes caused by anthropogenic as well as climatic impacts. The intense harvest of resources like lobsters (*Panulirus* spp.), sea cucumbers (*I. fuscus*) and several finfish species is already affecting biomass levels and productivity of these species, and oceanographic variability such as the El Niño/La Niña oscillation impacts on the Bolivar Channel and other marine communities of the Galápagos Archipelago, and their eco-systemic services (e.g. Bost and Le Maho 1993, Colinvaux 1972, Houvenaghel 1984, Glynn 1988).

Prior to this study, an Ecopath model was constructed for the Galápagos rocky reef system of Floreana Island by Okey et al. (2004), which allowed for a first characterization of food web structure of this system. The model was also used to explore hypotheses about system dynamics and to propose potential solutions to the overfishing problem in the Galápagos Marine Reserve with emphasis on the sea cucumbers (*I. fuscus*) fishery.

The objectives of the present study were to: 1) identify the main functional compartments of the BCE and their biological elements, 2) integrate the available ecological and fishery information of this system to build a quantitative trophic model, 3) compare global statistics of the flow structure with other tropical subtidal systems modeled, included the Floreana model, to identify the specific particularities of the BCE, 4) identify keystone groups of the system and to explore how biomass changes of these groups may impact on the system, and 5) provide a reference model for future simulations of the effects of different fishing and climate regimes on the system.

Methods

Model definition:

The model area chosen was the rocky habitat (< 30 m depth) fringing the eastern and western sides of Fernandina and Isabela respectively, covering 14% of the total BCE area (Fig. 1). For model construction the Ecopath module of Ecopath with Ecosim 6.0 (EwE) (Christensen et al. 2008) was used, which allows for the visualization and quantification of trophic flows within an ecosystem and to include the fisheries, and for evaluating specific ecosystem properties (http://www.ecopath.org).

30 functional compartments were identified (Table 1) and species were grouped into these if having similar diets, similar predators, similar body size and similar metabolic requirements. For each compartments key input parameters were biomass (B), production rate (P/B), consumption rate (Q/B) and catch (C) (in case of a fisheries resource). The compartments are linked trophically through a diet matrix. The model unit used was g wet weight per m^2. In Ecopath, the production of each group is balanced by losses in the ecosystem as given in the following equation (1):

$$P_i = Y_i + B_i * M2_i + E_i + BA_i + P_i * (1 - EE_i) \qquad (1)$$

Where P_i is the proportion of the total production of (i), Yi is the total of the fishery catch proportion of (i), B_i the biomass of the group (wet weight), $M2_i$ is the rate of total predation for the group (i), E_i the proportion of net migration (emigration-immigration), BA_i is the accumulated proportion of the biomass for (i), while P_i * (1 - EE_i), is "other mortality" ($M0_i$) which is not attributed to predation and fisheries mortalities, where EE_i is the ecotrophic efficiency, being the proportion of the production of each group which is consumed by the higher trophic levels or are extracted by the fisheries (for further details, see Christensen et al. 2000). To assure the balance of the mass among the groups, an energy balance equation is used (2):

$$\textbf{\textit{Consumption = production + respiration + unassimilated food}} \qquad (2)$$

For energy to flow in the model, a diet matrix has to be defined which connects all model groups with each other. In this diet matrix the fraction of each functional group that will serve as food to another group has to be defined. This diet matrix allows calculating the trophic level of each group according to equation (3).

$$TL_j = 1 + \sum TL_i * DC_{ij} \qquad (3)$$

Where, DC_{ij} is the fraction of the prey (i), in the predator diet (j). The trophic level of the predator TL_j is calculated as the average sum of the trophic level of its prey and its fraction in the predator's diet ($\sum TL_i * DC_{ij}$) added 1.0. The groups of primary producers and detritus are assigned as trophic level 1.0 as default (Christensen et al. 2008). The trophic level concept (Lindeman 1942) allows aggregated components of a food web into a discrete trophic level (Ulanowicz and Kay 1991), giving an indication of the average number of steps in food webs and their efficiencies of transferring material and energy. Transfer efficiency (TE) is the fraction of the total throughput at a discrete trophic level (TL), either exported or transferred to another TL (through consumption) (Shannon et al. 2003)

Input parameters

Biomass

Mean phytoplankton biomass was estimated and averaged for the period September1997 to December 2009, excluding El Niño 1982/83 and 1997/98, using the ESA Globcolour database (http://hermes.acri.fr/), which consists of SeaWiFS from before April 2002, and a merged product of SeaWiFs, MODIS, and MERIS thereafter. For converting chlorophyll a (Chl-*a*) concentrations (mg m^{-3}) to wet weight biomass the conversion factors were: Chl-*a* – Carbon (40:1; Brush et al. 2002) and Carbon – wet weight (1:14.25; Brown et al. 1991). The phytoplankton water column biomass was then estimated for a mixed layer depth of 20 m yielding 30g m^{-2}. Macro-algal biomass on the rocky reef at Isabela and Fernandida Islands was based on measured standing wet biomass at two low intertidal sites on Fernadina Island, and based on subtidal observations during the Subtidal Ecological Monitoring (SEM) that the Charles Darwin Foundation (CDF) has carried out at 14 sites in the BCE, between the years 2004 to 2008 (Banks et al. 2006).

Table 1. Functional groups and representative species for the steady-state model of the Bolivar Channel Ecosystem.

Functional group	Species
Phytoplankton	Cianobacters, Diatoms, Coccolithophorids, Dinophagellets
Macroalgae	*Ulva* sp. (90.8%), *Sargassum* sp. (8.0%), *Gymnogongrus* sp., *Pocockiella* sp., *Ralfsia* sp.
Herbivorous reef fish	*Girella freminvillei* (80.6%), *Prionurus laticlavius* (17.1%), *Microspathodon dorsalis* (2.3%)
Sea cucumbers and other	*Isostichopus fuscus*, *Stichopus horrens*, *Nodipecten magnificus*
Herbivorous zooplankton	*Sergestes* sp., *Pagurus* sp.
Sea turtles and marine iguanas	*Chelonia mydas* (64.0%), *Amblyrhynchus cristatus* (36.0%)
Small herbivorous gastropod	*Cerithium* sp., *Rhinoclavis gemmata*, *Tegula cooksoni*
Sponges and polychaetes	*Aplysilla sulphurea*, *Carmia* sp., *Hyotissa solida*, *Spirobranchus giganteus*
Gorgonians	*Muricea* sp. (91.3%), *Pacificgorgia* sp. (8.7%)
Parrotfish	*Scarus* spp.
Mullets	*Mugil galapagensis*
Benthic omnivorous fish	*Anisotremus scapularis* (70.8%), *Stegastes leucorus beebei* (21.0%), *Oplegnathus insignis* (5.6%), *Johnrandallia nigrirostris* (1.0%), *Ophioblennius steindachneri* (0.8%), *Chaetodon humeralis* (0.7%), *Canthigaster punctatissima*
Anemones and zoanthids	*Bunodosoma grandis*, *Parazoanthus* sp.
Sea stars and sea urchins	*Eucidaris galapagensis* (38.6%), *Nidorellia armata* (28.4%), *Diadema mexicanum* (20.4%), *Tripneustes depressus* (10.4%), *Lytechinus semituberculatus* (2.2%)
Planktivorous reef fish	*Chromis* sp. (2.7%), *Paranthias colonus* (97.3%)
Small planktivorous reef fish	*Xenocys jessiae* (67.8%), *Abudefduf troschelii* (26.7%), *Apogon* sp. (5.6%)
Lobsters	*Panulirus gracilis* (7.1%), *Panulirus penicillatus* (8.0%), *Scyllarides astori* (84.8%)
Predatory zooplankton	*Abylosis* sp., *Agalma* sp., *Atolla* sp., *Aurelia aurita*, *Beroe* sp., *Canthocalanus pauper*, *Carinaria* sp., *Cestum* sp., *Cladonemas* sp., *Creseis* sp.
Predatory macroinvertebrates	*Octopus* sp. (58.8%), *Asteropsis carinifera* (13.0%), *Heliaster cumingii* (27.7%)
Small predators gastropods	*Conus* sp. (43.6%), *Columbella* sp. (38.5%), *Muricopsis zeteki* (16.0%), *Thais* sp. (2.0%)
Small benthic predatory fishes	*Anisotremus interruptus* (29.6%), *Haemulon* sp. (21.3%), *Holacanthus passer* (16.7%), *Halichoeres* sp. (8.2%), *Calamus* sp. (7.7%), *Arothron meleagris* (5.1%), *Labrisomus dentriticus* (3.8%), *Scorpaena plumieri mystes* (2.7%), *Alphestes immaculatus* (2.6%), *Synodus lacertinus* (0.9%), *Thalassoma lucasanum* (0.6%), *Serranus psittacinus* (0.4%), *Cirrhitichthys oxycephalus* (0.1%), *Hippocampus ingens* (0.1%), *Plagiotremus azaleus* (0.1%), *Lepidonectes corallicola*, *Malacoctenus tetranemus*
Benthic predatory fishes	*Lutjanus* sp. (51.0%), *Paralabrax albomaculatus* (10.6%), *Epinephelus* sp. (8.5%), *Balistes polylepis* (7.3%), *Bodianus* sp. (5.4%), *Caulolatilus princeps* (4.8%), *Cirrhitus rivulatus* (3.4%), *Diodon holocanthus* (3.3%), *Dermatolepis dermatolepis* (1.9%), *Sufflamen verres* (1.1%), *Cephalopholis panamensis* (0.1%), *Chilomycterus reticilatus*
Barracudas	*Sphyraena idiastes*
Groupers	*Mycteroperca olfax*
Pelagic predatory fishes	*Scomberomorus sierra* (77.8%), *Seriola rivoliana* (22.2%)
Rays	Aetobatus narinari (81.0%), Dasyatis sp. (7.0%), Taeniura meyeni (11.5%)
Predatory marine mammals	*Arctocephalus galapagoensis* (1.4%), *Orcinus orca* (44.9%), *Tursiops truncatus* (47.1%), Zalophus wollebaeki wollebaeki (6.6%)
Seabirds	*Phalacrocorax harrisi* (74.1%), *Spheniscus mendiculus* (25.9%)
Sharks	*Carcharhinus galapagensis*, *Heterodontus quoyi*, *Triaenodon obesus*
Detritus	

Zooplankton biomass was estimated from the oceanographic monitoring that NASA, CDF, and Galápagos National Park have carried out during the period 2004 – 2006 (Tirado et al. 2009).

The biomass of benthic fish (g m^{-2} of wet weight) was calculated using the length-weight relationship and the relative abundance of the species registered during the SEM (Banks et al. 2006). To calculate the wet weight (g) of each species we applied the following equation (Fulton 1904, Ricker 1975) (4):

$$W = a * L^{b} \qquad (4)$$

Where W is the wet weight expressed in grams (g), L is the mode length of the species registered during the SEM, expressed in centimeters (cm), and, *a* and *b* parameters are the coefficients of linear regression length-weight relationship for each species obtained from FishBase (Froese and Pauly 2011).

The weight obtained for an average individual fish was then multiplied by the number of individuals counted per square meter (N m^{-2}), obtaining as result the biomass of each species in g m^{-2}.

The biomass of benthic macro-invertebrates (g m^{-2}) was calculated using the relative abundance of the species (sea stars, sea cucumbers, sea urchins and mollusks) registered at the same transects sites as the fish species (Banks et al. 2006).

The relative abundance of each species was multiplied by the wet weight (g) of this species obtained from field samples, and in some cases a dry to wet weight (g) conversion had to be done for some sampled specimens from the CDF Museum, using conversion factors as given in Opitz (1996).

Seabird, marine mammals and reptile biomasses (g m^{-2}) were calculated using the available information of visual census that CDF has carried out for these groups during the 2004 - 2006 period at the Bolivar Channel and literature resources (Snell and Márquez 2002, Vargas et al. 2005, Wilson et al. 2008, FCD Vertebrates Census Data Set). The relative abundance for each vertebrate species estimated for the study area was multiplied for the mean individual weights in g obtained from the literature.

Catch and predator-prey matrix

The catch estimates were based on the assumption that most target resources are heavily but not fully exploited with catch amounting to one third of annual production

(= 0.3 P/B ∗ B). This assumption had to be done since available catch data are not georeferenced, and the proportion of the BCE to the Galápagos catches can only be approximated (Castrejón Mendoza 2008, Hearn et al. 2004a, 2004b). The diet matrix was constructed based on general knowledge from the literature, using FishBase (Froese and Pauly 2011) and other published models (Opitz 1986, Taylor et al. 2008a, 2008b).

Production/Biomass (P/B) and Consumption/Biomass (Q/B)

These values for fish and vertebrates were obtained from other published models and literature sources (Pauly et al. 2003, Okey et al. 2004, Taylor et al. 2008a, 2008b). P/B was usually estimated by assuming that it equals total mortality (Z) under the assumption of population equilibrium (Allen 1971) and adjusted proportionally as weighted estimates for all species in a functional group whenever possible. For macro-invertebrates Brey's Multi-Parameter P/B Model was used (Brey 2001) setting mean water temperature at 15 °C. The values for aggregated groups were estimated as derived as averages of species-specific estimates weighted by relative biomass (B) or consumption (Q) as appropriate (Okey et al. 2004). Q/B was most commonly estimated from the empirical relationship proposed by Palomares and Pauly (1999), setting mean water temperatures at 15 °C.

A summary of the sources of information and population data used for the input parameters Biomass (B), the Production/Biomass (P/B), and the Consumption/Biomass (Q/B) is given in Table 2.

Mass balance, flow characteristics and summary statistics

The initial model constructed from input data was unbalanced, the estimated production of many of the model groups was insufficient to support the estimated consumption by other groups. Estimates of diet composition were uncertain due to lack of dietary data, which explains estimated predation on some components exceeding their production. Therefore, it was necessary to alter the contribution of some preys in the diet of certain predators.

The missing biomass values for three groups (small herbivorous gastropods, sponges and polychaetes, and anemones and zoanthids) were left in blank to be estimated by Ecopath fixing the ecotrophic efficiency (EE) values at 0.95.

Table 2. Input data source for estimate values by the Bolivar Channel Ecosystem model. *B* biomass, P_i/B_i production rate, Q_i/B_i consumption rate. *Nomenclature*: EM: Ecological monitoring OM Oceanographic monitoring, PM: Population monitoring, GE: guess estimate, EO: Ecopath output

Functional groups / Parameter	B_i (t km^{-2})	P_i/B_i (year^{-1})	Q_i/B_i (year^{-1})
Phytoplankton	ESA Globcolor database	Opitz (1996), Taylor et al. (2008)	-
Macroalgae	EM (2004 to 2008) and Vinueza et al. (no published data)	Macchiavello et al. (1987)	-
Herbivorous benthic fish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Okey et al. (2004), Pauly et al. (2003)
Sea cucumbers and other	EM (2004 to 2008)	Brey (2001), Opitz (1996)	Opitz (1996) and Okey et al. (2004)
Herbivorous zooplankton	OM (2004 to 2006)	GE	GE
Sea turtles and marine iguanas	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Opitz (1996), Okey et al. (2004)
Small herbivorous gastropod	EO	Brey (2001), Opitz (1996)	Opitz (1996), Okey et al. (2004)
Sponges and polychaetes	EO	Opitz (1996)	Opitz (1996), Okey et al. (2004)
Groupers	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Okey et al. (2004)
Gorgonians	EM (2004 to 2008)	Opitz (1996)	Opitz (1996)
Parrotfish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. 2003
Mullets	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Okey et al. (2004)
Benthic omnivorous fish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Okey et al. (2004)
Anemones and zoanthids	EO	Opitz (1996)	Opitz (1996)
Sea stars and sea urchins	EM (2004 to 2008)	Opitz (1996), Ortiz and Wolff (2002)	Opitz (1996), Taylor et al. (2008)
Planktivorous reef fish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003)
Small planktivorous reef fish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003)
Lobsters	EM (2004 to 2008)	Brey (2001), Okey et al. (2004)	Opitz (1996), Okey et al. (2004)
Predatory zooplankton	OM (2004 to 2006)	GE	GE
Predatory macroinvertebrates	EM (2004 to 2008)	Brey (2001), Taylor et al. (2008)	Opitz (1996), Taylor et al. (2008)
Small predators gastropods	EM (2004 to 2008)	Brey (2001), Opitz (1996)	Opitz (1996), Okey et al. (2004)
Small Benthic predatory fishes	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Okey et al. (2004)
Benthic predatory fishes	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Okey et al. (2004)
Barracudas	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003)
Pelagic predatory fishes	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011), Pauly et al. (2003)
Rays	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004)	Froese and Pauly (eds) (2011) and Pauly et al. (2003)
Predatory marine mammals	PM (2003 to 2007)	Opitz (1996), Okey et al. (2004)	Opitz (1996) , Okey et al. (2004)
Seabirds	PM (2005 to 2007), Wilson et al. (2008)	Opitz (1996), Moloney et al. (2005), Okey et al. (2004)	Opitz (1996), Okey et al. (2004)
Sharks	EM (2004 to 2008)	Opitz (1996), Okey et al. (2004)	Opitz (1996), Okey et al. (2004)

Once the model was balanced through a manual process of adjusting some of the input parameters, the model was subjected to the resampling routine Ecoranger, which draws a set of random input variables from normal distributions for each basic parameter. We started this process by allowing the routine to use confidence intervals around the different input parameters as derived from a pedigree of the data sources. However, since Ecoranger was unable to deliver successful runs, we decided to fix all confidence intervals at 20% and to re-run Ecoranger as was similarly done by Arias-González et al. (1997) and Taylor et al. (2007). We allowed resampling until 10 000 runs passed the selection criteria. The best run was chosen as that with the smallest sum of square residuals between the input parameters and the mean value of all successful runs (for more information, see Christensen et al. 2000). The resulting steady-state model is shown in table 4 and selected statistics were calculated with the output Ecoranger data that allow for a description of the system state (Table 3).

Table 3. Description of flow characteristics and summary statistics

Name of index	Meaning
Cycling index	Fraction of an ecosystem's throughput that is recycled (Finn 1976)
Connectance index	The ratio of the number of actual links to the number of possible links. Feeding on detritus (by detritivores) is included in the count, but the opposite links (i.e. detritus 'feeding' on other groups) are disregarded
System omnivory index	The average omnivory index of all consumers weighted by the logarithm of each consumer's food intake. It is a measure of how the feeding interactions are distributed between trophic levels. An omnivory index is also calculated for each consumer group, which is a measure of the variance of the trophic level estimate for the group (Pauly et al. 1993a)
Trophic transfer efficiencies	Calculated for a given trophic level as the ratio between the sum of the exports plus the flow that is transferred from one trophic level to the next, and the throughput on the trophic level. The transfer efficiencies are used for construction of trophic pyramids, and others.
Primary production required (PPR)	To estimate the PPR (Christensen and Pauly 1995) to sustain the catches and the consumption by the trophic groups in an ecosystem the following procedure is used: all cycles are removed from the diet compositions, and all pathways in the flow network are identified using the method suggested by Ulanowicz (1995). For each pathway the flows are then raised to primary production equivalents using the product of the catch, the consumption/production ratio of each path element times the proportion the next element of the path contributes to the diet of the given path element.
Mixed trophic impact (MTI):	Network mixed trophic impact analysis, based on Leontief's economic input–output analysis, allows expressing the relative change of biomasses in the food web that would result from an infinitesimal increase of the biomass of the observed group, thus identifying its total impact. It includes both direct and indirect impact (i.e. both predatory and competitive interactions) (Libralato et al. 2006)
Keystoneness	The "keystoneness" of a species in a given ecosystem may be formulated by considering: a) the total impact it causes on the different elements of an ecosystem resulting from a small change to the biomass of the species (based on MTI analysis) and b) its own biomass (Libralato et al. 2006)
Ascendency	EwE includes a number of indices related to the ascendency measure described in detail by Ulanowicz (1986). Ascendency is seen as a measure of ecosystem growth and development (Ulanowicz and Norden 1990).
Gross efficiency of the catch	Catch divided by the net primary production
Mean trophic level of the catch	Calculated by the proportions and trophic levels of all groups extracted by the fishery (Pauly et al. 1998)
Total System Throughput	Sum of all flows in the system, indicative for the size of the system (Christensen and Pauly 1993)
System Ascendency	Measure for System state of growth and development (Ulanowicz 1986)

Trophic impacts were estimated for each pair of functional groups (*preys* and *predators*, interacting directly or not) by means of the net impact matrix (Libralato et al. 2006). The net impact of preys on predators is given by the difference between positive effects, quantified by the fraction of the prey in the diet of the predator, and negative effects, evaluated through the fraction of total consumption of preys used by predator (Ulanowicz and Puccia 1990). The mixed trophic impact was then estimated by the product of all the net impacts for all the possible pathway in the trophic web that link the functional preys and predators groups.

The method for identifying keystone species is derived from the mixed trophic impact (MTI) analysis, proposed by Libralato et al. (2006). Keystone species are those that show relatively low biomass but have a structuring role in the ecosystem (Power et al. 1996).

The total flows within the ecosystem were quantified in terms of consumption, production, respiration, exports, imports and flows to detritus (t km^{-2} year^{-1}). The sum of all these flows is the total system throughput (T is a measure of ecosystem size (Christensen and Pauly 1993).

Based on our bi-annual monitoring, which did not reveal important seasonal changes in biomass pools, we assumed that immigration and emigration processes were generally balanced, maintaining relatively constant and characteristic biomass levels in the area. According to the classification of biogeographic zones by Edgar et al. (2004a) - based on the most comprehensive, on-going estimates of mobile macro-invertebrate and reef fish biodiversity within the GMR - the BCE lies within a characteristic zone of unique water-mass properties, and conditions of a quasi-consistent forcing regime that favors the introduction of particular colonizers and the development and maintenance of a distinct species assemblage (Houvenaghel 1978).

Results

A first balanced model was successfully constructed for the Bolivar Channel ecosystem through a manual balancing procedure that was guided by the pedigree values (Christensen and Walters 2000) given to the primary input data, which provided a range within which the original input data were allowed to be varied. Large biomass reductions to our initial estimates (> 40%) were needed for the groups mullets, benthic predatory fish, sea cucumbers, sea turtles, macroalgae, gorgonians, sea stars and sea urchins, barracudas, planktivorous reef fish, predatory mammals, rays and sharks, while positive biomass changes (> 40%) had to be done for herbivorous zooplankton, predatory macroinvertebrates, lobsters, predator zooplankton, seabirds and small planktivorous fish. In a second step, the Ecoranger resampling routine was applied and yielded a new balanced model, which was very similar to the original model, with the new parameters deviating less than 1% from the original input data (see Appendix 1).

Appendix 2 shows the diet matrix used and Table 4 gives the system summary statistics computed.

Table 4. Selected summary statistics estimated for the Bolivar Channel Ecosystem.

Parameter	Value
Sum of all consumption (t km^{-2} year^{-1})	6 940.401
Sum of all exports (t km^{-2} year)	13 024.699
Sum of all respiratory flows (t km^{-2} year^{-1})	4 076.784
Sum of all flows into detritus (t km^{-2} year^{-1})	14 653.101
Total system throughput (t km^{-2} year^{-1})	38 694.984
Sum of all production (t km^{-2} year^{-1})	18 577.021
Mean trophic level of the catch	2.448
Gross efficiency (catch/net primary production, %)	0.300
Calculated total net primary production (t km^{-2} year^{-1})	17 101.486
Total primary production/total respiration	4.195
Total primary production/total biomass	13.400
Total biomass/total throughput	0.300
Total biomass (excluding detritus) (t km^{-2} year^{-1})	1 276.278
Total catches (t km^{-2} year^{-1})	54.304
Connectance Index	0.169
System Omnivory Index	0.156
Primary Production Required/Catch (PPR/Catch)	53.890
Ascendency (%)	37.400
Finn’s cycling index (%)	1.290

The analysis of trophic flows in the BCE indicates high overall transfer efficiencies between trophic levels (17.4%). The Lindeman Spine (Fig. 2) shows the large proportion of flows from the first (primary producers: phytoplankton and macroalgae) to the second level and to the detritus pool. It also shows that most catches are derived from the second and third trophic levels, making the average trophic level of the fishery relatively low and its gross efficiency quite high (2.45 and 0.3%, respectively, see Table 5). The system size in terms of overall flows (consumption, exports, respiration and flows to detritus) is 38 695 t km^{-2} year^{-2}, with considerable proportions of flows allocated to Fisheries (33.7%) and to Detritus (37.9%).

Of the BCE biomass, macro-algae and detritus account for 44.6% and 28.4% respectively, while benthos feeding fish represent 3.1%, benthic predatory fish 1.7%, pelagic fish predators 1.1% and planktivorous fish 1.6% of the biomass. Echinoderms (sea stars and sea urchins) are a very important group representing 4.6% of the total biomass.

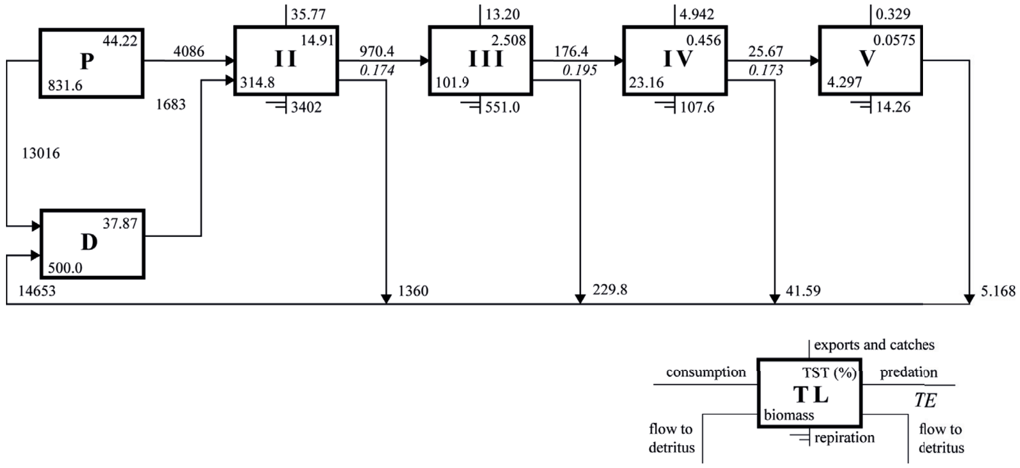

Fig. 2. Lindeman spine is showing the flows between discrete trophic levels of the Bolivar Channel Ecosystem during the period 2004-2008.

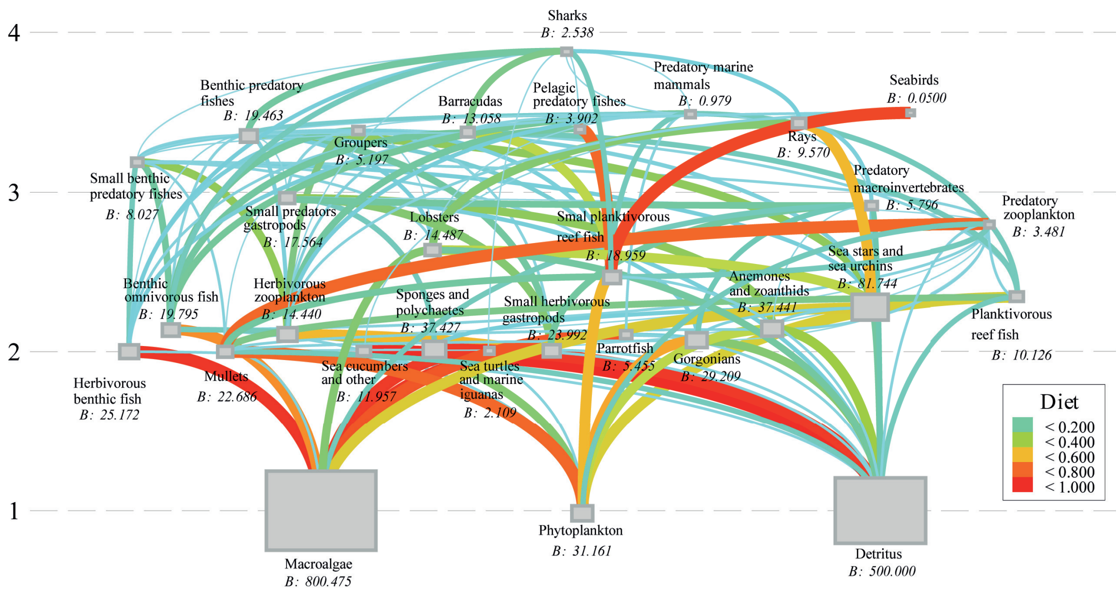

Fig. 3. Flow diagram of the Bolivar Channel Ecosystem. (B: Biomass in t km-2). The graph shows all flows and biomass into the system, the size of each box which represents a functional group is proportional to the biomass it represents, and the color and width of the lines connection between functional groups represent de relative proportion of the preys on the diet of a predator (y-axis is shown the trophic level of the functional group).

The flow diagram (Fig. 3) shows the prominent role of macro-algae in terms of biomass and flows to the second trophic level and also shows the important contribution of phytoplankton and detritus for the energy transfer from the first to the second level. As seen by the diagram, half of all model compartments are on or near the second trophic level making use of this enormous primary food production. These groups also include the emblematic sea turtles and marine iguanas. The plankton food web is also strongly articulated and large amounts of flows are channeled from the herbivorous plankton to the predatory plankton, to planktivorous fish, echinoderms and anemones. With regard to the dominant fish groups, both benthic (groupers and others) and pelagic predators (barracudas, jacks and sharks) are important in the system, as are seabirds and marine mammals.

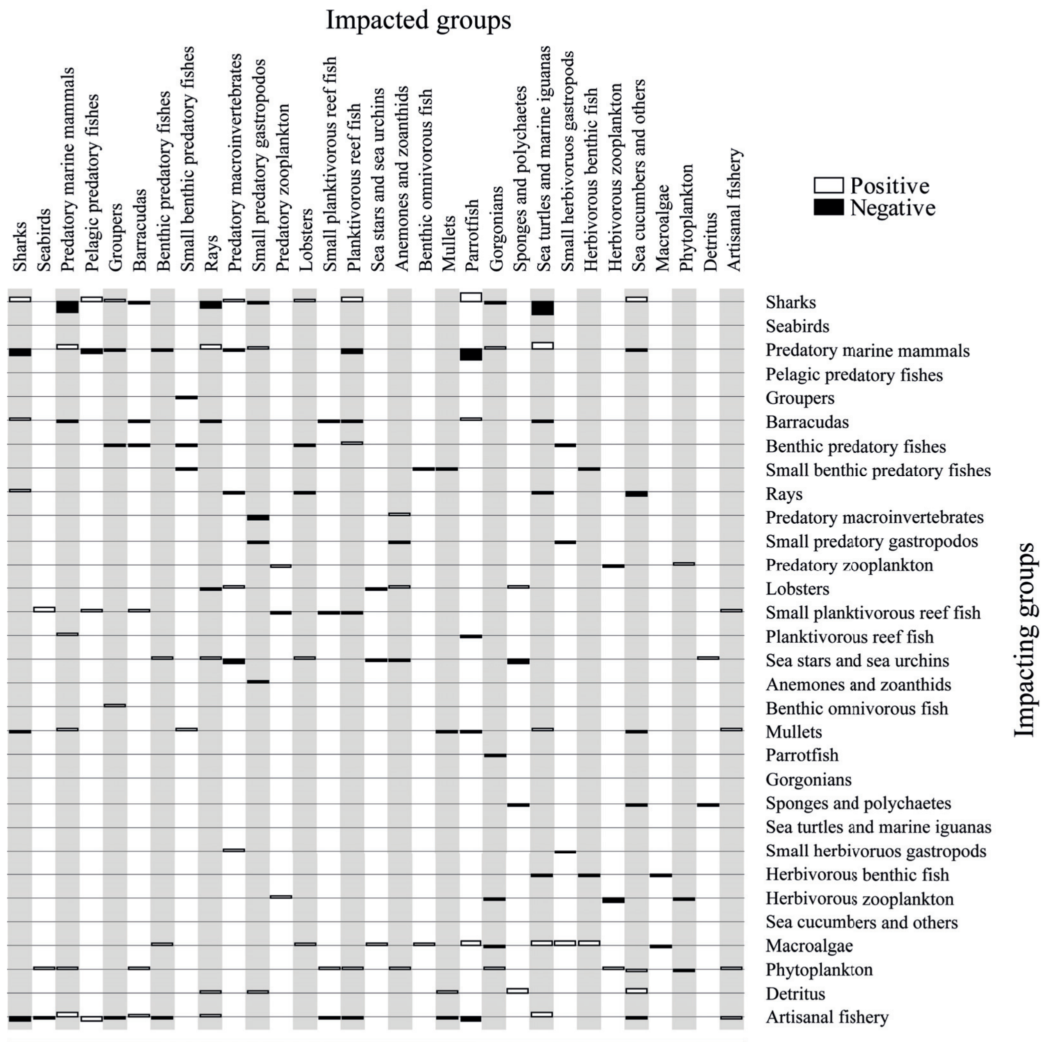

Fig. 4. Mixed trophic impact analysis of the Bolivar Channel Ecosystem. The figure shows the direct and indirect impacts on the living groups in the system caused by groups at the left. Positive impact shows by the bars above the base line and negative below. The impacts are relative but comparable between groups.

The mixed trophic impact (MTI) routine (Fig. 4) shows how an increase in biomass of a group would impact the other groups of the system. Most evident is the important role of macroalgae for the herbivorous group surgeonfish, benthic omnivorous fish, small herbivorous gastropods, parrotfish and others. Echinoderms (sea stars and sea urchins), greatly impact several benthic invertebrate groups, phytoplankton favors a great number of filter feeding benthic groups as well as zooplankton. Marine mammals impact several (mainly) pelagic fish groups. Sharks negatively impact marine mammals, rays, groupers and jacks, but through the reduction of these predator groups others are favored, such as parrotfish, barracudas, planktiovorous reef fish and even lobsters.

The figure also shows that a fishery increase would negatively impact most of the resources presently fished but would also cause indirect positive impacts on some other compartments.

The keystoneness index obtained through the network analysis (Fig. 5) shows highest values for the top predatory groups sharks and predatory marine mammals.

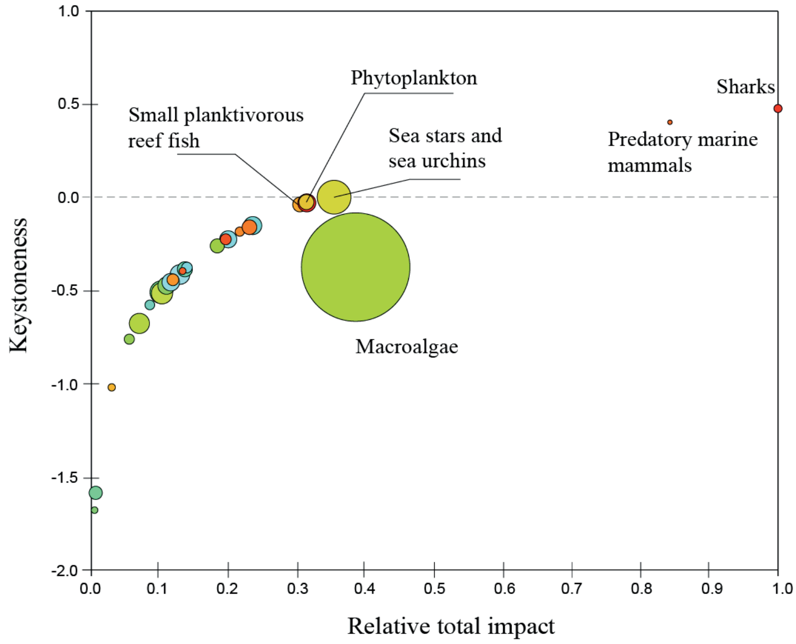

Fig. 5. Keystoneness and relative impact of model groups on the Bolivar Channel Ecosystem (groups with highest impact are named, circle sizes represent the group biomass). The keystoneness index (*y*-axis) is reported against overall effect (*x*-axis). Overall effects are relative to the maximum effect measured in each trophic web, thus for x axis the scale is always between 0 and 1. In the graph the functional groups are ordered by decreasing keystoneness, therefore the keystone functional groups are those up to the keystoneness index value > 0 (sharks and predatory marine mammals).

Discussion

The size of the BCE in terms of combined flows (T = 38695 t km^{-2} year^{-1}) is large and comparable to other upwelling systems. Jarre-Teichmann et al. (1998) report for the Peruvian upwelling system values of 29600 t km^{-2} year^{-1} (years 1953-1959), 29382 t km^{-2} year^{-1} (1963-1969), 33539 t km^{-2} year^{-1} (1973-1979) and even values of around

60000 km^{-2} year^{-1} for the period prior to the collapse of the anchovy fishery in 1972. Taylor et al. (2008) give values of T = 27820 t km^{-2} year-^{1} for Sechura Bay in Northern Peru and of T = 34208 t km^{-2} and T = 24827 t km^{-2} for Independence Bay in Central Peru for normal and El Niño conditions respectively, Wolff (1994) reports a value of T = 20594 t km^{-2} for Tongoy Bay in Northern Chile.

The large system size of the BCE may be explained by its complex bathymetry, the great heterogeneity of habitats (dominated by large boulders of great surfaces) and constant inflow of nutrients through topographic upwelling of nutrient–rich cold waters of the Cromwell current. However, comparing total throughput of the BCE with the Floreana system in Galápagos (T = 94850 t km^{-2} year^{-1}, Okey et al. 2004) shows great discrepancies with the latter suggesting methodological differences for the estimation of biomass within the Floreana rocky reef system. It is possible that Okey and collaborators overestimated the fish biomass, which is often the case in visual fish census (see further below).

BCE is known to be among the most productive fishing areas in the Galápagos Marine Reserve. It is assumed that approximately 40% of total sea cucumber (*I. fuscus*) catches and substantial catches of lobsters (*Panulirus* spp.) are derived from here (Hearn et al 2006, Toral et al. 2006). Our model estimates of fisheries catch (54.30 t km^{-2} year^{-1}) and gross efficiency (0.3%) confirms this assumption. Most of harvested biomass is from the lower trophic levels 2 and 3 (small planktivorous, mullets, lobsters and sea cucumbers) as shown by the low mean trophic level of the catch (2.45) estimated in our model., a value similar to the one estimated by Okey et al. (2004) for the Floreana system (2.27). This is also the reason why the mean PPR/Catch (53.89) is relatively low. If this value was computed on a specie by specie basis, high trophic level predators such as sharks, rays and groupers reveal much higher PPR/Catch values of 4 912, 331 and 327 respectively. Their overall harvest is, however, relatively low in this system.

The primary production of the BCE (17101 t km^{-2} year^{-1}) calculated by Ecopath is even larger then that estimated for the Floreana model (13250 t km^{-2} year^{-1}, Okey et al. 2004) and for the southern Benguela upwelling ecosystem during the period 1980-1989 (approximately 12000 t km^{-2} year^{-1}) (Shannon et al. 2003), and close to the values reported for the Peruvian ecosystem during the period 1972-1981 (about 17000 t km^{-2}

year^{-1}) (Jarre-Teichmann et al. 1998). Since our estimate of phytoplankton biomass was derived from an average of 13 years of satellite-derived Chl-*a* data for the area, we are quite confident about this value, which clearly confirms the upwelling characteristics of the BCE.

The relatively high value of the ratio of total primary production / total respiration (PP/R) of 4.19 suggests that the system is at a low maturity level with production largely exceeding respiration. (Christensen et al. 2000). Finn`s cycling index, another descriptor for ecosystem maturity (Odum 1969) is low (1.29%) compared to values reported by Baird and Ulanowicz (1993) for four estuaries and by Wolff (1994) for Tongoy Bay (Chile). However, it is comparable to the calculated values for the upwelling ecosystem of Central Chile (Arancibia et al. 2003). The low value is also indicative for a system that is far from maturity, a general feature of upwelling systems.

An explanation for these findings may be the great environmental stochasticity to which the BCE is subjected: during the past strong El Niño events 1982/83 and 1997/98, which caused great reductions in primary production, both in plankton and macro-algae, several of the emblematic species such as marine iguanas, sea lions and penguins suffered substantial population reductions (Wikelski and Wrege 2000, Salazar and Bustamante 2003, Vargas et al. 2005, Vinueza et al. 2006). The ENSO cycle thus seems to periodically "*reset*" the system (sensu, Bakun and Weeks 2008) keeping it at a relatively low (but highly productive) development state, also typically for the above-mentioned upwelling systems.

Due to the nature of the BCE as a semi-enclosed system, most energy is recycled locally, which could explain the high transfer efficiency (17.4%). On the other hand, our model suggests that a large part of the macroalgae biomass is not directly used by herbivores, (EE = 0.12), and feeds the detritus-based food chain.

The mixed trophic impact analysis and the biomass of the model groups shows the rank order of relative impact of the different groups, clearly revealing the importance of sharks and predatory marine mammals as keystone species and the important contribution of macro-algae, sea stars and sea urchins, phytoplankton and planktivorous reef fish for their great biomass and large flow contributions to the BCE.

The BCE has an enormous diversity and biomass of fish species of different habitats (open water, rocky reef, sand bottom) and trophic guilds (predators, detritivores, planktivores, omnivores), whereas in the coastal upwelling systems of the south east Pacific fish diversity is low with a clear dominance of one or two pelagic planktivores (anchovy and sardine) and just a handful of other, much less abundant, fish species. The BCE system also comprises large biomasses of non bivalve filter feeders (there is, however, a very rare endemic scallop species, *Nodipecten magnificus*), such as gorgonians, zoothanthids, sponges and endemic ahermatypic corals (*Tubastraea faulkneri* and *T. tagusensis*), while bivalve filter feeders use to dominate the shallow upwelling systems along the South East Pacific shore. An interesting feature of the BCE is the lack of large cangrid or xanthid crabs, well know benthic predators of the south east Pacific. Their niche seems to be occupied by three species of spiny lobsters (*Panulirus penicillatus*, *P. gracilis*, and *P. femoristriga*) and one species of slipper lobster (*Scyllarides astori*). The proportion of endemic species is high in the BCE and exceeds the level of endemism in the north-western and south-western regions of Isabela and of western Fernandina. For this reason, and because several invertebrates species have only been recorded here, the BCE area is considered unique for its mix of tropical and temperate species (Edgar et al. 2004a)

If we look into the weaknesses of our model, we must consider that a substantial part of the data input used to balance the model had to be taken from literature sources and our fishery estimates were based on the assumption of a fishing regime that harvests 1/3 of biomass production of the resource groups. We had to make this assumption because of a complete lack of standardized fisheries monitoring in this area. The more reliable input corresponds to the biomass data from our subtidal ecological monitoring and population monitoring that CDF is carrying out for several years. However, during the model balancing process substantial biomass changes (mainly reductions) had to be done for some fish groups that were visually censed during our surveys. We thus have to consider the possibility of an overestimation of fish biomass by our visual census, which has already been reported by several authors (DeMartini and Roberts 1982, Thompson and Mapstone 1997, Kulbicki and Sarramega 1999, Edgar et al. 2004b). In addition, the length-weight relationships for fish species may vary between areas and our Fishbase-derived conversion factors may have not been adequate for the Galápagos region, which could have introduced another bias into our estimates. However, we strongly believe that the order of magnitude of our fish group estimates and the biomass proportions between groups are realistic, since the estimates were based on the average of several years of monitoring using a standard methodology. For the benthic groups we assume a rather low estimation bias since monitoring is straightforward and less error-prone. We have not taken into account migration of species or groups within our model of Bolivar Channel due to the reason that biomass of none of the groups considered in our model had shown substantial intra-annual variations in our survey data. And it is believed that most groups have strong biogeographical affinities to the model area (Edgar et al. 2004a). However, significant migrations could occur in the BCE in response to changing upwelling intensities and associated with primary production variability during ENSO events. Further work should focus on the simulation of El Niño/La Nina induced variations in primary productivity and their effect on the ecosystem. Time series of the subtidal monitoring data taken over the past decade could be compared to model simulations in order to elucidate the factors responsible for observed changes in compartment biomasses over time.

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2

El Niño induced changes to the Bolivar Channel Ecosystem (Galápagos): comparing model simulations with historical biomass time series

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Abstract: During a strong El Niño event, nutrient and phytoplankton concentrations around the Galápagos Archipelago greatly decrease, while sea surface temperature increases (> 7 °C). Several species suffer under these conditions, while some benefit and new species appear. To understand the mechanisms behind observed changes, a trophic reference model of the Bolivar Channel Ecosystem was forced by a 16-year (1994-2009) satellite - derived time series of phytoplankton biomass including the El Niño period 1997/98. Emergent changes in model compartment biomasses, as derived from dynamic simulations, were compared to *in–situ* observations of the subtidal communities and marine vertebrates over the study period. Observed population reductions of seabirds (penguins and flightless cormorants) and of several fish groups were well predicted by the simulations, suggesting that bottom-up effects largely control the system during an El Niño event. Observational data also allowed modifying the reference model to an El Niño state model. In this El Niño model, ecosystem size (total energy throughput) was reduced by 70.1 %. Overall system characteristics show great similarities with other coastal upwelling systems of the Peruvian coast in that strong El Niño events cause disruptions to trophic flows and keep them at a low (but highly productive) development state.

Keywords: Trophic modelling, El Niño, Bolivar Channel Ecosystem, Galápagos, Ecopath with Ecosim.

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Introduction

The Bolivar Channel ecosystem (Fig. 1) is located in the western part of the Galápagos Archipelago and belongs to a biogeographic “cold water sub region” of the Archipelago, which is heavily shaped by the Pacific Equatorial Undercurrent (Cromwell current) that impinges on the Archipelago from the west, frequently causing strong topographic upwelling of cold nutrient-rich waters to the surface (Eden and Timmermann 2004, Houvenaghel 1978). For this reason the mean sea surface temperature (SST) in this region is between 14 – 20 °C, generally lower than in the “mixed water zone” around the central islands (18 – 24 °C) and far lower then in the northern warm zone around the islands of Wolf and Darwin (24 – 26 °C). The shallow habitats of the Bolivar Channel are heavily influenced by these cold and nutrient-rich upwelling waters. They are comprised mainly by rocky reef areas of bedrock, boulder, cobbles and sand patches that spread along the coastline, interrupted by small sandy beaches and mangrove patches.

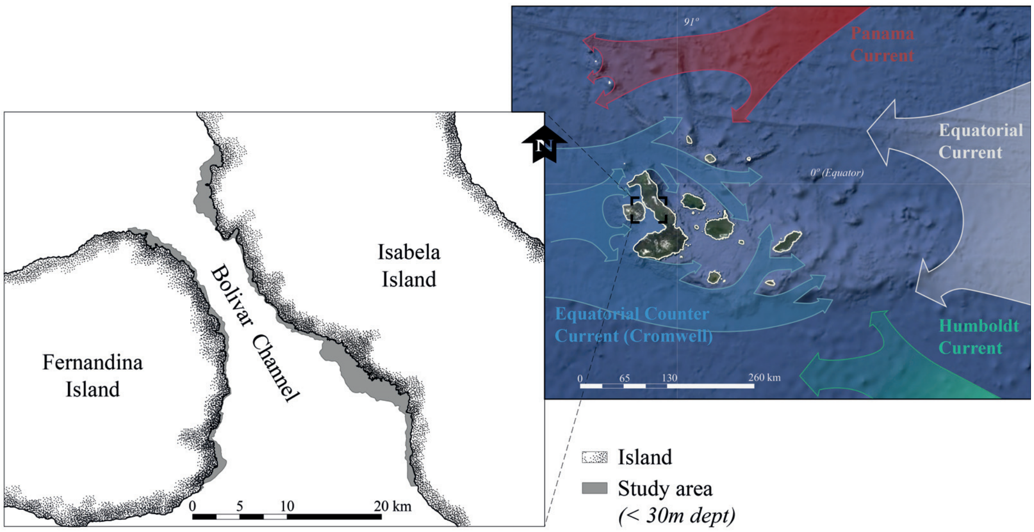

Fig. 1. Left: Bolivar Channel ecosystem, located between Isabela and Fernandina islands, in the west of the Archipelago. Grey areas show the zones considered for the model, upper right: the Galápagos Archipelago and its prevailing currents

The system supports high resource biomasses of small pelagic fish such as sardines, thread-herrings, anchovies, pompanos (*Trachinotus stilbe*) and mackerels (*Scomberomorus sierra*), which are in turn prey for substantial populations of top predators such as sharks (*Carcharinus galapagensis, Trianodon obesus*), tunas

(*Thunnus albacares*), wahoo (*Acanthocybium solandri*), barracudas (*Sphyraena idiastes*), dolphins (*Tursiops truncates*), seabirds (*Spheniscus mandicultus*, *Phalacrocorax harrisi*), and marine pinnipeds (*Zalophus wollebaeki*, *Arctocephalus galapagensis*) (Feldman 1985, 1986), many of which actively visit the rocky reefs to feed.

This productive pelagic system surrounds and interfaces with the benthic rocky reef habitats of Fernandina and Isabela islands on the west and east sides of the Bolivar Channel respectively. High production and accumulation of phytoplankton through the confluence of ocean currents as well as dense macroalgae beds provide elevated levels of primary production to these reefs. Many planktivorous fish that inhabit the rocky reef areas feed in the water column. The plankton also supports a high biomass of benthic filter-feeders that include gorgonians (*Muricea* spp. and *Pacifigorgia* spp.), zoanthids (*Parazoanthus* spp.), sponges (*Aplysilla* sp. and *Carmia* sp.), polychaetes (*Spirobranchus giganteus*) and ahermatypic corals (*Tubastraea* spp.). In some parts of the Bolivar Channel macroalgae (*Ulva* sp., *Sargassum* sp., *Spatoglossum* sp. and *Heterosiphonia* sp.) form dense beds while other areas are covered with filamentous algae, encrusting algae or/and benthic diatoms. These primary producers are important food sources for several invertebrates, fish and other vertebrates, including marine iguanas (*Amblynchus cristatus*). Two abundant species of sea urchins (*Lytechinus semituberculatus*, *Eucidaris galapagensis*) are the dominant herbivores on the benthic primary producers (Ayling 1981, Breen and Mann 1976, Himmelman and Lavergne 1985). Herbivore fish include damsels, surgeon and parrotfishes. Marine turtles (*Chelonea mydas*) are also present and in some cases in high abundance. Several species of sea cucumbers (*Holothura* spp. and *Isostichopus fuscus*) are also highly abundant and make use of the rich detrital material in the area. Omnivorous reef fishes are mainly comprised of butterflies, damsels and chopas, which consume both algae and small invertebrates. Grunts, wrasses and angelfishes are mainly invertebrate feeders, as are the carnivorous invertebrates, which include lobsters (*Panulirus* spp. and *Scyllarides astroi*), crabs, sea stars (*Pentaceraster cumingi* and *Asteropsis carinifera*) and gastropods (*Pleuroploca princeps*).

In the higher trophic levels we find piscivorous fishes like the endemic grouper of Galápagos (*Mycteroperca olfax*) and the snappers, octopuses (*Octopus* spp.), sea lions

(*Z. wollebaeki*), the Galápagos penguin (*S. mandicultus*) and the flightless cormorant (*P. harrisi*). Several sharks like the Galápagos shark (*C. galapagensis*) and the white tip-reef shark (*T. obesus*) also feed in these areas and interact with the pelagic system.

During the past 30 years, the Galápagos Archipelago, and particularly the zone of the Bolivar Channel, was greatly affected by both natural and anthropogenic impacts. Among the natural disturbances were the severe El Niño events 1982/83 and 1997/98, which brought extended periods (of about 8 months) of unusually warm (> 26 °C, occasionally up to 29 °C) and nutrient depleted waters to the surface layer (Chavez et al. 1999, Enfield 2001, Glynn et al. 2001, Wellington et al. 2001) and phytoplankton biomass was reduced by 50 to 70% of the mean quantity during periods of normal conditions (Kogelschatz et al. 1985, Jiménez 2008).

Monitoring surveys conducted by the Charles Darwin Foundation before, during and after these warm El Niño periods revealed changes in the abundance of subtidal organisms and also showed that the emblematic penguin and flightless cormorant populations suffered greatly during the warming events. It was hypothesized that among the main causes for the changes in the bird population numbers were shortage of food (mainly small pelagic fish), resulting from the disruption of the trophic structure of the system through the bottom-up effect of reduced primary production (Vargas et al. 2005).

For this study a trophic model was constructed for the El Niño period 1997/98 and compared to the Bolivar Channel reference model (Ruiz and Wolff 2011) covering the period 2004-2008 to search for differences in energy flow structure and the role of key compartments. To elucidate the mechanisms behind the observed changes we explored the impact of El Niño 1997/98 on the system by forcing changes in primary producer biomass as derived from remote sensing (phytoplankton) and Charles Darwin Foundation surveys of macroalgae. The model response in terms of biomass changes of other model groups was then compared to functional group's biomass estimates as derived from subtidal ecological monitoring and the marine vertebrate population monitoring (1994-2009) carried out by the foundation.

Methods

Comparing the reference model and the El Niño state model of the Bolivar Channel

All modeling explorations were conducted with the software Ecopath with Ecosim 6.0 (EwE) (Christensen et al. 2008). Biomass inputs for the 30 groups of the reference model (Ruiz and Wolff 2011; Appendix 3) were based on mean compartment biomasses in the Bolivar Channel derived from surveys during the period 2004 – 2008. Production rates (*P/B*), consumption rates (*Q/B*), catches (*C*) (in case of a fisheries resource) and diets were derived from various data sources as described by Ruiz and Wolff (2011).

In order to create an El Niño model, each group's biomass data were adjusted to values representative for the El Niño period in 1998. For some groups (macroalgae, herbivorous zooplankton, small herbivorous gastropods, gorgonians, anemones and zoanthids, lobsters, predatory zooplankton and small predatory gastropods) biomass estimates were not available and these values were left blank in the input matrix. Ecotrophic efficiencies (the fraction of total production that is consumed within the system) of these groups were fixed at 0.95 and the model computed the missing biomass values. Catch values were adjusted to the new biomass values maintaining the original catch to biomass ratio (= 0.3 * P/B * B). This ratio was based on the assumption that the stocks were moderately exploited (30% of annual biomass production removed) (Ruiz and Wolff 2011). The diet matrix of the reference model was modified considering that many groups are rather unselective opportunistic feeders and that available food item proportions had changed due to the El Niño caused changes in biomass of most groups. Following the method of Taylor et al. (2008a) the diet proportions were thus adjusted to reflect predatory groups' consumption habits as well as the available production of prey groups. Moreover, an increased base percentage of detritus feeding (10%) was assumed for most benthic feeders, which is proximate to values given in Ortiz and Wolff (2002) and Taylor et al. (2008b) for benthic compartments in a Chilean and Peruvian bay system respectively.

The P/B and Q/B values for the functional compartments were maintained due to lack of information about these values during El Niño.

Once the El Niño model was balanced through a manual process of adjusting some of the input parameters, it was subjected to the resampling routine Ecoranger, which draws a set of random input variables from normal distributions for each basic parameter. All confidence intervals around the input parameters were fixed at 20% as was similarly done by Arias-González et al. (1997) and Taylor et al. (2008a). Resampling was allowed until 10000 runs passed the selection criteria. The best run was chosen as that with the smallest sum of square residuals between the input parameters and the mean value of all successful runs (for more information, see Christensen et al. 2000). The resulting steady-state model inputs and outputs are shown in Appendix 3 and selected system summary statistics were calculated and compared with those of the reference model (Table 1).

For both system state models, trophic impacts were estimated for each pair of functional groups (*prey* and *predators*, interacting directly or not) by means of the net impact matrix (Libralato et al. 2006). The net impact of prey on predators is given by the difference between positive effects, quantified by the fraction of the prey in the diet of the predator, and negative effects, evaluated through the fraction of total consumption of prey used by the predator (Ulanowicz and Puccia 1990). The mixed trophic impact (MTI) was then estimated by the product of all the net impacts for all the possible pathway in the trophic web that link the functional prey and predator groups. Negative elements of the matrix MTI indicate a prevailing negative effect of the predator on the prey, analogously, positive elements of MTI indicate prevailing positive effects of the prey on the predator. Therefore, negative elements of MTI can be associated to prevailing top-down effects and positive ones to bottom-up effects (Libralato et al. 2006).

To visualize the major differences in flow structure between both systems states the Lindeman spine routine of EwE was used, which aggregates the entire system into discrete trophic levels (Baird and Ulanowicz 1993, Lindeman 1942). This routine, based on an approach suggested by Ulanowicz and Kay (1991) visualizes the biomass of each (aggregated) trophic level and allows showing all flows into and out of each trophic level.

Time series analysis

ECOSIM, basic equations

In Ecosim, the biomass dynamics of all ecosystem components that occupy trophic levels above the primary producers are determined by the following equation:

$$\frac{\partial B_i}{\partial t} = g_i \sum C_{ki} - \sum C_{ji} + I_i - (M_i + F_i + e_i) B_i \qquad (1)$$

Where $\partial B_i/\partial t$ is the rate of change in biomass of group i, g is the growth efficiency (proportion of food intake converted into production), F is fishing mortality, M is natural mortality rate (excluding predation), e is emigration rate, I is immigration rate, and the first sum represents the food consumed, over prey types k of species i, and the second sum represents the losses due to predation summed over all predators j of i. In our model, immigration and emigration were assumed to be equal and thus were not considered. In Ecosim, the biomass of component i that is vulnerable to predation by component j (V_{ij}) is a function of a vulnerability rate (v).

$$\frac{\partial V_{ij}}{\partial t} = v(B_i - V_{ij}) - vV_{ij} - a_{ij} V_{ij} B_{ij} \qquad (2)$$

Where, a_{ij} is the effective rate at which predator j searches for prey i. The vulnerable biomass increases from exchange with a pool of invulnerable biomass, $v(B_i$ - $V_{ij})$, and decreases when prey return to the invulnerable condition (vV_{ij}) or by predation ($a_{ij}V_{ij}B_j$). When v is small, the flows between predators and prey are controlled mostly by variations in prey biomass, *i.e.* control is bottom-up. When v is large, these flows are controlled mostly by variations in predator biomass, *i.e.* control is top-down. We started the simulations using the default vulnerability value (*v=2.0*) and then did a second run with the vulnerability search routine of the program to see if the fit can be improved. The resulting changes in the vulnerability values for the model groups were then explored in the light of possible control mechanisms operating in the system.

Table 1. Ecosystem indicators to compare between the reference state model (Ruiz and Wolff 2011) and the El Niño 1997/98 state model.

Ecosystem indicator	Value		
	Reference state	El Niño (EN)	Difference
Trophic indicators			
Total system throughput (t km^{-2} per year)	38694.98	11578.55	-70.07%
Total net primary production (t km^{-2} per year)	17101.49	4093.07	-76.07%
Total biomass (excluding detritus) (t km^{-2} per year)	1276.28	492.30	-61.43%
Mean transfer efficiency (%)	17.40	17.60	1.15%
Connectance index	0.17	0.18	9.05%
Fishery indicators			
Total catches (t km^{-2} per year)	54.30	23.99	-55.82%
Mean trophic level of the catch	2.45	2.62	7.03%
Gross efficiency (catch/net primary production, %)	0.30	0.60	100%
Primary Production Required/Catch (PPR/Catch)	53.89	47.80	-11.30%
Energy indicators			
System primary production/respiration	4.20	1.41	-66.44%
System primary production/biomass	13.40	8.31	-37.95%
System biomass/throughput	0.30	0.04	-85.83%
Network indicators			
Finn's cycling index	1.29	4.19	224.80%
Relative ascendency	37.40	24.50	-34.49%

Simulating ecosystem response to the El Niño caused reduction in primary production

To force the model with changes in phytoplankton biomass, a phytoplankton biomass time series for the period 1994-2009 was derived by the following steps: (1) For the period September 1997 to December 2009 monthly satellite data of Chl-*a* and sea surface temperatures (SST) were used to reconstruct the time series for the Bolivar Channel area. Chl-*a* estimates are from the ESA Globcolour database (http://hermes.acri.fr/), which uses estimates from the SeaWiFS sensor before April 2002, and a merged product of SeaWiFS-, MODIS-, and MERIS-derived estimates thereafter. SST estimates come from the AHRSS Pathfinder product (level 3) (NOAA). Both time series were converted to annual means and a regression between both variables was computed. (2) For the years 1994, 1995, 1996, (those with SST data but without available satellite data for Chl-*a*), the annual means of SST were used to calculate the corresponding Chl-*a* values (3) The resulting time series of Chl-*a* (mg m^{-3}) for the period 1994 - 2009 was then converted to wet weight biomass using the following conversions: Chl-*a* – Carbon (40:1; Brush et al. 2002) and Carbon – wet weight (1:14.25; Brown et al. 1991). In addition, a uniform mixed layer depth of 20 m was assumed (de Boyer Montegut et al. 2004) to derive biomass values per square meter.

Macro-algae biomass was used as a second forcing variable. Its time series was based on biomass estimates obtained during the Subtidal Ecological Monitoring (SEM) carried out by the Charles Darwin Foundation in the Archipelago between the years 1997 to 2009 (Banks et al. 2003, Banks et al. 2006, Edgar et al. 2004, Edgar et al. 2011).

With these two forcing variables, the model was run for the period of 16 years (1994 – 2009) and the resulting changes in compartment biomasses were compared to the biomass estimates of the subtidal monitoring surveys conducted by the Charles Darwin Foundation since 1994 – 2009 and to the population census data of penguins and flightless cormorants available from the foundation's data base (Vargas et al. 2005). The performance of the simulation was evaluated by the sum of squared differences (SS) between the simulated and reference (log) biomass time series. The significance of improvements in SS (i.e. decrease) for individual functional groups was assessed by the correlation of observed vs. simulated time series of (log) biomass.

Results

Comparing system characteristics between "Normal" and El Niño state

Fig. 2 shows the biomass change of the model compartments from the normal state to the El Niño year 1998. Most model groups (21 of 29) largely decreased in biomass, while some - sea cucumbers and others, sea stars and sea urchins, lobsters, benthic predatory fish, barracudas, groupers, rays and sharks - increased. This increase was noticeable only for the group sea cucumbers and others, however. Besides the primary producers (phytoplankton and macroalgae), several groups decreased by over 50% including herbivorous zooplankton, mullets, small planktivorous reef fish, jacks and mackerels, predatory marine mammals and seabirds. Figure 3 (Lindeman spine) summarizes these differences in biomass and flows between trophic levels for both system states and shows the highest biomass reduction for the first trophic level (88.1%). The decrease is lowest on the second and third level (11%) and above 13% for the levels 4 and 5. Table 1 gives a summary of system descriptors for both states that will be discussed further below. The net impact analysis sensu Libralato et al. (2006) for both system states is shown in Fig.4, revealing the positive impact of primary producers during both states and the great increase in negative impact during the El Niño state of the predator groups sharks, barracudas, benthic predatory fish but also of the groups sea stars and urchins. The impact of predatory mammals changes from strongly negative during the reference state to strongly positive during the El Niño state.

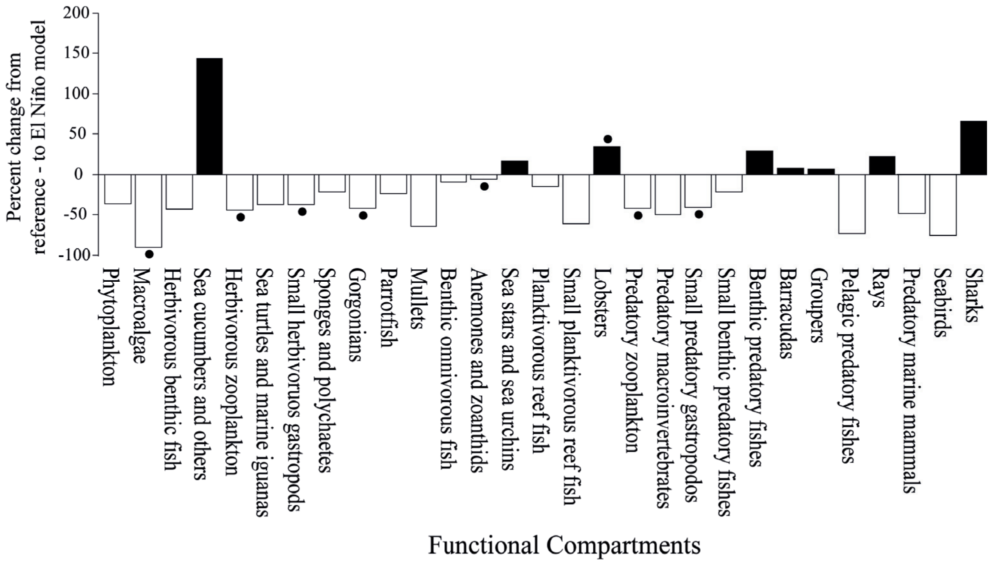

Fig. 2. Biomass changes of model groups from Reference- to El Niño state (in %) (Biomass for groups with black dots was estimated by the model during the balancing process).

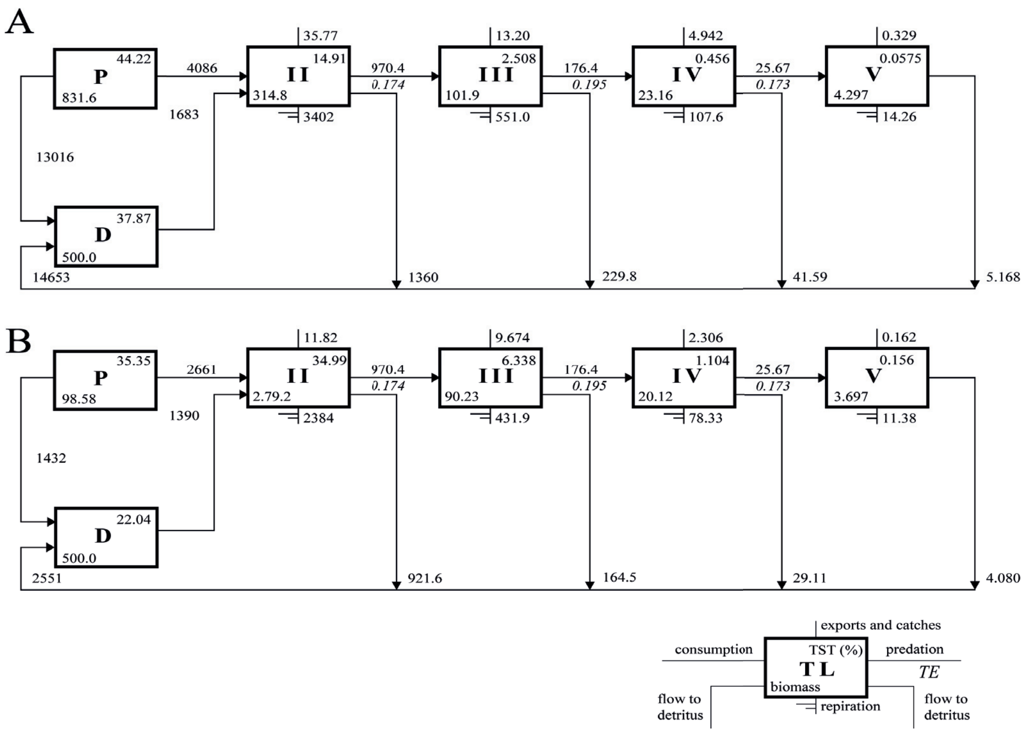

Fig. 3. Aggregated energy flow charts (Lindeman spine) for reference model (A) and El Niño model (B) of Bolivar Channel (only the first 5 trophic levels are considered, which comprise > 99.9% of total throughput). P and D stand for primary producers and detritus respectively, see little flowchart in the right corner for the description of numbers in boxes and on connecting lines.

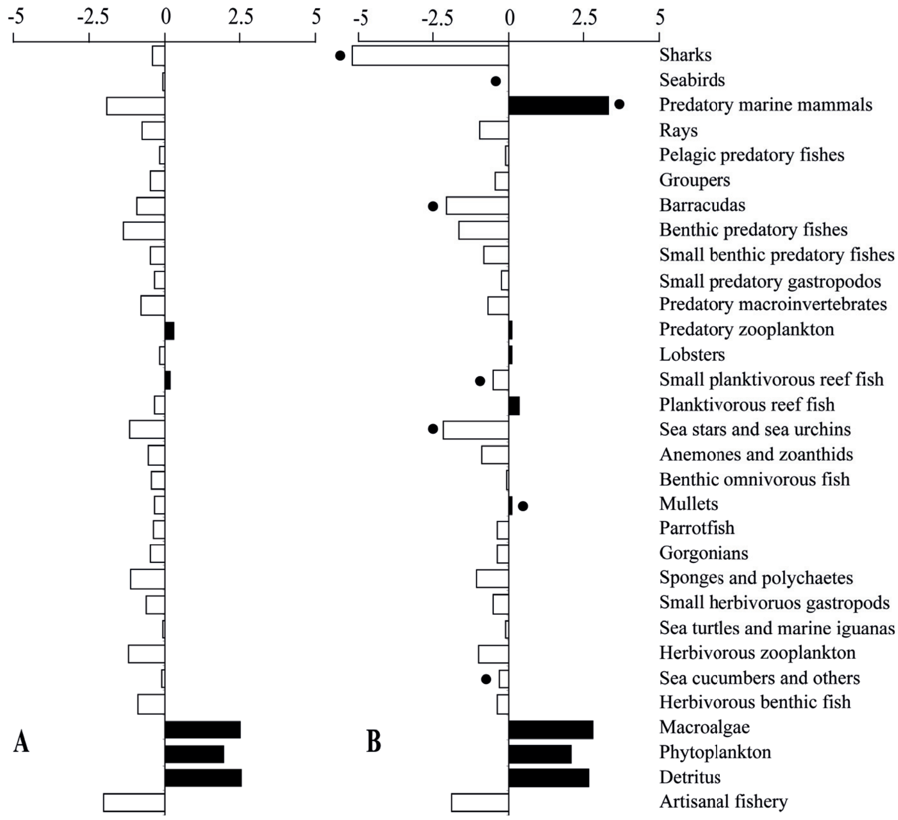

Fig. 4. Net Impact Analysis sensu Libralato et al. (2006) for reference (A) and El Niño (B) system states, positive impacts are indicated by the black bars, while the white bars show negative impacts (bars for groups with black dots show increase or decrease in impact strength by > 50% during the El Niño 1997/98 event).

System response to El Niño reduced primary production

Phytoplankton biomass was above average during the years preceding the El Niño (predicted based on colder average temperatures for that period), greatly decreased for the El Niño years 1997 and 1998 (by 46% and 33% respectively) and increased to maximum values over the post El Niño period 1999-2004. Thereafter values decreased and remained below average until the end of the study period.

Fig. 6 shows the observed and simulated trajectories of model group biomasses over the study period forced by the time series of primary producers (phytoplankton and macro-algae). Sum of squared differences between observed and simulated (log) biomasses changed from SS = 239.5 (no forcing) to SS = 207 (forcing using v = 2) and to SS = 169.5 (using vulnerability search). Fig. 6 only presents those sensitive groups, whose biomass changes over the El Niño cycle were > ± 20% (either in the *in-situ* survey data or as outcome of the simulations). Table 2 provides the results of the fitting for each group separately.

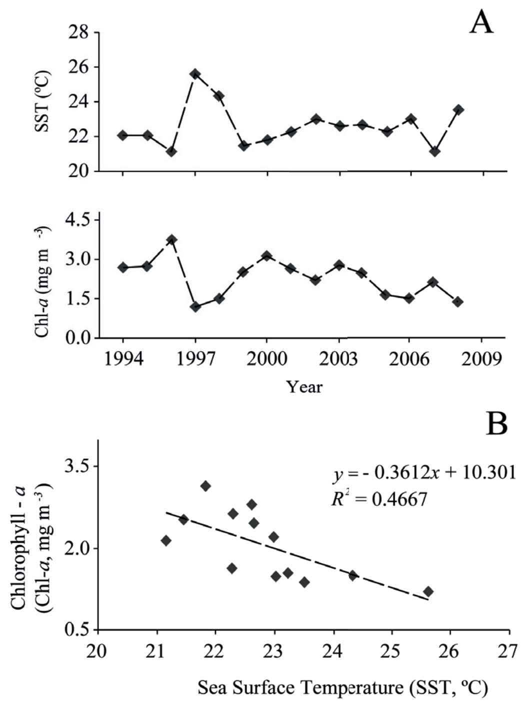

Fig. 5. (A) Annual means of Sea Surface Temperature (SST) and (B) Chlorophyll-*a* (Chl-*a*) in the Bolivar Channel area and regression between both variables. The Chl-*a* values for the years 1994 - 1997 were approximated by this regression and SST data for these years

Simulated dynamics of seabirds (penguins and flightless cormorants), mullets, small benthic predatory fishes and benthic predatory fishes were significantly correlated with observed data. Additional positive correlations exist for surgeonfish, benthic omnivorous, groupers and planktivorous reef fish, although the correlations were not significant at the p >0.05 level. A biomass decrease during the El Niño period 1997/98 for the group sponges and polychaetes and jacks and mackerels also correlated significantly to the observed data and was also evident (but not statistically significant) for the groups parrotfish, and barracudas, as seen by both simulation and observational data. The groups sea turtles and marine iguanas, and predatory marine mammals decrease only very slightly in the simulations (however, statistically significant in the first case), whereas great density reductions were observed in the field surveys. While in the case of the turtles and iguanas post El Niño population counts remain relatively high, marine mammal populations vary greatly between annual surveys. The observed stock proliferation of lobster following the El Niño event is also simulated by the model, which provides a statistically significant correlation between observed and simulated values. Since small planktivorous fish were not surveyed quantitatively their in-situ abundance could not be compared with the model simulations, which suggest strong decreases during the El Niño period.

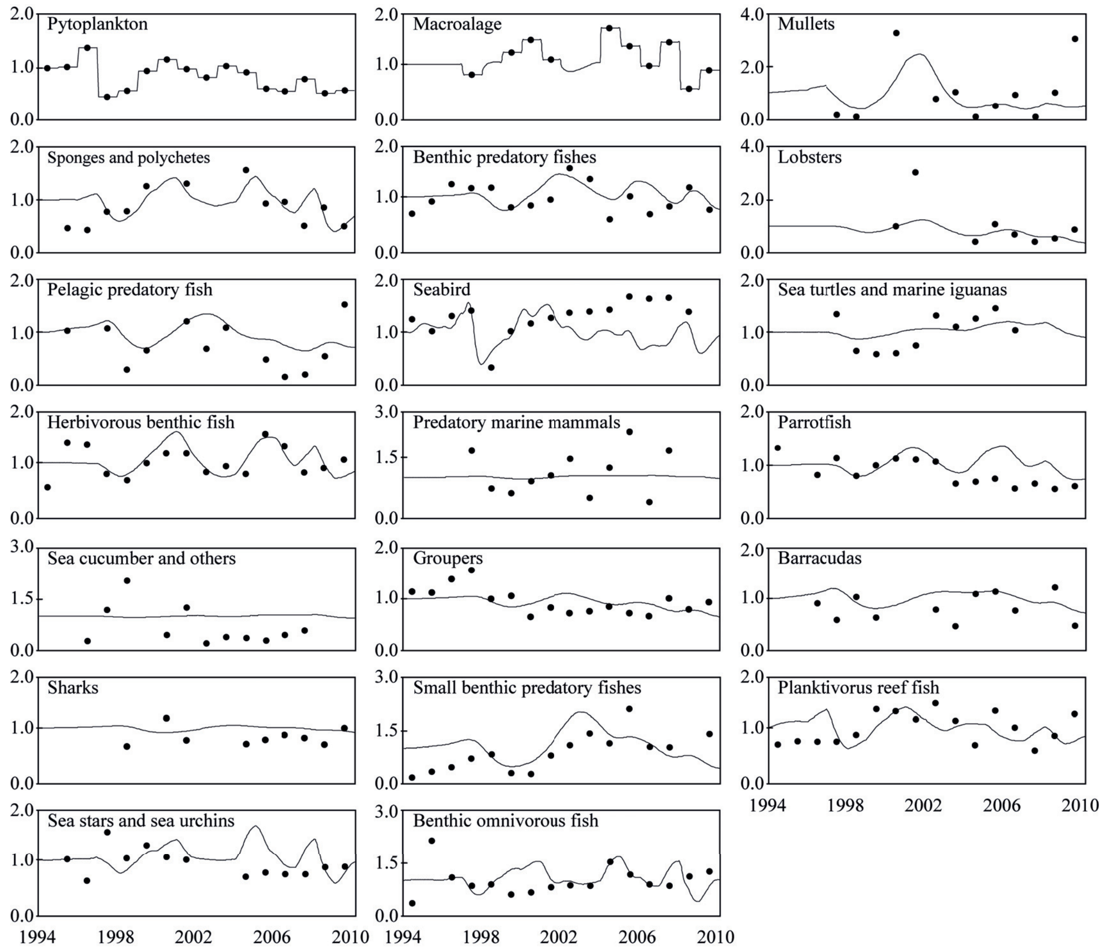

Fig.6. Simulated (lines) and measured (points) compartment biomasses over the time period 1994-2009, vertical axis stands for biomass relative to EwE reference model (B=1), Continuous lines represent results following the fitting of vulnerability settings, the model was forced with satellite – derived time series of Chl-*a* biomass and with *in-situ* observations of macro-algae biomass. Through the vulnerability search the sum of squares (SS) between observed and simulated values were reduced from 239.5 to 169.5 (-29 %).

Table 2 shows ranges of the vulnerability values computed for the predator-prey matrix during the vulnerability search. While for the groups benthic predatory fish, groupers, small predatory gastropods, pelagic predator fish, and sharks high vulnerability values of $v > 2.0$ point to their role as top-down controller in the system, the low values of $v = 1.0$ for the groups phytoplankton, macroalgae, and herbivorous zooplankton suggest bottom-up control of these groups of their consumers.

Table 2. Correlation of observed vs. simulated (log) biomass time series for each model functional group. *na* = Not applicable, no decrease in sum of squares, *r* = correlation coefficient, *t* = student-*t* value, *p* = significance level.

Functional group	% decrease in SS	*r*	*t*	*p*	*n*
Lobsters	48.00	0.465	2.280	0.038	8
Sponges and polychaetes	30.03	0.287	2.100	0.030	13
Sea turtles and marine iguanas	25.30	0.436	2.480	0.019	10
Small benthic predatory fishes	23.10	0.193	1.828	0.040	16
Pelagic predatory fishes	22.30	0.270	2.110	0.027	14
Seabirds	20.10	0.206	1.833	0.043	15
Benthic predatory fishes	4.35	0.147	1.550	0.070	16
Mullets	4.30	0.247	1.720	0.060	11
Herbivorous benthic fish	na				
Sea cucumbers and other	na				
Parrotfish	na				
Benthic omnivorous fish	na				
Sea stars and sea urchins	na				
Planktivorous reef fish	na				
Barracudas	na				
Groupers	na				
Predatory marine mammals	na				
Sharks	na				

Discussion

System characteristics of "Normal" and El Niño state

As shown by the Lindeman spine (Fig. 3), all trophic levels had largely reduced biomasses during the El Niño event, but it is interesting that the decrease was lowest for the levels II and III (11%). This can be explained by the fact that several of the groups on these two levels (sea cucumbers and others, sea stars and sea urchins, and predatory macroinvertebrates and other sea stars) use detritus as an important food source, which means that they were not as affected by the El Niño induced reduction in phytoplankton and macro-algae. As to be expected, the highest reduction in energy flow occurred between the primary producers and the primary consumers (trophic level I to II) (about 88%), but the decrease cascades through all trophic levels showing the profound effect of the El Niño on the whole food web.

The net impact analysis (Fig. 4) predicts that the roles (overall system impact) of some of the model groups change during the El Niño state. The most pronounced change is for predatory marine mammals from a strong negative impact during the reference state to a strong positive impact during the El Niño event. This can be explained by the assumed change in diet from several groups of small fish prey (whose biomasses decreased during the El Niño) to large predators such as benthic omnivorous fish, groupers and sharks (Appendix 4). The great increase in negative impact of sharks during the El Niño state seems due to increased proportions of many groups in their diet, such as sea turtles, parrotfish, benthic omnivourous fish, planktivorous fish, groupers and rays and to the biomass increase of sharks during this period. In general it appears that the relative system impact changes more strongly for predatory groups during the El Niño event, since overall prey biomass decreases and predators make use of any prey they can access, which also implies switching to unusual food items. If, like in the case of sharks and predatory marine mammals, top predators invade the already debilitated system, their impact on the flow structure of the system is thus very strong. The overall positive mixed trophic impact of mammals during the El Niño state as revealed by Fig.4 seems due to their assumed switch from low trophic level fish species to sharks, groupers and benthic omnivorous fish, thereby greatly releasing the consumption pressure of these species over their prey.

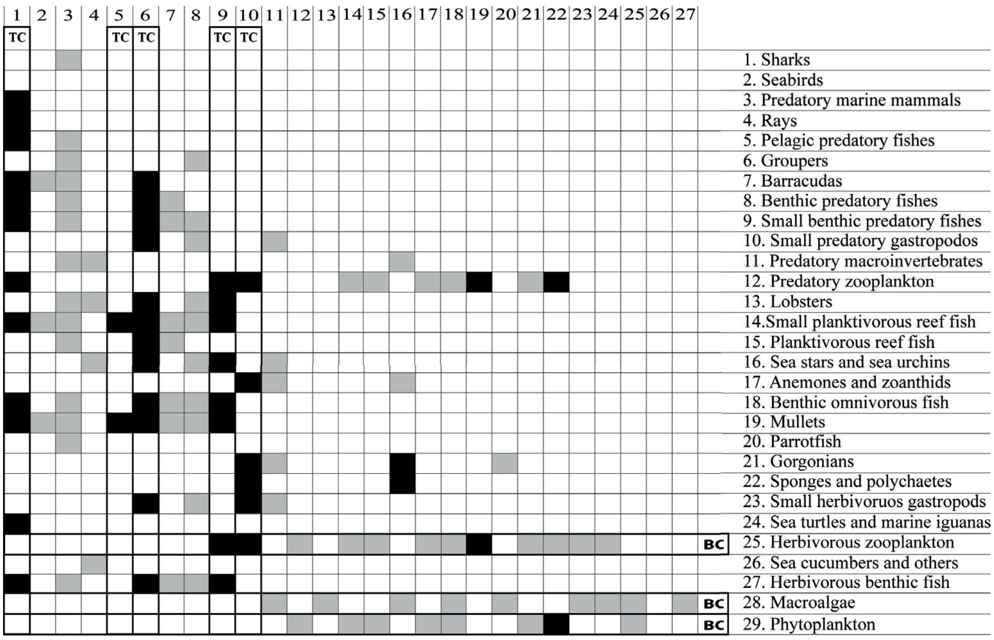

Fig. 7. Vulnerability ranges as derived from vulnerability search, grey cells: *v* = <1.5 (bottom-up control, TC), black cells: v = > 2 (top-down control, TC). Black boxes along rows and columns highlight groups, which in the vulnerability search show and important control effect over other groups when their biomass changes.

Overall system biomass as well as energy throughput are reduced to about one third during the El Niño (Table 1), and explain why catches were also reduced by 55.8%. The parallel increase in the gross efficiency of the catch (catch/primary production) by 100%

can be explained through the fact that the reduction in primary production greatly exceeded the reduction in catch. The reduction of the system P/R ratio by 66.4% is indicative for the relative increase in respiration since overall production greatly decreased. Finn's cycling index (FCI) increased by 224.8% during the El Niño event showing that a larger fraction of the ecosystem's throughput was recycled, which would, in addition to the reduced P/R ratio, also suggest that the system became more mature during the El Niño event. It seems, however, that this value increase can be explained by the very strong reduction in exports (catches) and primary production, which largely reduces the overall throughput and thus inflates FCI. Taylor et al. (2008b) found a similar increase in FCI in the Independence ecosystem (Peru) during the El Niño state, but showed that FCI decreased below the reference level, when the phytoplankton reduction effect was considered. The lower relative ascendency obtained for the El Niño state can mainly be attributed to the reduction in overall throughput (T) and, possibly to a lesser extent, to the changes in the diet matrix/flow structure of the model. The slight increase of mean transfer efficiency between trophic levels during the El Niño state as well as the great decrease in the system primary production to biomass and system biomass to throughput ratios suggests that energy flow efficiency was improved during the El Niño state. A slightly elevated mean trophic level of the catch during the El Niño state reflects the fact that small planktivorous fish were disproportionally reduced in the catches.

If we compare the Bolivar Channel ecosystem with other tropical shallow water ecosystems it appears that it has more features of an upwelling system than of a classical tropical system. Its system size (throughput) (for non El Niño years) of almost 40000 t km^{-2} year^{-1} greatly exceeds tropical systems of the East Pacific Seascape Region and elsewhere, such as Nicoya Gulf, Costa Rica (T = 3049 t km^{-2} year^{-1}, Wolff et al. 1998), Golfo Dulce, Costa Rica (T = 1404 t km^{-2} year^{-1}, Wolff et al. 1996), Campeche Bank, Mexico (T = 2049 t km^{-2} year^{-1}, Arreguín-Sánchez et al. 1993), Celestun Lagoon, Mexiko (T = 8969 t km^{-2} year^{-1}, Chavez et al. 1993), South China Sea (T = 2934 t km^{-2} year^{-1}, Silvestre et al. 1993), Venezuelan shelf (T = 7621 t km^{-2} year^{-1}, Mendoza et al. 1993), Caete Estuary, Brazil (T = 10559 t km^{-2} year^{-1}, Wolff et al. 2000) among others. Instead, it much more resembles coastal ecosystems of the Humboldt current along the Peruvian/Chilean coastline such as Sechura Bay, North Peru (T = 27820 t km^{-2} year^{-1}, Taylor et al. 2008a), Independencia Bay, Central Peru (T = 34208 t km^{-2} year^{-1} versus

24827 t km^{-2} year^{-1} for normal and EN conditions respectively, Taylor et al. 2008b) or Tongoy bay, Northern Chile (T= 33579.3 t km^{-2} year^{-1}, for the sand-gravel habitat, Ortiz and Wolff 2002). However, in the center of the Northern Peruvian Upwelling system, throughput is 55689t km^{-2} year^{-1} (Tam et al. 2008), significantly higher then for the above – mentioned coastal systems, the Bolivar Channel included. Note that in this Peruvian system, during the past EN event 1997/98, this throughput was reduced by approximatively 50% (Tam et al. 2008), similar to the Bolivar Channel of Galápagos, suggesting that the El Niño impact was very similar between these systems. In this context it is worth noting that the comparison of SST time series between Galápagos and coastal sites of the Eastern tropical Pacific also revealed higher similarities between the Galápagos with upwelling sites in Peru then with other tropical sites of the ETP region (Wolff 2010).

Maturity as based on the P/R ratio computed (4.20) as well as the relative ascendency (37.4%) also suggest similarity with a highly productive upwelling system of low to intermediate development, with biomass production exceeding respiration and a rather low complexity of flows. The very low Finn's cycling index (1.29%) is also indicative for a system of little recycling and low development.

An explanation for these system characteristics is the great environmental stochasticity (on inter-annual and intra-annual timescales) to which the Bolivar Channel is subjected. During the strongest two El Niño events of the last century, as was shown in this paper, system size was greatly reduced through a bottom-up disruption of the food web, as has also been described for the abovementioned systems of the Peruvian/Chilean coast. The El Niño - Southern oscillation cycle thus seems to periodically "reset" the system (sensu Bakun and Weeks 2008) keeping it at a relatively low (but highly productive) development state, also typically for the above-mentioned upwelling systems.

On the other hand there are other system features, which need to also be emphasized. The Bolivar Channel has an enormous diversity and biomass of fish species of different habitats (open water, rocky reef, sand bottom) and trophic guilds (predators, detrivores, planktivores, omnivores), whereas in the coastal upwelling systems of the South East Pacific fish diversity is low with a clear dominance of one or two pelagic planktivores (anchovy and sardine) and just a handful of other, much less abundant, fish species. The Bolivar Channel system also comprises large biomasses of non-bivalve filter feeders (there is, however, a very rare endemic scallop species, *Nodipecten magnificus*), such as gorgonians (*Muricea* spp. and *Pacificgorgia* spp.), zoanthids (*Parazoanthus* spp.), sponges (*Aplysilla* sp., *Carmia* sp.) and endemic ahermatypic corals (*Tubastraea faulkneri* and *T. tagusensis*), while bivalve filter feeders typically dominate the shallow upwelling systems along the South East Pacific shore. An interesting feature of the Bolivar Channel is the lack of large cangrid or xanthid crabs, well-known benthic predators of the South East Pacific. Their niche seems to be occupied by three species of Spiny lobsters (*Panulirus penicillatus*, *P. gracilis*, and *P. femoristruga*) and one species of Slipper lobster (*Scyllaride astori*). The proportion of endemic species is high in the Bolivar Channel and exceeds the level of endemism in the north-western and south-western regions of Isabela and of western Fernandina. For this reason, and because several invertebrates species have only been recorded here, the Bolivar Channel area is considered unique for its mix of tropical and temperate species (Edgar *et al*., 2004).

Time series / simulations

For the fish groups benthic predatory fish, small benthic predators and mullets observed reductions in average density following the El Niño event 1997/98, correlated well with those predicted by the model (Table 2), which also predicted observed reductions in surgeonfish (including chubs and giant damselfish), benthic omnivorous fish, and groupers. In case of the Galápagos grouper (*Mycteroperca olfax*), the data suggests that densities were quite high in 1997 and decreased later during El Niño. This is congruent with high catches of this species, during the first months of the event, before they dropped. According to Nicolaides et al. (2002), the Galápagos grouper and several other benthic fish (including serranids such as camotillo, *Paralabrax albomaculatus* and norteño, *Epinephelus cifuentesi*) may have migrated to deeper and colder waters during the event and returned, when conditions normalized. If this was the case, the model prediction of increased mortalities due to food shortage of these fish during the El Niño event would not mirror reality. Stein-Grove (1985) lists the Galápagos grouper and the camotillo among those fish species that were also less observed during scuba dives during the El Niño 1982/83. He also reports a density reduction in the plankton feeding damselfish *Chromis atrilobata* and *Azurina* sp. and

Labrisomidae as well as in the algae feeding parrot fish (*Nicholsina denticulate*), which confirms our findings for the El Niño 1997/98. Our survey data for open water predatory fish groups suggest a certain biomass decrease for barracudas and jacks during the El Niño 1997/98, while according to *in-situ* observations, sharks appeared to have increased during the same period. The simulations confirm the negative trends for the predatory pelagic fish (statistically significant for Jacks and mackerels), while for sharks a rather neutral population response is predicted. Landings of some species of the group jacks and mackerels were lower during and immediately after the El Niño 1997/98 (e.g. sierra, *Scomberomorus sierra*), which would confirm the survey data, while others, like that of wahoo (*Acanthocybium solandri*) and palometa *(Seriola rivoliana*), greatly increased (Nicolaides et al. 2002). These latter two species are large open water species, however, and were not included as part of this model. There is no other available information on the shark response to the El Niño warming. If the survey data reflect reality, the model prediction of a neutral or slightly negative response would be wrong. An explanation could be that sharks successfully switch between preys when the food spectrum changes during El Niño conditions. It could also be that shark onshore movements into the Bolivar Channel area increased during the El Niño period, when open water resources are reduced, to make use of the different coastal and more easily accessible prey. In this context it seems important to note that the Bolivar Channel area has, on average, a 5-fold higher average phytoplankton biomass than the Galápagos Marine Reserve (GMR) as a whole and that even during the El Niño period 1997/98, Chl-*a* – values in the BC area never dropped below levels of average Chl-*a* for the greater GMR (approx. 0.4 mg m^{-3}), while they were near zero in the GMR (ESA Globcolour database at http://hermes.acri.fr/). These data thus suggest that the Bolivar Channel ecosystem may still be used by the large predators to search for food during this critical period, so that relative sharks abundance may have increased in this area as was revealed by the survey data.

The seabirds (penguins and flightless cormorants) monitoring data show great reductions in population numbers during the El Niño event 1997/98. Valle-Castillo (1985) also reports decreases of 45% and 78% for cormorants and penguins respectively during the El Niño 1982/83 with similar reductions recorded during the El Niño in 1997/98 (Vargas et al. 2005). The model simulations confirm the direction and also the magnitude of change (Table 2), which strongly suggests that El Niño induced food shortage was the main reason for the increased mortalities within both populations, as has been hypothesized before (Valle-Castillo 1985, Vargas et al. 2005). Possible reasons for the much higher observed than simulated seabird biomasses for the last decade following the El Niño event are not clear. However, since this decade is considered as an extended period of strong upwelling in the study area (Wolff 2010) it is possible that small pelagic fish (including those outside the Bolivar channel) were abundant and contributed to the population increase in sea birds. Sea turtle and marine iguana populations decreased during the El Niño 1997/98 as clearly revealed by the Charles Darwin Foundation monitoring data and as also, but to a lesser extend, predicted by our model simulations (Table 2). A similar decrease was also reported for El Niño 1982/83 by Laurie (1985), who attributed it to the great reduction of macroalgae, which forms the basis of their diet. The larger decrease of reptile biomass (as compared to the simulation) maybe explained by additional food competition effects with other herbivores under conditions of macroalgae shortage. It appears that the foraging arena for marine iguanas is limited to close – shore algae beds, which require little swimming effort. If algae cover decreases and algae become more patchily distributed, swimming capacity may not suffice for successful feeding under these conditions.

The marine mammals monitoring time series, which shows a significant decrease during the El Niño event (by as much as 50%, Salazar 2002, 2003), is not congruent with the model simulation, which predicts < 10% population decrease. Explanations may be: 1) the data we used for the time series were extrapolated from surveys conducted in the central and southern areas of Galápagos and did not include the Bolivar Channel area. These other areas may have experienced higher impacts from the El Niño, and thus the extrapolation of their population dynamics may overestimate the reduction in the Bolivar Channel, 2) our diet matrix for marine mammals considers that only about 30% of ingested food stems from small pelagic and plankton feeding fish groups, so that food reduction due to the El Niño changes is quite small and 3) the biomass of marine mammals in our model (and thus the amount of food ingested) is comparatively low at 1 g m^{-2}.

For most invertebrate groups the model simulation predicts substantial population reductions during the El Niño period, but since biomass time series were only available for the group's sea stars and sea urchins, sponges and polychaetes and lobsters, a comparison between simulated and observed trends can only be done for those groups. In the case of the group's sea stars and sea urchins and of sponges and polychaetes a decrease is seen from 1997 to 1998, as also predicted by the model. The simulation suggests an increase in Lobster biomass following the El Niño impact. This trend is even more pronounced in the survey data (Fig.6) and also confirmed by the fisheries catches, which increased during the post El Niño years 1999 and 2000 (Toral et al. 2002). An explanation for the lobster proliferation following the El Niño event could be the relative low levels of predator (predatory marine mammals and large benthic predatory fish) and high levels of prey (sea urchins, sponges, etc.) biomasses immediately after the event, favoring population increase of the lobsters. The very high biomass value found during the monitoring in 2001 (see Fig. 6), which surpasses the prediction of the model, may have resulted from El Niño (warm water) induced high recruitment levels. This was also reflected in high lobster catches during the year 2001.

The fitting of the time series data based on the model forcing using the Phytoplankton and macro-algae time series allowed for 29.2% a reduction in the sum of squares (from 239.5 to 169.5), which is a substantial improvement in the fit of the curves to the data and clearly shows the importance of the bottom up-regulation of the system during El Niño periods. While the observed and simulated biomass trends of the different model groups agreed well in most cases, a statistically significant correlation was only obtained for some groups, (Table 2) for which the abovementioned regulatory mechanisms are postulated. Longer observational time series will thus be needed to redo the model simulations and find out if the observed biomass trends for the other groups also resulted from the postulated mechanisms.

The vulnerability values computed by the program during the search procedure yielded *v*-values indicative of top-down control of their prey (benthic predatory fish, groupers, small predatory gastropods, pelagic predatory fish, and sharks), The very low *v*-values of lower trophic levels (phytoplankton, macroalgae and herbivorous zooplankton) point to their role as bottom-up controllers to higher predators. While most of the *v* -values calculated made ecological sense, a note of caution is needed here, since our time series are relatively short (16 years) and assembled from different sources, which may limit their value for the vulnerability search routine of EwE.

Strength and weaknesses of approach

The approach used in this study is based on several assumptions and has limitations. As for all ecosystem system scale models, many biotas had to be lumped into manageable functional units such as marine iguanas and sea turtles or different species of seabirds and of fish. This simplification means a loss in realism, since none of the species lumped can be considered of having an identical (redundant) function in the system. However, the species grouped in our functional compartments have similar population dynamics, preys and predators and can be expected to respond in a similar way to disturbances such as El Niño caused shortage in food and changes in predator abundance. The coupling of our model to time series of environmental drivers and observational data offered a great opportunity to explore the model's capacity to reproduce observed trends. We think that the exercise here presented was worthwhile and shines new light on the trophic functioning of this unique marine ecosystem and the role of El Niño in shaping the system configuration and modulating the system's bottom-up and top-down regulation over time.

The trophic modeling approach also allowed for system scale comparisons with other shallow water areas of the ETP region, revealing that the Bolivar Channel system, although often considered a typical tropical rocky reef system, exhibits many features of an upwelling system of the Humboldt Current despite a unique species composition and high degree of endemism.

The simulation exercise, while revealing the cascading effect of El Niño reduced primary productivity through the food web, also evidenced that some model groups did not respond as anticipated by the observational data. This is no surprise since other drivers not captured by our model may also play important roles in the regulation of population sizes over time. Examples include lobsters and sea cucumbers, whose proliferation during and shortly after the El Niño period may have been possibly due to El Niño triggered recruitment events. It may also be assumed that food shortage has not played a crucial role for their population survival during the El Niño warming, since population of both species have greatly been reduced by the fishery over the past decades, so that food may not even be a limiting factor under conditions of reduced food supply during the El Niño. It seems advisable to combine this kind of ecosystem scale trophic modeling with population scale models in order to better understand the role of different factors in regulating population sizes.

Does the data quality allow for the construction of such a complex trophic model as the one here presented? If we rank our pedigree index (Christensen and Walters 2004) of the input data (0.54 with a measure of fit of 3.34) with that of other published models (Arreguín-Sánchez et al. 2004, Coll et al. 2006, Morissette 2006, Taylor et al. 2008a), our model can be considered of intermediate data quality. The additional use of the resampling routine of EwE (Ecoranger) provided new parameter value – estimates, that were very similar to the original input values (<5% in most cases), indicating that our basic input was very reasonable. However, the fact that the model is balanced thermodynamically and physiologically plausible (parameter ranges are realistic) does not necessarily mean that all input values used are correct.

However, considering that the system modeled is that of a remote Archipelago in the tropics, where international research and monitoring standards are difficult to achieve, the data volume and quality is quite astonishing. This holds especially for the long time series of in-situ observations of compartment biomasses used for the model construction and time series simulations. We had to make assumptions with regard to the exploitation rate of the fishing targets and had to adjust the diet matrix of the El Niño state to drastic changes in producers and consumers caused by this warming event. While we achieved a mass balanced El Niño model that obeys to the general rules of physiology and trophodynamics of its groups, we cannot exclude the possibility of some biased model inputs. However, the general system properties described and trends observed should be real and meaningful and provide an important basis for future studies.

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3

Elucidating fishing effects in a large-predator dominated system: the case of Darwin and Wolf Islands (Galápagos)

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Abstract: Fifty years of artisanal fishing history in Galápagos with boom and bust of resources have changed the natural marine community structure close to fishing ports in the Galápagos Marine Reserve. However, remote regions are still in "*near natural state*". Darwin and Wolf, the most remote islands in Galápagos, show an unusual species richness with a complex trophic structure. The artisanal fishery in this island system is aimed at extracting principally large demersal predators of high trophic level. The nature of Darwin and Wolf allowed construction of a trophic steady-state model using the EwE software that connects benthic and pelagic communities through a predator-prey matrix. This Ecopath model was then used to simulate the responses of system components to a potential increase in artisanal fishing effort. The total biomass estimated was 937 t km^{-2}. Large fish aggregations accounted for 55%, primary producers for 9.0% and non-fish (seabirds, marine mammals and sea turtles) for less than 0.1% of the total living biomass. The total system size in terms of overall flows was 16652 t km^{-2} $year^{-1}$, four times larger than a generic seamount model but smaller than the Bolivar Channel – Galápagos upwelling system. Consumption and respiration dominated the flows in the system with 53.4 and 31.8% respectively. The Darwin and Wolf system shows a mature state, dominated by respiration, and top-down controlling sharks and benthic predatory fish as the most important system compartments. Model simulations highlighted the strong effect on the system, when natural predator densities are reduced through fishing: the system clearly tends to become to a sea urchin-dominated state. However, system impacts due to increased fishing differ significantly for the different gears. Darwin and Wolf have attracted attention by Galápagos National Park, Charles Darwin Foundation and other no-governmental organization due to their unique marine diversity dominated by large-predators that occur nowhere else in Galápagos. Fishing activities are very sparse and it is difficult to accurately estimate the present day fishing effort in Darwin and Wolf. Our model simulations show, however, that heavy fishing pressure on top predators would greatly change the system's trophic structure.

Keywords: Trophic modeling, Darwin, Wolf, Galápagos, Ecopath with Ecosim, Oceanic islands.

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Introduction

Seamounts represent natural obstacles to water flow, modifying local currents, which can enhance topographic upwelling to support a wide variety of marine life (Roger 1994). Darwin and Wolf, the most remote islands in the Galápagos Marine Reserve (GMR), are small islands with abrupt bathymetry similar to seamounts that sustain a unique marine biodiversity (Edgar et al. 2004a) with a complex structure (Hearn et al. 2010).

The benthic shallow habitats off Darwin and Wolf (<30 m depth, Fig. 1) are mainly dominated by a heterogeneous rocky reef substrate and extensive corals that maintain high system biomasses attributed to local upwelling of deeper nutrient-rich water (e.g. Kunze and Llewellyn Smith 2004, Lueck and Mudge 1997), plankton advection by the "island mass effect" (Emery 1964, Hamner and Hauri 1981, Palacios 2002), and the elevated productivity of corals through nutrient recycling (Muscatine and Porter 1977).

Due to the islands small size and abrupt bathymetry, the pelagic and benthic reef biota interact closely, resulting in a complex trophic structure, of highly abundant benthic and pelagic predators such as snappers (e.g. *Lutjanus* spp., *Hoplopagrus guentherii*), jacks (e.g. *Caranx* spp., *Seriola rivoliana*), mackerels (e.g. *Scomberomorus sierra*) and sharks (*Carcharhinus galapagensis*, *Triaenodon obesus* and *Sphyrna lewini*).

Similar large-predator dominated systems have been described for remote islands where fishing intensities are low (e.g. DeMartini et al. 2008, Sandin et al. 2008, Stevenson et al. 2006). Here the marine ecosystems are considered to be in a quasi-pristine state. However, the mechanism behind to this unusual large-predator accumulation in those systems is still unclear.

The artisanal fishery in Darwin and Wolf is principally targeting large demersal predators of high trophic levels. They includes the endemic Galápagos grouper *Mycteroperca olfax*, three species of cabrilla *Cephalopholis panamensis*, *Epinephelus cifuentesi* and *E. labriformis*, the camotillo *Paralabrax albomaculatus*, and one spiny lobster species *Panulirus peniscillatus*, all of which are the principal exploited resources in this system (Comisión Técnica Pesquera de la Junta de Manejo Participativo 2009). However, spatial explicit fishery information for the GMR is difficult to obtain and estimates of the real magnitude of the captures in and around the Darwin and Wolf islands are poor. Most of the captures are directly traded between fishermen and tourism boats in the same area and not completely reported during their landings to the Galápagos National Park (authors *per. obs.*).

Darwin and Wolf have very high conservation significance due to their distinctive biota and anomalously high species richness of fish and corals dominated by large-predators that occur nowhere else in the Galápagos Archipelago in these high abundances (Edgar et al. 2004a). However, under the zonation scheme declared in 2000 and physically demarcated in 2004, Darwin and Wolf have been allocated a low protection category in the GMR, with only 26.8 % of its benthic habitats completely excluded from artisanal fisheries (Edgar et al. 2010, 2011). The effects of the fishery in this remote part of the Archipelago during the last 5 decades of small-scale fishing history remain as yet unclear.

Commercially important species has been fished down to a vulnerability level in the most fished areas in the Galápagos Marine Reserve (Ruttenbergs 2001, Sonnenholzner et al. 2009) because their continue demand in the local market to supply the necessities of the population and tourism sector. In respond to this, fishermen have extended their fishing areas to remote island such as Darwin and Wolf where fishing resources are still abundant. Experience from now unsustainable artisanal fisheries effects in Galápagos suggest an urgently needed of understanding the direct and indirect effects of increase fishing effort in the trophic structure and functionality of such quasi-pristine system to formulate and implement appropriated conservation strategies.

The present study intends to: 1) integrate the information of ecological surveys and fisheries to construct a holistic quantitative energy flow model by identifying and quantifying main biological compartments and energy flow pathways in this large-predator dominated system, and 2) explore a potential scenario of increase in fishing effort in this remote region of the Galápagos Archipelago and its consequence in the community structure, system development and stability.

Methods

Area description

Darwin and Wolf are 36 km apart and over 250 km distant from fishing ports (Puerto Ayora – Santa Cruz Island, Pto. Baquerizo Moreno – San Cristobal, Pto. Villamil – Isabela, and Pto. Velasco Ibarra – Floreana), in the central south-east of the Galápagos Archipelago (Fig.1). Edgar et al. (2004a) describes these two islands as a system with a predominantly Indo-Pacific and Panamic fauna which is self-sustaining in the Archipelago but with intermittent recruitment from the mainland and North-eastern Pacific Islands (Isla del Coco – Costa Rica, and Malpelo – Colombia, Edgar et al. 2004a, Edgar et al. 2011). Compareing the water temperatures to other areas of the Galápagos Archipelago, it is relatively high throughout the year (between 22 – 27 ºC, Banks et al. 2009) showing a peak during February (Banks 2002). The assemblages of fish, macroinvertebrates and benthic sessile organisms of both islands are very similar, but differ substantially from other regions in the Archipelago (Edgar et al. 2004a, 2011).

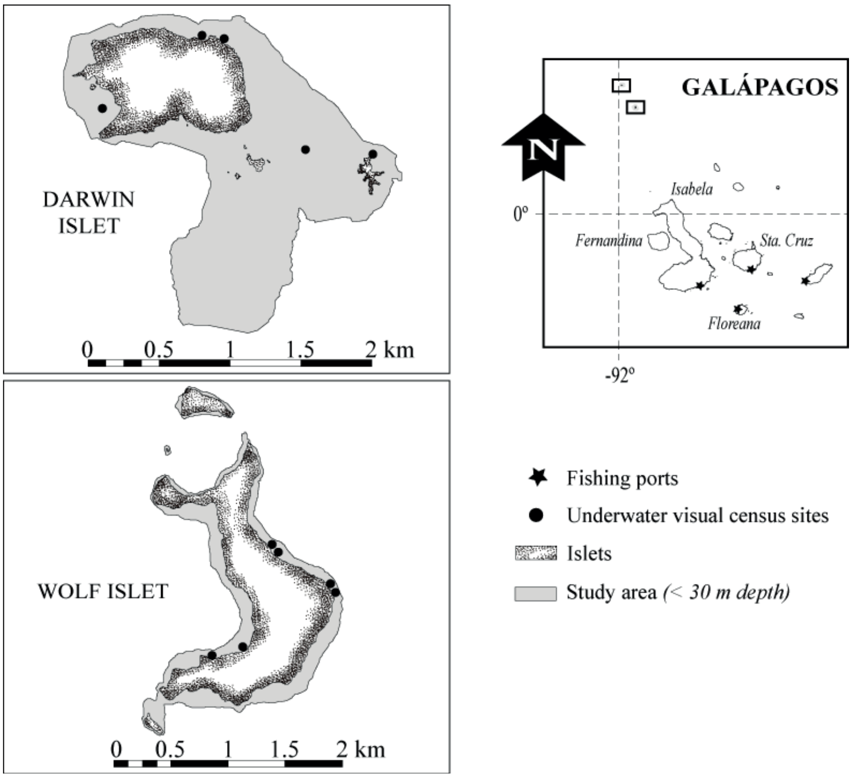

Fig. 1. Darwin and Wolf islands locate in the northern region of Galápagos Archipelago. Gray areas around the Darwin and Wolf islands represent the areas chosen for the model (<30 m depth) approach. Black circles in the left graphs shown the monitoring sites during the period 2004 – 2008 were the data used for the model was recorded, black stars in the right graph show the location of fishing ports.

Model definition and construction

The model area chosen was the shallow habitat (<30 m depth) around Darwin and Wolf, covering a total area of 2.8 km^2 (Fig. 1). For model construction the software Ecopath with Ecosim 6.0 (EwE) (Christensen et al. 2008) was used (http://www.ecopath.org – Christensen and Walter 2004). The model was organized into 32 functional compartments, including detritus. The species were grouped into compartments using the same criteria described in Ruiz and Wolff (2011). The model contains 13 fish functional groups, 10 benthic invertebrates groups, 3 primary producers groups, 2 zooplankton, 2 marine mammals, reptiles, seabirds, and detritus (Appendix 5). The model interconnects both benthic and pelagic communities through a predator-prey matrix, given the relative contribution of each prey group in the diet of its predator groups (Appendix 6). In Ecopath (Christensen et al. 2000), the production of each group is balanced by losses in the ecosystem, shown in the master equation (1) that follows:

Production = catches + predation mortality + net migration + biomass accumulation + other mortality (1)

For a detailed description of the equation and parameters used by Ecopath see Christensen et al. (2000). To assure mass balance among groups, a second master equation is used (2):

Consumption = production + respiration + unassimilated food (2)

Key input parameter for functional groups

Biomass of primary producers

Estimates of mean phytoplankton biomass were obtained from Globcolour GSM (http://hermes.acri.fr/) merged product (2003-2013), which combines SEAWIFS, MERIS, and MODIS. For converting chlorophyll *a* (Chl-*a*) concentrations (mg m^{-3}) to wet weight biomass the conversion factors used were: Chl-*a* - Carbon (40:1) (Brush et al. 2002) and Carbon – wet weight (1:14.25) (Brown et al. 1991). The phytoplankton water column biomass was then estimated for a uniform mixed layer depth of 20 m (Ruiz and Wolff 2011) yielding 1.87 g m^{-2}. Macroalgae biomass estimates on the rocky reef at Darwin and Wolf Islands were based on underwater observations during the underwater surveys that the Charles Darwin Foundation (CDF) has carried out each year at 5 sites in Darwin and 6 sites in Wolf, between the years 2004 to 2008 (Banks et al. 2006, Edgar et al. 2011, Ruiz and Wolff 2011).

Fish biomass

Fish abundance (N per m^2) and size classes were obtained from underwater counts and visual size estimation along a 50 m transect line (Banks et al. 2006, Edgar et al. 2011, Ruiz and Wolff 2011). The bias in divers' perceptions of underwater fish size was corrected using relationships proposed by Edgar et al. (2004b). Biomass (g per m^2, wet weight) was then calculated for each species using its abundance estimates and the length-weight relationships (Fulton 1904, Ricker 1975) provided for each species in Fishbase (Froese and Pauly 2013).

Invertebrate biomass

Abundances of benthic macro-invertebrates (sea stars, sea cucumbers, sea urchins and mollusks) (N per m^2) were calculated from transect sampling at the same transect sites as the fish species (Banks et al. 2006, Edgar et al. 2011, Ruiz and Wolff 2011).

Biomass (g per m^2 of wet weight) was then calculated by multiplying the abundance by the corresponding weight (g) of the specimens derived from previously computed length-weight relationships (authors unpublished data). In some cases a dry to wet weight (g) conversion had to be made for some collection specimens from the CDF Museum, using conversion factors as given by Opitz (1996).

For the groups, herbivorous zooplankton, predatory zooplankton, small herbivorous gastropods, stony corals and zooxanthella biomass estimates were not available and these values were left blank in the input data matrix. Ecotrophic efficiencies (EE) of these groups were estimated in term of the minimum biomass needed to satisfy predation mortality given the groups' biomass, productivity rates, and consumption rates of their predators and fixed as an input parameter to allow the model to compute the missing biomass.

Other vertebrate biomass

Seabird, marine mammals and sea turtle biomass estimates (g m^{-2}) were based on observations of the authors during the boat trips between 2006 and 2008 in both islands. The relative abundance for each vertebrate species estimated for both islands was then multiplied by the mean individual weights in *g* obtained from the literature (Fischer et al. 1995, Zavalaga1 et al. 2007, Dunning 2007).

Catch and diet matrix

The catch estimates were based on the data obtained from the Project "Shifting Baselines": INCOFISH WP2 Data Pages (Castrejon and Moreno, www.hull.ac.uk/incofish, see Table 1). The basic structure of the diet matrix connecting predator and prey within the system was adopted from the developed Bolivar Channel model (Ruiz and Wolff 2011) and modified considering the relative production of available food items (Appendix 6). Diet proportions were thus adjusted to reflect predatory groups' consumption rates as well as the available production of prey groups. Given that Darwin and Wolf ecosystem is an open system, a "*diet import*" approach (Christensen et al. 2008) was needed for some groups. This approach take in account the consumption of preys that are not part of the system as was defined and it was entered as a fraction of the total diet for species/compartments that continuously move in and out of the boundaries of the model area to feed, such as sharks (Klimley 1987, Torres-Rojas et al. 2006), transitory marine mammals such as dolphins, seabirds (Ashmole and Ashmole 1967) and some predatory fishes (Appendix 6).

Table 1. Catch estimates by functional group in the Darwin-Wolf system. Last column shows the proportion of the functional group production harvest by the artisanal fishery. HL, Handline, HS, Hawaiian spear, TL, Trolling line/Lure, HD, Hookah diving, P, Functional group production used in Ecopath equations 1 and 2, * Illegal fishing. (based on Castrejon and Moreno, www.hull.ac.uk/incofish)

Group name	Landings by fleet (tonnes km^{-2} year^{-1})					P (tonnes km^{-2} year^{-1})	% Catch of total production
	HL	HS	TL	HD	Total		
Lobsters				0.270	0.270	11.626	8.92
Small benthic predatory fishes	0.005	0.006	<0.001		0.011	34.145	0.02
Benthic predatory fishes	2.510	0.003			2.513	29.181	7.64
Groupers	0.310	0.017			0.327	0.591	57.54
Pelagic predatory fishes	0.300	0.003	0.147		0.450	10.003	1.43
Sharks	*0.180				0.180	2.969	2.30
Total	**3.395**	**0.029**	**0.147**	**0.270**	**3.841**		

Production/Biomass (P/B) and Consumption/Biomass (Q/B)

For fish compartments P/B and Q/B values were calculated based on parameters obtained from FishBase ((Froese and Pauly 2013). For other vertebrates as seabirds, marine mammals and sea turtles, those values were calculated based on similar compartments from other published models (e.g. Okey et al. 2004, Ruiz and Wolff 2011, for more details see Appendix 7). For macro-invertebrates Brey's Multi-Parameter P/B Model was used (Brey 2001) using a value of the mean water temperature of 24 °C (Banks 2002). The values for model compartments comprised by several species were estimated as averages of species-specific values weighted by their relative biomass (B) or consumption (Q) as appropriate.

Mass balance, flow characteristics and summary statistics

The initial input data produced an unbalanced model and a manual process was needed for model balancing. Small changes on the initial predator-prey matrix were made considering the availability of prey groups in the model. Therefore, it was necessary to alter the contribution of some preys in the diet of certain predators and to reduce the consumption of some predator groups in the system, in order to reach the mass-balance, adjusting values on the diet matrix to obtain $EE_i < 1$ for every group.

Additional increases of import values in the predator-prey matrix were needed for some groups such as hammerhead sharks (95%), predatory marine mammals (90%), seabirds (85%), other sharks (61%), dolphins (50%) and planktivorous reef fish (50%), in order to reach mass-balance. Some predators such as dolphins, sharks and seabird (e.g. Ashmole and Ashmole 1967, Klimley 1987, Torres-Rojas et al. 2006) are part-time residents in this system only obtaining substantial portions of their diets from outside of the model area

Once the model was balanced, selected system summary statistics provided by Ecopath were used for general description of the Darwin and Wolf model including: (i) total throughput (T) – measure of the total sum of flows within the system and considered here as indicator of ecosystem size, (ii) contributions to T from different flows – production, consumption, export, respiration and flows to detritus, and (iii) total primary production (PP) to total respiration (R) ratio (PP/R), biomass (B) supported by total primary production (PP/B) and biomass supported by total throughput (B/T) which are related to ecological succession and relative system maturity, according to Odum (1969) and Margalef (1987) succession can be assumed to be an orderly process of community development towards the mature stage, while maturity is the last state in the process of succession.

Network analysis and output indices of inters for fishing impacts

In the predator–prey matrix, the fraction of each functional group that will serve as food to another group has to be defined. The predator–prey matrix allows calculation of the trophic level of each group according to equation (3).

$$TL_j = 1 + \sum TL_i * DC_{ij} \qquad (3)$$

where, DC_{ij} is the fraction of the prey ($_i$), in the predator diet ($_j$). The trophic level of the predator TL_j is calculated as the average sum of the trophic level of its prey ($\sum TL_i$ * DC_{ij}) added 1.0. The groups of primary producers and detritus are assigned as trophic level 1.0 as default (Ulanowicz 1995). The omnivory index (OI) of each consumer group was calculated as:

$$OI_i = 1 + \sum_{j=1}^{N} (TL_j - (TL_i - 1))^2 * DC_{ij} \qquad (4)$$

where, TL_j is the trophic level of prey *j*, TL_i is the trophic level of the predator *i*, and, DC_{ij} is the proportion prey *j* constitutes to the diet of predator *i*. The OI represents the trophic specialization of the predator, values close to zero are assuming for specialized consumers, that feed on a single TL, and higher values when the predator feeds on several TLs (Christensen et al. 2008, Pauly et al. 1993)

Following the trophic level concept of Lindeman (1942), Ecopath calculates the average number of steps in the food webs and integrates the components into discrete trophic levels, in order to compute their transfer efficiencies. Transfer efficiency (TE) is the fraction of the total throughput at a discrete trophic level (TL), either exported or transferred to another TL (through consumption).

The keystoneness of a species can also be computed using the mixed trophic impact (MTI, Libralato et al. 2006). The Mixed Trophic Impact (MTI) analysis is derived from economic theory (Leontief 1951, Ulanovicz and Puccia 1990) and allows for the quantification of directs and indirect trophic interactions among trophic groups for the model. This analysis provides a quantification of the positive and negative impact that a hypothetical increase in biomass of a group would produce on the other groups in the ecosystem, including the fishery activities in the system. Therefore, keystone species can be identified by plotting the relative overall effect (ε_i), calculated from the MTI, against the keystoneness (KS_i). The Overall effect (ε_i) is described according to equation (5):

$$\varepsilon_i = \sqrt{\sum_{j \neq i}^{n} m_{ij}} \qquad (5)$$

where, m_{ij} calculated from the MTI analysis as the product of all net impacts for all the possible pathways in the food web linking prey (i), and predator (j). The keystoneness (KS_i) of a functional group is calculating according to equation (6):

$$KS_i = log\left[\varepsilon_i \left(1 - p_i\right)\right] \qquad (6)$$

where, p_i is the contribution of each functional group to the total biomass of the food web. This index is higher when the functional group or specie has both low biomass proportion within the ecosystem and high overall effect (Libralato et al. 2006).

Artisanal fishery impacts simulation using Ecosim

In Ecosim, the biomass dynamics of all ecosystem components that occupy trophic levels above the primary producers are determined by equation (7):

$$\frac{\partial B_i}{\partial t} = g_i \sum C_{ki} - \sum C_{ji} + I_i - \left(M_i + F_i + e_i\right) B_i \qquad (7)$$

where, ∂B_i represent the growth rate of group i during the interval ∂t, g_i is the net growth efficiency (proportion of food intake that is converted into production), M_i is natural mortality rate (excluding predation), F_i is fishing mortality rate, e_i is the emigration rate, I_i is immigration rate, the first sum represents the food consumed, over prey types *k* of species *i*, and the second sum represents the losses due to predation summed over all predators *j* of *i*. In our model, immigration and emigration were assumed to be balanced and thus were not considered.

Additional parameter requires for Ecosim dynamic simulation as vulnerability were kept as Ecosim default values. Changes in vulnerability parameters can benefit to fit the model response to biomass time-series, but in this case, time-series of data were not available for the 50 years of model simulation. With the default value of 2 for vulnerability we assumed a mixed control type in the system where bottom-up and top-down area acting simultaneously.

Simulations

Effects of a potential increase on fishing effort in Darwin and Wolf model were simulated with Ecosim to elucidate the response of the system components to a predators mortality increase. An indirect estimation of the magnitude that reached the fishing effort in Galápagos during the last 50 years in areas near to fishing ports was made based on literature, reports and personal communication with fishermen that point to assume an "*estimated value*" of 10 times greater as a guested increases on fishing effort. Simulations for each artisanal fishing gear (see, Table 1) were run independently in Ecosim. Additionally, a cumulative simulation combining all fishing gears was also done to identify the overall effect of all combined artisanal fishing gears in the system.

Results

Darwin and Wolf model characteristics

Input and output parameters of the model's compartments are listed in Appendix 5 and the predator-prey matrix defined is shown in Appendix 6. The total biomass of the modelled system, excluding detritus, was estimated as 937 t km^{-2}. Of this biomass, large fishes account for 55% of the total. Primary producers at the base of the food web contribute 9.0% of the total biomass (85.2 t km^{-2}), whereas the non-fish at the highest trophic levels (seabirds, marine mammals and dolphins) contribute less than 0.1% to the total system biomass.

Summary statistics, energy flows and trophic structure

Total system throughput (T) was 16652 t km^{-2} year^{-1}, four times larger than the value estimated by Fulton et al. (2007) for a generic seamount model, but it represents the smallest system in the Galápagos Archipelago (Floreana model 94850 t km^{-2} year^{-1}, Okey et al. 2004 and Bolivar Channel model 38695 t km^{-2} year^{-1}, Ruiz and Wolff 2011). Lowest primary production (3408.8 t km^{-2} year^{-1}) and biomass (937.8 t km^{-2} year^{-1}) are also characteristics of this remote islands system. Total consumption dominated the T with 53.4% of the total flows, followed by total respiration (31.8 %) and flow to detritus 12.8%.

The system net primary production of 3408 t km^{-2} year^{-1} estimated for the Darwin and Wolf was the lowest estimated for a system in the GMR (17101 t km^{-2} year^{-1} – Bolivar Channel; Ruiz and Wolf 2011, and 13250 t km^{-2} year^{-1} – Floreana; Okey et al. 2004). The relatively low values of PP/R (0.65) and PP/B (3.64) reflect a system dominated by respiration due to the accumulation of biomass in higher trophic levels and supported per unit of flows (Christensen 1995, Fulton et al. 2007). Just as for the low PP/R ratio, the estimated negative value of 1898 t km^{-2} year^{-1} for the net production in this system, results from the fact that a large part of the food ingestion flows is derived from outside the system boundaries. Higher positive values of net system production are expected in immature systems (Christensen 1995).

The high system omnivory index of 0.32 point to a system's relatively high level of interaction diversity (Fulton et al. 2007), here concentrated in the high trophic level groups such as sharks, seabirds, predatory marine mammals and pelagic predatory fish, which are as top predators more generalist feeders in this system, as has also been observed in other coral reefs systems (e.g. Ruttenberg et al. 2011). However, the hammerhead sharks are an exception here, since they show the lowest omnivory index of all large predators in this system (see Appendix 5) large proportion of the diet for this group (95%) is obtained from outside the system.

Main flows are included within TL I, II and III. TL I generated 23.7% of the T, less than TL II (48.4%, Fig. 2). The figure also shows the important link between detritus and TL II. Groups in the TL II (mainly composed of benthic fishes and invertebrates) consume a large proportion of detritus realized during predation activities of large predators in this system as has commonly been observed in tropical coral reefs.

The system's mean transfer efficiency was 10.2%, with a large proportion of flows originating from detritus (41%). The transfer efficiency was greater at the lower levels of the food web due to intrinsic differences in metabolic characteristics of organisms at different levels in the food web. Transfer efficiency in the Darwin and Wolf system is at the low end of the range (8 – 17%) reported by Christensen and Pauly (1993), Wolff (1994), Wolff et al. (1996) and also lower than the Bolivar Channel system of Galápagos modelled Ruiz and Wolff (2011).

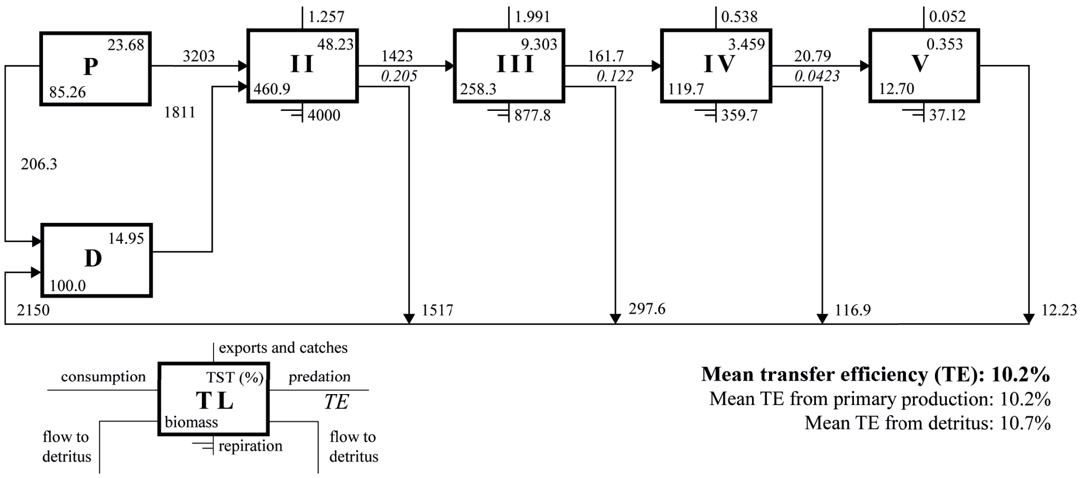

Fig. 2. Lindeman spine schematically represents the Darwin and Wolf system flows organized by integer trophic levels (TLs), where primary production (P) and detritus (D) are separated to clarify its representation in the model (both with TL = I). See flow chart at low left for description of numbers in boxes and on connecting lines

Mixed trophic impact analysis

The analysis of direct and indirect interactions within the ecosystem based on the mixed trophic impact (MTI) routine is shown in Fig. 3. Numerous functional groups in the model would be positively impacted by a biomass increase of the groups at the base of the food web such as detritus, phytoplankton and macroalgae. Sharks produce the highest negative impact on most of the other groups in this system, by way of a direct competition for resources with other large predators and fisheries.

Unexpectedly, predatory zooplankton ranks second in negatively impacting the system. As Pitcher et al. (2007) suggested that zooplankton biomass is not enhanced by advection in seamount systems, but has a significant influence on the community composition. The continued turnover of predatory zooplankton maintains the predation of phytoplankton in the system. Pelagic predatory fish and rays show similar net negative impact in this system. The competition for preys between large predators is the most important factor for the negative impact of pelagic predatory fishes, while a direct predation on macroinvertebrates is by rays.

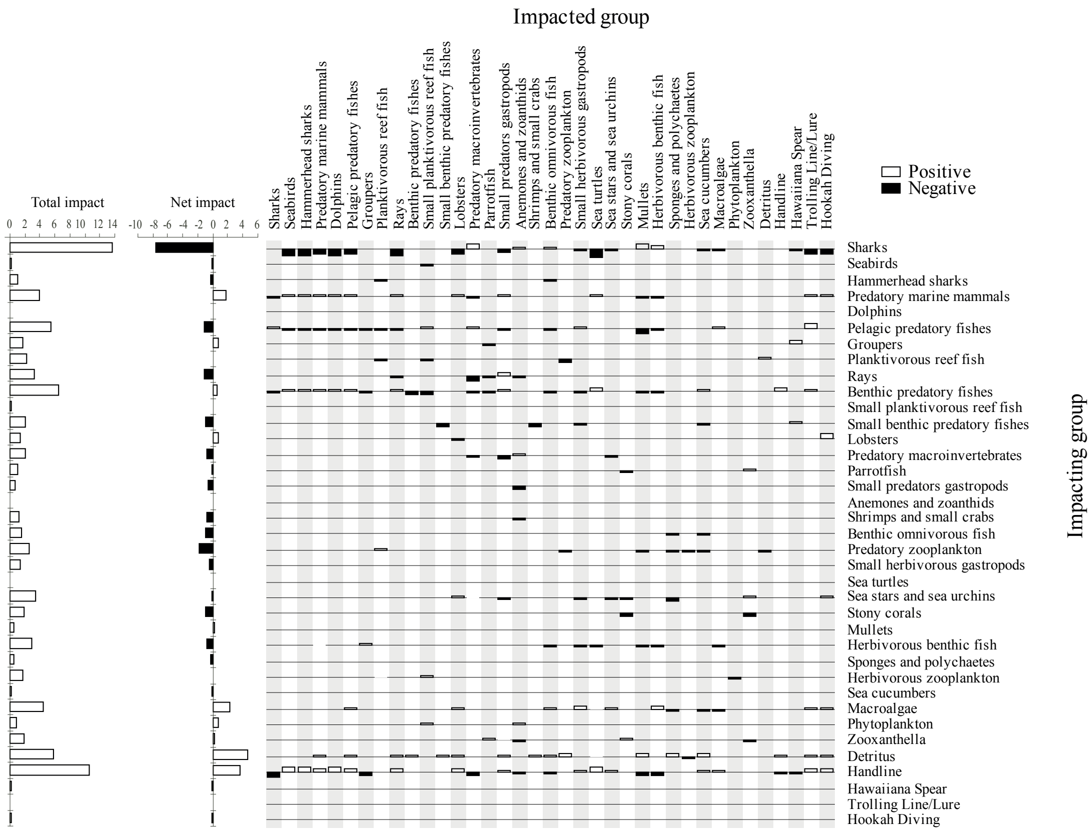

Fig. 3. Mixed trophic impact analysis of the Darwin and Wolf seamount ecosystem showing the direct and indirect impacts that changes in compartment biomass of each group cause on the other groups of the system. Positive impacts are shown by shaded bars above the base line, and negative by black bars below. The impact is relative but comparable between groups. Impacted groups are placed along the horizontal axis and impacting groups alone the vertical axis. Total and net impacts are shown on the left side of the graph.

Trolling and hookah fisheries have very low impact on other system functional groups, due to their very small scale and low effort in Darwin and Wolf. However, handline fishery shows a wide range of negative and positive impacts on the food web as result of direct and indirect effects of both benthic and pelagic predator fish removal.

Impact of artisanal fisheries in the system

The role of artisanal fisheries in this remote system is equivalent to a top-down control by a predator occupying a mean TL of 3.02 (Fig. 4), taking out of this system a substantial biomass (3.8 t km^{-2} $year^{-1}$) of high trophic level species. The benthic and pelagic predatory fishes represent 65% and 11% of the total catch, respectively.

The primary production required to sustain fisheries in the system (%PPR, Wallace 1998) was 76.5%, which is high when compared with the Bolivar Channel upwelling system (53.9 %; Ruiz and Wolff 2011), but lowest when compared with the Floreana model (356.9%; Okey et al. 2004). Exploitation rate (fishing mortality/total mortality – F/Z) for the endemic Galápagos grouper (0.57) is slightly higher in the Darwin and Wolf system, indicating that more that 50% of the total production of this resource is extracted from the system (see Table 1 and Appendix 5).

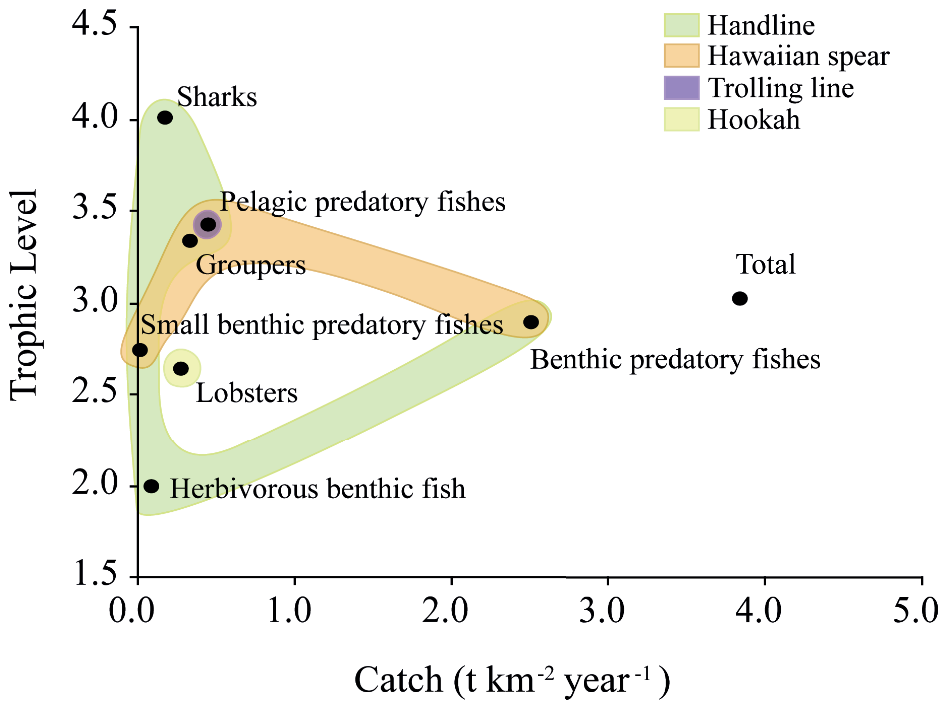

Fig. 4. Trophic level by functional compartment for the catch estimated by the Darwin and Wolf model.

The MTI analysis highlights the direct and indirect impacts that the fishing activities would have on the other groups (Fig. 3). The handline has a wide impact in the system (88% of the total catches). Conversely, Hawaiian spear, trolling line and hookah diving cause the least impact, less than 12% of the annual capture (see Fig. 3 and Table 1).

Simulating system impact of an intensification of artisanal fisheries

The model simulation showed strong system changes when benthic predator biomass is reduced due to fishing (Fig. 5). Under conditions of increased hookah (diving) fisheries on lobster (*P. peniscillatus*), biomass of sea urchins and other macroinvertebarte increase (hookah simulation, Fig. 6). Also, an inverse relation between sea urchin biomass and herbivore fish biomass was evident in this simulation.

An increase in the Handline fishery would cause a strong decline in grouper (*M. olfax*) biomass in a relatively short period (collapsing in 2 years) concomitant with a > 70% reduction in benthic predatory fishes and a strong positive response of lobsters and small planktivorous. Small benthic predatory fish also show a moderate increase in this scenario. In contrast to the hookah simulation, increased handline fishing would not cause an increase in sea urchin biomass in the system. However, herbivore fishes showed a negative response that can be related to direct and indirect effects of competition for algal resources with other groups such as sea urchins, in the system.

Hawaiian spear and trolling have a relative small effect on the system compared to handline and hookah. However, due to the selectivity of the Hawaiian spear (that mainly target the grouper *M. olfax*) the biomass of other large pelagic predators would increases in the system, through concomitant increases of their preys. In addition, Hawaiian spear has a positive effect on the herbivore fish biomass, through decreasing predation pressure on primary consumers fish and grazing competition through predatory effects of spiny lobster (*P. peniscillatus*) and other potential benthic predator fishes (*Diodon holocanthus*, *Sphoeroides annulatus*, *Sufflamen verres* and *Bodianus diplotaenia*, Ruttenbergs 2001) on sea urchins.

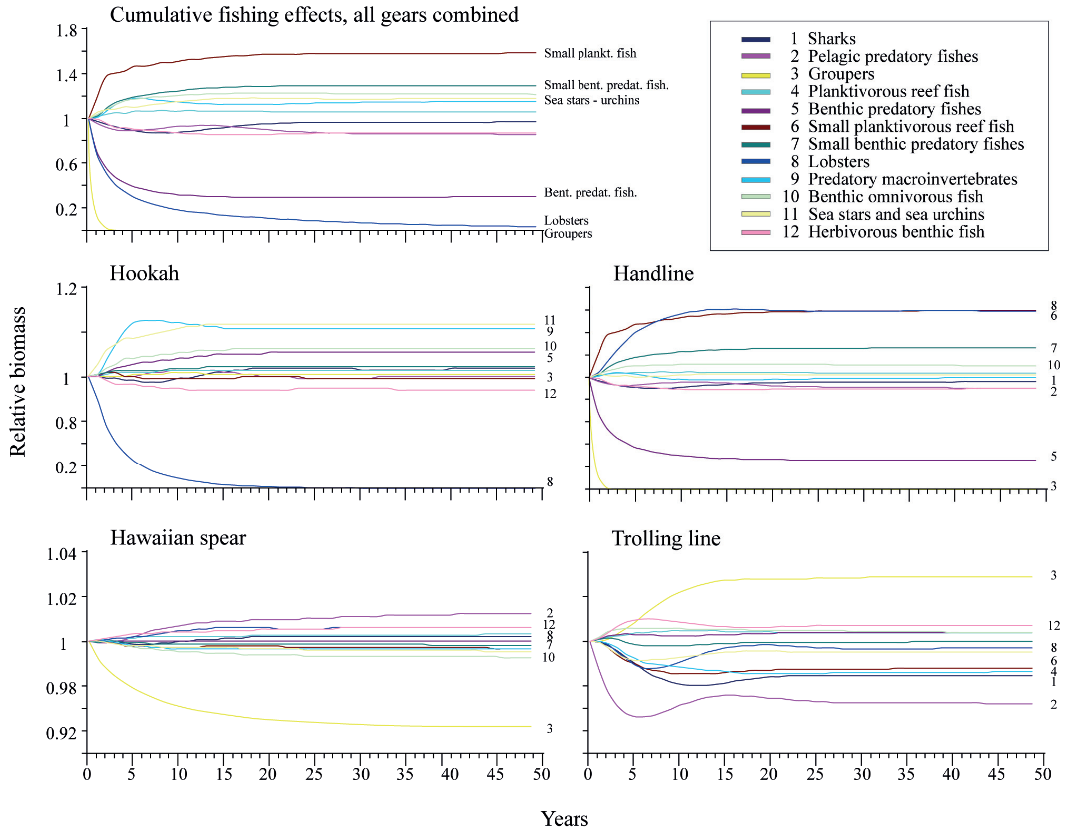

Fig. 5. Small-scale artisanal fishery effects on selected system compartments in the Darwin and Wolf system. Fishing effort was increased ten times from the base line. Effects of accumulative fishing gears and effects of each gear are presented as changes in relative biomass estimated for the model for a period of 50 years. Scales on *y*-axis of hookah, Hawaiian spear and trolling line are expanded.

Discussion

Structure and functionality of the ecosystem

One of the biggest challenges for modelling the Darwin and Wolf system was to capture the complex interactions between open, deep and shallow waters. The complex topography of Darwin and Wolf makes these islands hotspots of productivity and diversity, at all trophic levels, and for both pelagic and benthic zones. The islands attract and concentrate marine live from the surrounding ocean making them interesting and unique systems in terms of community structure and trophic interactions. Import of nutrients by topographic upwelling and plankton advection from oceanic surrounding water plays an important role in enhancing primary and secondary production available to large-predators. Moreover, import of food-energy to the system through consumption of pelagic and benthic preys from outside of the system boundaries by marine mammals, large pelagic fish and sharks with a great home range appears to be another important mechanism to maintain the system functionality, reducing the excessive predation pressure exerted by the great fish accumulation and the complex species structure that can be observed in Darwin and Wolf.

Those mechanisms and the large amount of energy that is lost through respiration of large predators in the higher trophic level results in low mean transfer efficiency that contrasts to the situation in the rest of the Archipelago. The high species diversity (Edgar et al. 2004, 2011), and a system dominated by respiration due to the accumulation of biomass in higher trophic levels clearly reveals a more mature system state (e.g. Christensen 1995, Heatwole and Levins 1972, Odum 1969, Odum 1983) when it is compared with the rest of the Archipelago (Okey et al. 2004, Ruiz and Wolf 2011).

In agree with some authors (e.g. Peréz-España and Arreguín-Sanchez 2001, May 1980, Pimm 1979), we believe that the mature state and complexity in the food web of Darwin and Wolf does not make this system stronger against human perturbation. More stable ecosystems are those maintained simple, such as the Bolivar Channel (Ruiz and Wolff 2011, Wolff et al. 2012), were organisms are adapted to temperature variability and upwelling processes. The Darwin and Wolf system may thus be more fragile due to its higher biodiversity and longer lifetime species present in this remote area of the Galápagos Archipelago.

Potential fishing impacts

The Galápagos Islands have increasingly been influenced by a growing human population (Merlen 1995) and by an increase of effort in the local small-scale artisanal fisheries (Camhi 1995, MacFarland and Cifuentes 1996, Merlen 1995), which has mainly targeted benthic predators. The high level of removal of these in some areas over the past three decades has resulted in cascading effects throughout the food web (Ruttenbergs 2001, Sonnenholzner et al. 2009).

The cumulative effect, of an exponential removal of both benthic and pelagic predators, observed in the dynamic simulations of the Darwin and Wolf model, clearly reveals the down of the food web driven by the collapse of important predators in the system. The Galápagos grouper (*M. olfax*) would collapse after 2-3 years of intensive harvest, followed by large declines of lobster and other benthic predator fishes. The high fishing effort assumed during the simulation clearly exceeds the natural biomass fluctuation in the system. The Darwin and Wolf system, after 10-15 years of simulation, shows an alternative state dominated by planktivorous reef fishes and sea urchins due to lack of predatory control.

Our simulations thus show that system impacts would differ significantly for the different gears. Benthic predator fishes are more affected by Hawaiian spear, whereas pelagic predatory fishes are more affecting by trolling lines. The direct removal of pelagic predators such as sharks (*Carcharhinus galapagensis, Trianodon obesus*) and pelagic fishes (*Caranx* spp., *Dermatolepis dermatolepis, Seriola rivoliana*) clearly show a change in the biomass of several system compartments. Direct and indirect effects of the decline of top predators result in a small (~2% - 3%) biomass increase of *M. olfax*, which drives a downward trend in the biomass of prey species of the system. However, a slight recovery in pelagic predatory fish, lobsters and shark groups in terms of biomass can be observed in the first 15 years of the simulation, the recovery of partial biomass observed in pelagic predatory fishes – including sharks in the system (Fig. 5), can be attributed to external energy flow through "*diet imports*".

The selectivity of handlines for higher predators caused the most dramatic response (Fig. 6). The model simulations also show an increase of urchin biomass. When natural predator densities are reduced through fishing, sea urchins proliferate in both tropical and temperate systems (McClanahan and Muthiga 1988, McClanahan et al. 1999, Ruttenbergs 2001, Sala and Zabala 1996, Sonnenholzner et al. 2009, Watson and Ormond 1994). Increases in sea urchin densities can also affect herbivorous fishes by the direct competition for algal resources (McClanahan et al. 1996, McClanahan and Kaunda-Arara 1996).

The results of the Ecopath with Ecosim approach presented here therefore suggest that artisanal fishing has not only direct effects over the target resources but also secondary cascading effects throughout the benthic community. Our simulations also showed that the changes in biomass precipitated in a 5 to 15 years period of exploitation, making the system look like those that have been subjected to prolonged exploitation (Jackson 1997, Jennings and Polunin 1996) and also observed in shallow rocky reef areas near to fishing ports in the Galápagos Archipelago (Ruttenbergs 2001, Sonnenholzner et al. 2009).

A complete recovery of the system has not been observed during the simulation period, which support our assumption that Darwin and Wolf is a fragile system and increases in fishing effort can have important consequences in restructuring the system and should be taking in count for ecosystem management.

Conclusions

The remote rocky reef system of Darwin and Wolf has largely maintained their – ¨*natural state*¨, with a well-developed community of large-predators. Unfortunately, this remote area is not excluded from the influence of a fluctuating artisanal fishing effort due to a weakness protection in the actual marine spatial management planning. Fishing effects can change the community structure and trophic interactions, resulting in an alternative state of the system dominated by sea urchin aggregation, where biogeographical differences could be reduced by coral habitat degradation and loss of species diversity.

The information about fishing activities is very sparse and it is difficult to accurately estimate the fishing effort in Darwin and Wolf. Current fishing monitoring program must be evaluated and adapted to produce better spatial information to provide a robust input to evaluate the fishing effect and improve the effectiveness of the actual marine spatial management in the Galápagos Marine Reserve.

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4

Trophic flow structure of shallow rocky reef systems of the Galápagos Marine Reserve along a biogeographical gradient

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Abstract: Quantitative energy flow models (Ecopath) were used to describe the food web differences between the three major biogeographical regions of the Galápagos Marine Reserve (GMR): cold western-upwelling ("*Bolivar Channel*"), mixed south-central ("*Floreana*") and tropical far-northern ("*Darwin and Wolf*"). Model inputs were based on 5 years of ecological monitoring carried out by the Charles Darwin Foundation (CDF). Results indicate that total biomass is similarly high for the Bolivar Channel and Floreana systems, but significantly lower for the far-northern ("*Darwin and Wolf*") system. In addition, biomass distribution by trophic level differs greatly between tropical and cold systems. In the more tropical, far-northern system, most biomass is allocated to high trophic levels where large apex predators and other piscivorous fish dominate. The magnitude of observed predatory biomass differences between the "*Darwin and Wolf*" and "*Floreana*" systems is consistent with other studies comparing remote and populated islands in the Pacific and Indian Ocean where the level of geographic isolation prevents fishing impact in remote areas within the Galápagos Marine Reserve. Phytoplankton is the most important primary source of energy in the cold-upwelling ("*Bolivar Channel*") system, while detritus occupies this role in the non-upwelling systems ("*Floreana*" and "*Darwin and Wolf*"). According to several system indices calculated, maturity and development is higher in the remote far-northern system, where the anthropogenic and natural-oceanographic perturbations are lowest. Species identified as keystone differed between bioregions. Sharks are keystone species in the remote northern tropical system where they control intermediate predators, which are far more abundant in the remote northern tropical system, while sea lions dominate the cold-upwelling system.

Keywords: Trophic modelling, Galápagos Islands, Biogeographic regions, Ecopath with Ecosim, Eastern Tropical Pacific.

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Introduction

The Galápagos Marine Reserve (GMR) is located in the Equatorial Eastern Pacific Ocean, about 1000 km west of mainland Ecuador, between 01°40'N and 01°25'S and 89°15'W and 92°00'W. Its location at the confluence of warm and cold surface currents and its geomorphology makes it a complex transition zone between tropical, subtropical, and upwelling nutrient-rich waters. In addition, the isolation of the Galápagos Archipelago has led to the development of unique and diverse marine communities (Bustamante et al. 2008, Colinvaux 1972, James 1991, Wellington 1984). More than 90% of the shallow benthic habitats in Galápagos are formed by volcanic lava formations (Banks et al. 2009, Bustamante et al. 2002, pers. obs.) of high complexity that provide refuge and substrate for complex marine communities. Shallow reefs also comprise the habitat most affected by the local artisanal fishery and other human activities (Edgar et al. 2004). Coral reefs are rare amongst of all of the shallow habitats in Galápagos, being restricted to just a few islands in the north and central-south (e.g. Darwin, Wolf and Floreana, Bustamante et al. 2008).

Characteristics of biogeographic regions of the GMR

The cold and warm currents in the complex and dynamic environment of shallow waters (> 30 m depth) have resulted in discrete biogeographic regions (Abbott 1966, Glynn and Wellington 1983, Harris 1969, James 1991). Edgar et al. (2004a) reviewed previous bio-region classifications and suggested three major biogeography regions based on the community composition of benthic sessile organisms, reef fishes and mobile macro-invertebrates (Fig 1): the "*Western*" temperate-cold region, the "*Far-northern*" tropical-warm region, and "*South-central/eastern*" mixed temperate-subtropical region. These three biogeography regions are defined by their distinctive and unique biota, with components of the Indo-Pacific, Peruvian, Panamic-Caribbean, and local endemic species (Edgar et al. 2004a, Kay 1991, James 1991, McCosker and Rosenblatt 1984).

The "*Western*" temperate-cold bio-region, which includes the Bolivar Channel, is located, is heavily influenced by cold nutrient-rich upwelling waters from the Equatorial under-current (EUC). Peruvian species are restricted to this region where macroalgal-communities dominate as foundation species. The water temperatures are relatively low throughout the year (16 – 18 ºC), and high primary productivity can be observed (2.3 – 4.4 mg Chl-*a* m^{-3}, see Fig.2). Most of the endemic and emblematic species of Galápagos are present in this bio-region, such as the marine iguana (*Amblyrhynchus cristatus*), the Galápagos penguin (*Spheniscus mendiculus*), the flightless cormorant (*Phalacrocorax harrisi*), the Galápagos fur seal (*Arctophalus galapagensis*), and the endemic kelp of Galápagos (*Eisenia galapagensis*).

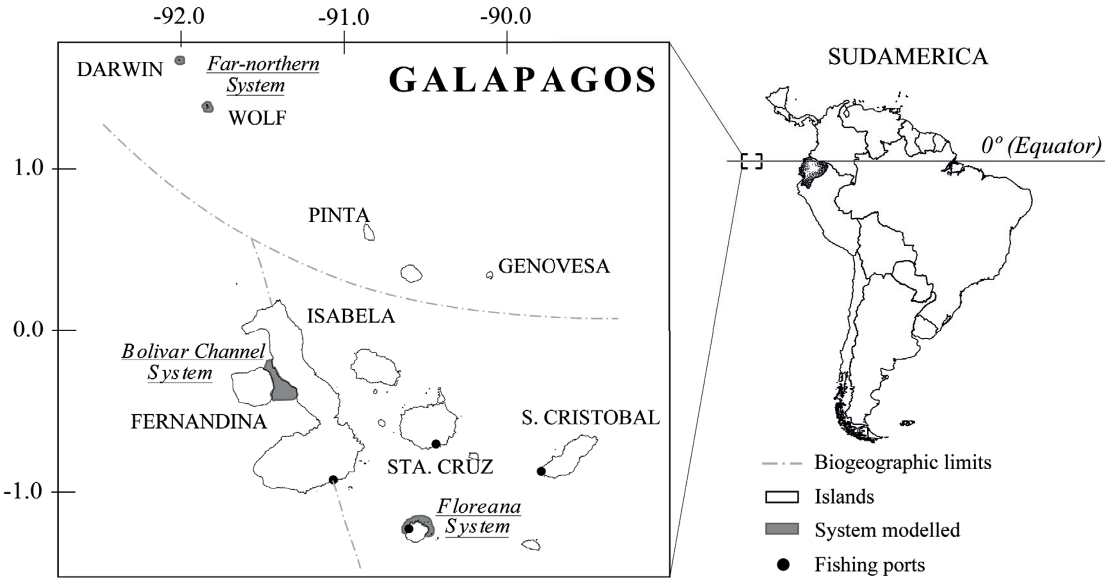

Fig. 1. Left: Galápagos Archipelago, dashed lines delimit the three major biogeographic regions of the Archipelago, Light gray areas around the islands show the location of the modelled systems compared in this study.

The "*Far-northern*" tropical bio-region, where the Darwin and Wolf Islands are located, has been reported to show closer affinity with the Pacific Oceanic Islands, such as Isla del Coco (Costa Rica) and Isla Malpelo (Colombia) than with the Galápagos southern and western islands (Edgar et al. 2004a 2011). Both of these far-northern islands show a predominance of fauna from the Indo-Pacific and Panamic regions with no endemic species of Galápagos, and a continuous larval supply from outside external sources (Edgar et al. 2011). Here, water temperatures are relatively high throughout the year (22 – 27 ºC) and the primary productivity is quite low (0.4 – 1.9 mg Chl-*a* m^{-3}, see Fig. 2). The assemblages of fish, macroinvertebrates and benthic sessile organisms off the two northern Darwin and Wolf islands are very similar, and differ in their species composition substantially from other bio-regions in the Archipelago (Edgar et al. 2004a, 2011). The total number of macroinvertebrate species described for the Darwin and

Wolf here is about 25% lower than elsewhere in the Archipelago (Edgar et al. 2004a, 2011).

The "*South-central/eastern*"- mixed water bio-region, which includes the Floreana Island, contains species from a variety of different sources, particularly 'Panamic' species such as the damselfish *Abudefduf troschelii*, wrasse *Halichoeres nicholsi*, grunt *Haemulon scudderi*, and sea urchin *Tripneustes depressus*. The water temperatures is the result of a mix of cold and warm waters masses and range between 20 – 24 ºC in this south-central/eastern region, with small zones of higher productivity (0.6 – 2.1 mg Chl-*a* m^{-3}, see Fig 2) due to the incursion of the EUC.

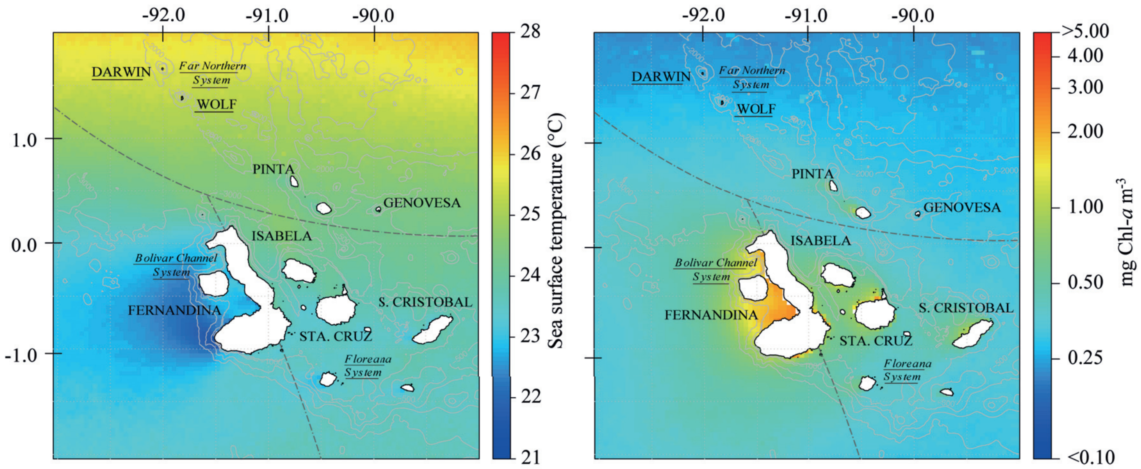

Fig. 2. Average spatial values for remote sensing estimates of sea surface temperature (SST) (MODIS-Aqua daytime SST, 2003-2013, 4 km resolution) and Chlorophyll-*a* (Chl-*a*) (Globcolour merged product, 2003-2014, 4 km resolution). Dashed lines delimit the three major biogeographic regions.

Existing food web models in the GMR

Prior to the comparative study presented here, a trophic model was constructed for the south-central Floreana Island system by Okey et al. (2004) to evaluate fisheries and conservation strategies in the shallow rocky reef areas around this island, by the period 2001-2002. The model used the Ecopath software and was constructed initially with 43 functional compartments, which was later simplified by Bustamante et al. (2004) to a 24 compartments.

A further Ecopath model was then constructed by Ruiz and Wolff (2011) for the most productive zone in the GMR, the Bolivar Channel upwelling system. This model consisted of 30 functional compartments and represents the period 2004-2008, characterized by normal oceanography conditions. This model was later used as a reference to simulate the effect of El Niño using historical observational data (Wolff et al. 2012). For the El Niño period, an adjusted model was also constructed considering the relative biomass of the different compartments estimated for El Niño 1997/98 period (for more details, see: Wolff et al. 2012).

Study approach

Due to differences in data and model architecture, a standardized methodology was used to comparatively analyze the food web structure of three representative shallow rocky reef systems from each bio-region following a three-step procedure: 1) re-estimate species/compartments biomasses by the period 2004 – 2008 using the Charles Darwin Foundation (CDF) subtidal ecological monitoring data base, 2) compare aggregated biomasses per trophic level and flows between systems, and 3) standardize the regional models to a similar structure (i.e. functional compartments) when possible.

The complexity and functioning of Galápagos rocky reef food webs in the different biogeographic regions of the Galápagos Archipelago are, as yet, poorly understood. The comparative trophic modelling approach followed in our study focus on identifying and comparing important energy pathways as result of top-down or bottom-up trophic interactions in each system. The standardized approach used provides a basis for the global interpretation of the trophic differences between the shallow rocky reef systems of the three biogeographic regions and is expected to lead to advice for ecosystem based resource management and ecosystem conservation.

Methods

Comparing fish and macroinvertebrates assemblages between major biogeography regions

To quantify between system differences in the community structure of fish and macroinvertebrates the mean biomass estimated for each species were pooled into two categories: i) "all groups together", and ii) "trophic/taxonomic" groups. Four trophic groups were considered for fish aggregations: primary consumers (including herbivores and detritivores), secondary consumer (including invertivores, corallivores, and omnivores), planktivores, and piscivores based on diet information taken from FishBase (Froese and Pauly 2013) and previous trophic classification of similar species proposed by Sandin and Williams (2010).

For macroinvertebrates a similar "trophic/taxonomic" grouping as described for fishes was done based on diet information taken from previous models (Okey et al. 2004, Ruiz and Wolff 2011, Wolf et al. 2012) and field observations by the authors. The four groups defined were: carnivores, omnivores, herbivores and detritivores.

Species level analysis was not done because model system locations are spread into the three different regions, and previous publications were showing species differences between biogeographic regions (Edgar et al. 2004a, 2011).

Comparing ecological system indicators between biogeographic regions

In order to allow for a comparison of the energy flow structure between the three biogeographic regions (Table 1), the published Floreana model (Okey et al. 2004) had to be reconstructed to make it comparable. Details are presented below.

Table 1. Characteristics of the areas modelled in each biogeographic region of the Galápagos Marine Reserve.

Bio–region	Modelled Area	Author	Area (km^2)	Modelling Period	Functional compartments
Far-northern	Darwin and Wolf Islands	Ruiz and Wolff, in review	2.8	2004-2008	32
Western	Fernandina and Isabela Island - Bolivar Channel-	Ruiz and Wolff 2011, and Wolff et al. 2012	6.2	2004-2008	30
South-central	Floreana Island	Okey et al. 2004	6.4	2001	43
	Floreana Island revisited	In the present study		2004-2008	30

Standardization of Floreana model

The initial Floreana rocky reef system model (Okey et al. 2004) was revisited to update the biological information to the period 2004-2008 (Banks et al. 2006, Edgar et al. 2011) for the fish and macroinvertebrate groups. Moreover, the species were grouped into compartments using the same criteria described in Ruiz and Wolff (2011). Biomass values of other vertebrates groups (predatory marine mammals, seabirds and sea turtles) and primary producers (macroalgae and phytoplankton) were taken from Okey et al. (2004), since information for the 2004-2008 period was not available. The P/B and Q/B for most of the compartments were obtained from the original model (Okey et al. 2004). In the case of new aggregated compartments, the P/B and Q/B values were derived as averages of the original groups, and specific estimates were weighted by relative biomass (B) or consumption (Q) as appropriate as was described by Ruiz and Wolff (2011). For groups such as shrimps and small crabs, anemones and zoanthids, stony corals, small herbivores gastropods, sponges and polychaetes, zooxanthella and two groups of zooplankton, biomass estimates were not available and these values were estimated based on the minimum biomass needed to satisfy predation mortality (for further details see Christensen et al. 2000).

Network analysis

System summary statistics provided by Ecopath fall under the categories of system organization, community energetics, cycling indices and fishery indicators. Further general descriptive statistics from the calculated outputs of the models included: (i) total throughput (T) – measure of the total sum of flows within the system and considered here as indicator of ecosystem size, (ii) contributions to T from different flows – production, consumption, export, respiration and flows to detritus, (iii) breakdown of biomass and flows from different components of the system – biomass and production.

System organization

Global measurements of system organization are calculated according to a network analysis based on flows among elements in the system as defined by Ulanowicz (1986). Indices include the above-mentioned statistics estimated as tonnes per km^{-2} per year.

Community energetics

Several indices of community energetics allow for the comparison of ecological succession and relative maturity according to Odum (1969) and include: total primary production (PP) to total respiration (R) ratio (PP/R) and biomass (B) supported by total primary production (PP/B), which are related to the ecosystem development theory, biomass supported by total throughput (B/T), and energy transfer efficiency (TE) between discrete trophic levels and estimated as the ratio between the sum of exports plus the flow that is transferred from one trophic level to the next (Margalef 1968).

Cycling indices

The Finn's cycling index (FCI) (Finn 1976) is calculated as T_c/T, where T_c is the amount of system flows that are recycled compared to the total system throughput (T). According to Odum (1969) recycling increases in more mature and less stressed systems. Predatory cycling index corresponds to the cycling index but is computed without consideration of cycles involving detritus groups (Christensen and Walters 2004), connectance index (CI), which is the ratio of the number of existing trophic links with respect to the number of possible links (Christensen et al. 2005), system omnivory index (OI), as average of the omnivory index of all consumer, which is a measure of feeding interactions between trophic levels (Christensen et al. 2005).

Fishery indicators

Other statistics allow for the assessment of the fishery activity such as Total catches in the system, gross efficiency (catch/net PP), mean trophic level of the catch, and primary production required for the catch (PPR/catch). According to Christensen (1995), ecosystems disturbed by fishing will cause a decrease in its maturity.

The aforementioned ecosystem 'health' indicators are based on quantified descriptions of the system food webs by Ecopath as part of a series of network analyses, facilitating the comparison of system properties between models. Our assumption in agreement with sensu Odum is that an undisturbed ecosystem should be more mature. Implications of this include that in a more mature system more niches should tend to be filled, that a larger part of the energy flows should be through detritus-based food webs, that primary production should be more efficiently utilized, that the total system biomass/energy throughput ratio should be higher, etc. (Christensen 1995).

Additionally, the mixed trophic impact (MTI) analysis (Leontief 1951, Ulanowicz and Puccia 1990) was used to quantify the positive and negative impacts that a hypothetical increase in biomass of a group would produce on the other groups in the ecosystem, including fishery activities.

Results and Discussion

Biogeographic differences in community biomass structure

Fishes

Large differences in total fish biomass can be observed between bio-regions. In Darwin and Wolf fish biomass (277 t km^{-2}) was higher then that estimated for the Bolivar Channel and Floreana systems (200 and 185 t km^{-2}, respectively). Direct effects of fishing appeared higher near to fishing ports, in the south-central area of Galápagos, compared with the remote far-northern area as indicated by the lower biomass contribution of piscivores in the Floreana system, where their contribution was 27%, while for the remote far-northern Darwin and Wolf system, piscivores contribute 59% to the overall fish biomass. A comparison of these observations with other studies (DeMartini et al. 2008, Friedlander and DeMartini 2002, Jennings and Polunin 1997, Stevenson et al. 2007) reveals similar patterns resulting from the effects of fishing pressure: total fish biomass and the proportion of biomass in top trophic levels decreases from the lightly to intensely fished areas. However, in the absence of intense fishing (as assumed for the Galápagos Archipelago in general), this pattern may also reflect bioregional differences in fish aggregation due to the contrasting oceanographic conditions between bio-regions.

The unique characteristics of the Bolivar Channel, which includes low temperature and high primary and secondary productivity, provide a favorable habitat for marine mammals such as the Galápagos fur seal (*Arctocephalus galapagoensis*) and the Galápagos sea lion (*Zalophus wollebaeki wollebaeki*), due to their high food density requirements. In this case out competition for fish prey by mammals over piscivores fish may be the reason for the low contribution of piscivores (10%) in this system.

Low fishery impact may be one factor explaining the high apex predator biomass in the far-northern area of Galápagos. However, the trophic structure of the food web and great differences in primary and secondary productivities between the systems compared is another important factor (see below – "*differences in system attributes*"). This is clearly seen for the Bolivar Channel system, which was dominated by planktivore fish (with 50% of the fish system biomass), in response to the high primary productivity in this upwelling-system.

Thus, the food web structure/species adaptations to food availability and the top-down and bottom-up controlling affects of the artisanal fishery effects should thus both be considered important in structuring fish communities in the GMR (Fig. 3).

Macroinvertebrates

Total macroinvertebrates biomass was slightly higher in the Bolivar Channel then in the other two systems. Interestingly, the characteristic of a system dominated by predator biomass was not restricted to the fish aggregation in Darwin and Wolf. Carnivores also dominated the macroinvertebrate biomass in this remote northern system. Relative biomass of those groups was more than 3 fold higher when compared with the Bolivar Channel and Floreana systems. The difference observed here can be attributed to the direct effects of fishing on important target species (e.g. *Pleuroploca princeps* and *Hexaplex princeps*) near to fishing ports for the local market (Fig. 3).

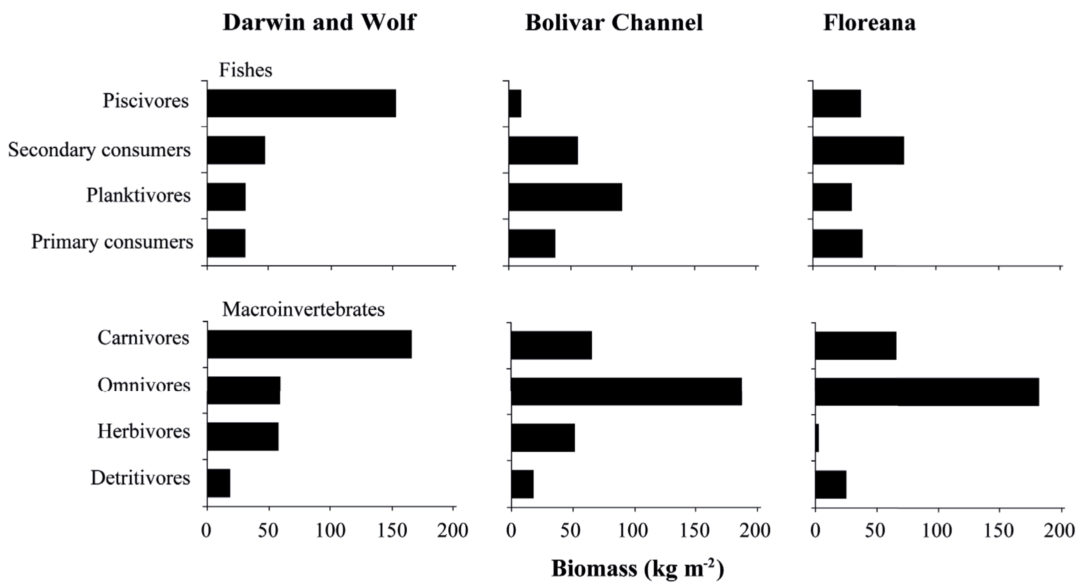

Fig. 3. Fish and macroinvertebrates biomass differences between biogeographic systems. The values are given in kg per m^{-2}. Biomass values presented in this figure were used as estimates for Ecopath models construction and comparisons rather than considered accurate absolute species or functional compartments biomass estimates.

As has been widely described, that secondary-fishing effects can lead to an increase in the macroinvertebrates preys of target species (see: Ruttenbergs 2001, Sonnenholzner et al. 2009). Such effect was found in the Bolivar Channel and Floreana systems with the proliferation of sea urchins following the depletion of spiny lobster (*Panulirus* spp.).

An inverse correlation between the biomasses of these two components apparently exists in the three systems.

Cumulative biomass by trophic level

Using Ecopath, a simplification of biomass distribution on each system was organized according to discrete trophic level (TL) aggregation (Lindeman spine), and is presented in Fig. 4. The system components are organized from TL 1.0 to TL 4.0, the highest values correspond to the top predators in the system as shark, seabirds, marine mammals and predatory fishes. Primary production and detritus are separated to clarify the representation of primary producer group in the systems (both with TL = 1.0, t km^{-2} year^{-1}).

Cumulative biomass by TL and species biomass contribution widely differed between the models of the bio-regions. The magnitude of those differences clearly reflects the unique structure of each system, with the greatest contrast being between Darwin and Wolf and Bolivar Channel (Fig. 4).

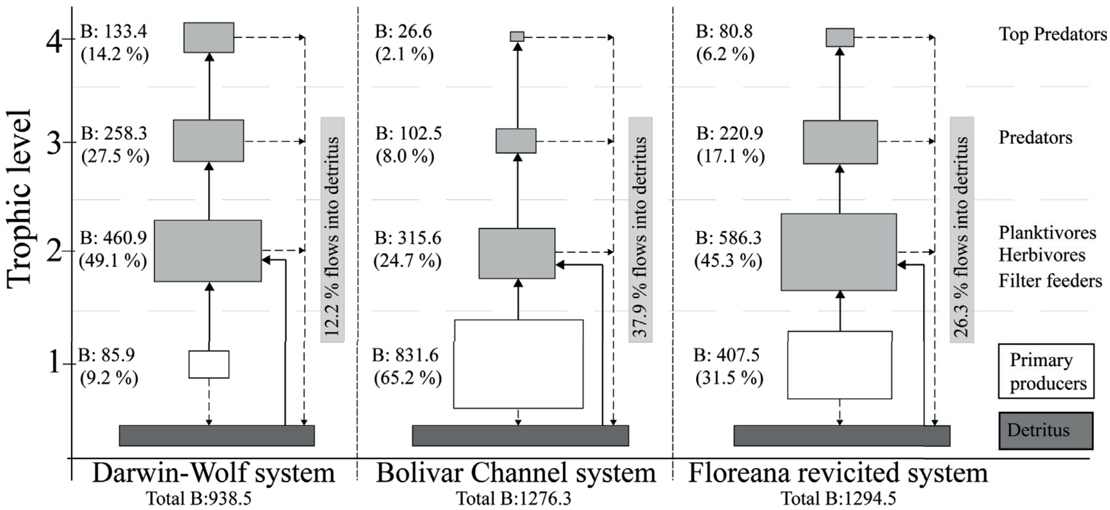

Fig. 4. Schematically representation of trophic aggregation analysis of the representative systems in the three major biogeography regions in the Galápagos Marine Reserve organized by discrete TL (B: Biomass in t km^{-2}). The size of each box represents the biomass accumulated by TL, which is proportional to the biomass it represents (y-axis is shown the tropic level of the functional group).

In all three systems, the highest biomass is to be found on TL 2.0, but the contributing species differ between systems: corals prevail in the Darwin and Wolf system, planktivorous, herbivorous and detritivorous in the Bolivar Channel, and sea urchins in the Floreana system. These structural differences are likely due to both differing fishing dynamics and oceanographic features among bio-regions.

Differences in general system attributes between biogeographic regions

Summary statistics and energy flows

Systems attributes in terms of system organization, community energetics, cycling indices, and fishery indicators are shown in Table 2. The Floreana revisited model has a 63% reduced total system size (Total throughput) when compared with Okey's original Floreana model (Table 2). The revisited Floreana model presented here contains much more locally derived biomass estimates mainly based on 5 years of standardized underwater census data collection through the Archipelago. Input and output parameters and the defined predatory-prey matrix of the Floreana revisited models are given as supplementary material in Appendix 8 and 9, respectively.

The highest total biomass (excluding detritus) correspond to the Floreana revisited model (1295 t km^{-2}), and is close to the value of in the Bolivar Channel model (1277 t km^{-2}, Ruiz and Wolff 2011) and corresponds to about half of the (apparently greatly overestimated) value of the previous Floreana model by Okey et al. (2004).

Important system differences were found in primary production, which amounts to 17101 t km^{-2} in the Bolivar Channel system and 13250 t km^{-2} in the Floreana system while it is substantially (4 – 5 times) lower in the far-northern tropical region (3400 t km^{-2}). The higher phytoplankton and macroalgae production in the Bolivar Channel can be attributed to the strong differences in nutrient inputs due to upwelling processes. However, the net primary production is dominated by macroalgae in all three systems.

Indices of community energetics indicate that Darwin and Wolf is the most mature bio-region system (Table 2). The relatively high amounts of respiration and biomass in relation to its primary production (PP/R, PP/B) reflect a system dominated by respiration due to the accumulation of biomass in higher trophic levels and supported per unit of flows (B/T). The differences observed between the three ecosystems attributes, clearly confirms a gradient in the system development state as influenced mainly by the contrasting oceanographic conditions within the Galápagos Archipelago.

Table 2. Selected summary statistics and indices estimated by Ecopath, showing differences between biogeography shallow rocky reef systems in the Galápagos Marine Reserve. Bold values show the highest values and italic the lowest values.

Ecosystem indicators	Darwin-Wolf 2004-2008	Bolivar Channel 2004-2008	Floreana-revisited 2004-2008	Floreana 2001
System organization		*(tonnes km^{-2} per year)*		
Total system throughput	*16652.94*	**38694.98**	24954.88	94850.00
Total net primary production	*3408.76*	**17101.49**	8730.05	13250.00
Total living biomass	*937.83*	1276.28	**1294.50**	2620.00
Net system production	*-1868.81*	**18577.02**	3514.50	-14388.00
Sum of all consumption	8880.16	6940.40	**9296.07**	51600.00
Sum of all exports	*334.84*	**13024.70**	3695.39	-5412.00
Sum of all respiratory flows	**5277.57**	*4076.78*	5215.55	27638.00
Sum of all flows into detritus	*2151.37*	**14653.10**	6747.88	21024.00
Sum of all production	*5235.31*	**18577.02**	10951.35	17337.00
Community energetics				
System primary production/total respiration (PP/R)	*0.64*	**4.20**	1.67	0.48
System primary production/total biomass (PP/B)	*3.64*	**13.40**	6.77	5.06
System biomass/throughput (B/T)	**0.06**	*0.03*	0.05	0.03
Mean transfer efficiency (TE – %)	*10.20*	**16.80**	14.30	7.70
Cycling indices				
Finn's cycling index (%)	4.58	*1.29*	**4.65**	6.89
Predatory cycled index (%)	0.35	*0.03*	**0.82**	0.00
Connectance Index	*0.15*	**0.17**	**0.17**	0.16
Proportion of flows originating from detritus	0.41	**0.43**	**0.43**	0.62
System Omnivory Index	**0.32**	*0.16*	0.25	0.25
Fishery indicators				
Total catches	3.84	**54.31**	*2.18*	4.15
Mean trophic level of the catch	**3.02**	*2.45*	2.77	2.27
Gross efficiency (catch/net primary production %)	*0.10*	**0.30**	0.00	0.03
Primary production requirement/catch (PPR/Catch)	**76.90**	53.89	*27.89*	356.84

The cycling of material and energy in marine systems can be considered as an important process in ecosystem function (Odum 1969), contributing to the autonomous behavior of each system (Ulanowicz 1986). One aspect of cycling is discussed in this bio-region system comparison, the Finn's cycling index (FCI), which represents the proportion of the throughput that is recycled in the system. The values estimated for the three systems range from 1.2 to 4.7 %, with highest values for the Floreana and Darwin and Wolf systems (Table 2), meaning that a higher proportion of material is thus retained within those two system that in the Bolivar Channel. The cycling structure in the remote far-northern system, apparently, does not differ markedly from the more anthropogenic affected Floreana system, and this can be attributed to the low primary productivity in both systems that cause a strong dependence on detritus, however the transitory nature of top predators in Darwin and Wolf is reflected through a decreased predatory cycling index in this remote system, indicating poorer cycling and transfer of energy in the higher trophic levels. Maturity of ecosystems have been associated with a higher degree of cycling (Odum 1969), however the highest value of FCI estimated for the Floreana system should not necessarily be an indicator of maturity in this highly exploited system as was suggested by Odum (1969), in this contradicting cases the increase of the FCI may be a response to stress within the system as argued by Ulanowicz (1984), and this could explain the high value of FCI in the Floreana system.

In terms of connectivity (CI) no differences were found between bio-regions, which can be possibly attributed to our standardized approach for the model construction (see also Christensen and Pauly 1993). However, the omnivory index can be used for identify differences between system structure. The high value for the omnivory index of the Darwin and Wolf model (0.32), (followed by the Floreana model, 0.25), suggesting that these two systems are more complex in terms of trophic interactions when compared with the Bolivar Channel (0.16). Consumer groups' apparently have a wide diet spectrum in those two bio-regions possibly in responses to a more dynamic prey assemblage shaped by of top-down pressure from consumers. This can be probably be most relevant to Galápagos marine systems where relatively large, mobile predators forage widely, and have strong average impacts on prey species.

The measurement of energy flows between system components and the efficiency with which energy is assimilated, transferred and dissipated provides also a significant insight into of the structure and function of the system (Baird et al. 1991, Ulanowicz 1986, Wulff and Ulanowicz 1989). Differences in the system mean trophic efficiency (TE) may be explained by the trophic structure of each bio-region system. The lowest value estimated for Darwin and Wolf is likely due to the complex interactions between the pelagic and benthic sub-systems components, the low efficiency of the food web in high trophic levels due to the high species richness of top predators that increase the trophic redundancy (Angelini and Agostinho 2005), the nutrient and plankton imports, and the large amount of energy that is lost through respiration in the higher trophic level (Burns 1989, Lalli and Parsons 1993, Lindeman 1942). In contrast to this situation, in the rest of the Archipelago, energy transfer is more efficient through the direct utilization of primary production and detritus by low trophic level consumers. Besides these causes for the differences in TE between systems, stress has been associated with low system efficiency (Baird et al. 2012, Christian et al. 2009, Libralato et al. 2004).

However, this relationship is critically being discussed and should possibly not be generalized (Niquil et al. 2014).

Trophic interactions, keystone species and trophic control

The results of the mixed trophic impact (MTI) routine show that the groups at the base of the food web such as phytoplankton, macroalgae and detritus have an important net positive impact in all three systems, suggesting bottom-up control mechanism operating in these ecosystems. Detritus was more important in the Floreana and Darwin and Wolf models, while phytoplankton in the Bolivar Channel (Fig. 5).

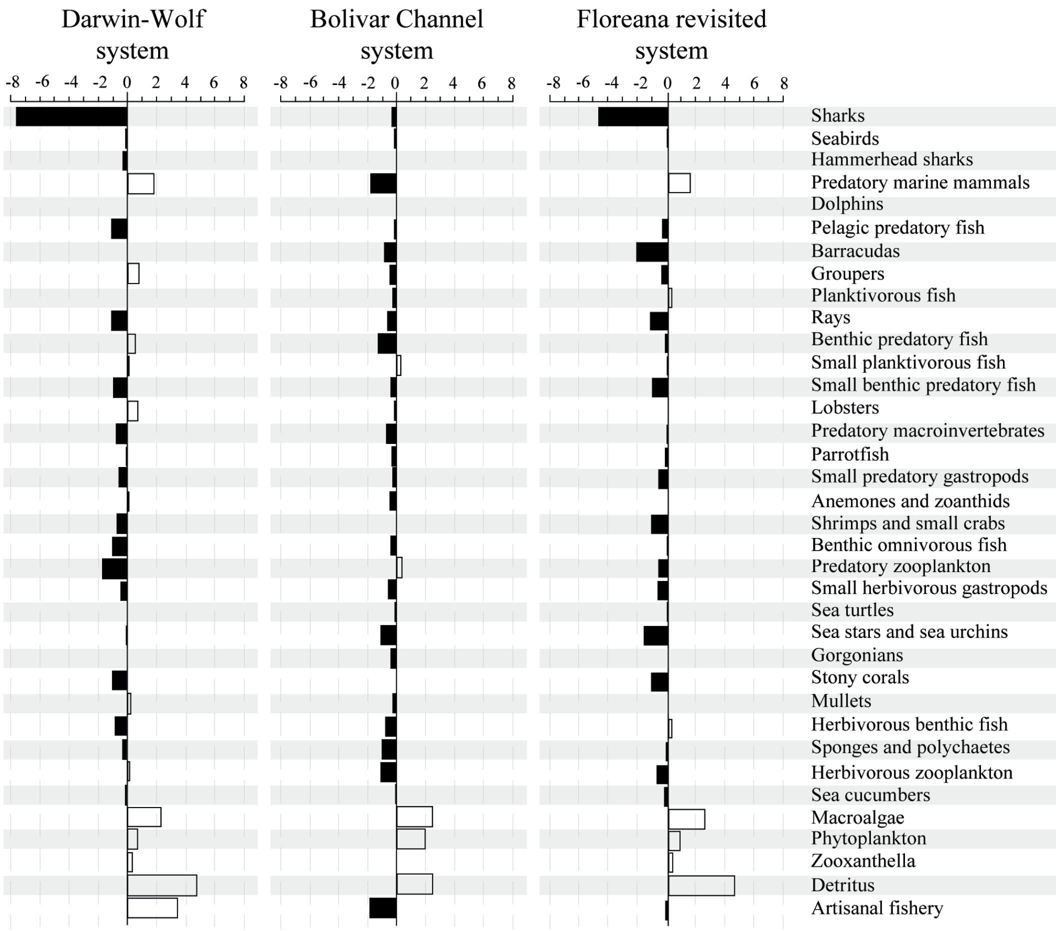

Fig. 5. Net mixed trophic impact analysis shows positive impacts by white bars on right of the vertical base line and negative on left. The trophic impact is relative but comparable between systems. Impacted groups are placed along the vertical axis sorted by their TL (high to low TL).

The influence of top predators (sharks) is more relevant in low productivity systems were the top-down control plays an important role structuring the system. Besides the low intensity of artisanal fishery in the Darwin and Wolf model, it shows a considerable positive impact through reducing natural effects of predation mortality when part of the predators are extracted from this system (Fig. 5). Despite the high biomass of top predators, a top-down control of low trophic levels by transitory predators (hammerhead sharks and dolphins) is poorly identified in the GMR due to the fact that these groups aggregate around the islands but they obtained large parts of their diet from outside of the system model boundaries. Top-down control was only observed in the case of the more "stationary" sharks species (*C. galapagensis* and *T. obesus*), with a high negative net MTI value observed in the Darwin and Wolf and Floreana shallow rocky reef systems. This is thus a more characteristic feature for the tropical and sub-tropical systems, while marine mammals controling the upwelling systems in the GMR.

The analysis of the mixed trophic impacts presented here allowed for the estimation of keystoneness for the functional groups in each biogeographic region. Keystone species are defined those having a structuring role within the system despite a relatively low biomass and hence food intake (Power et al. 1996). According to Libralato et al. (2006), it may be noted that the low biomass requirement for the definition of keystone species eliminates those that structure ecosystems by virtue of their high biomass, such as the case of phytoplankton and macroalgae in the Bolivar Channel system.

Differences in species abundance and diversity between biogeographic regions have been identified mainly as response to the oceanographic conditions in the GMR. However, losses in biodiversity resulted from the exceptional severity of El Niño events, coupled with the development of intensive artisanal fishery, could play an important role restructuring the trophic structure and the role of keystone species in the actual state of the Galápagos marine systems (e.g. Chapin et al. 1995, Lawton and Brown 1993, Tilman and Downing 1994). Species identified as keystone slightly differed between bio-regions (Fig. 6). In all three systems sharks are the most important keystone component. While marine mammals were identified as second important keystones in the Bolivar Channel and Floreana system, in Darwin and Wolf benthic predatory fishes occupy this position. Seabirds appear to have high keystoneness in shallow waters environments (Libralato et al. 2006) but the close link between shallow and open waters decreases the keystoneness of the seabirds in the GMR as show the models outputs. An important characteristic of the Floreana system is the role of herbivorous benthic fishes as a keystone component, the relative low biomass estimated for herbivorous macroinvertebrates in this system (see, Fig. 3) may have caused a compensatory increase in functionally similar herbivorous fishes in the Floreana system (Frost et al. 1995). Nevertheless the importance of each species should not only be evaluated on the basis of its trophic role, and an integral analysis should also consider other conservation relevant aspects. From a conservation perspective of the Galápagos Archipelago it is important if keystone, endemic and threatened species persist and thrive, and whether ecosystem processes and services are maintained.

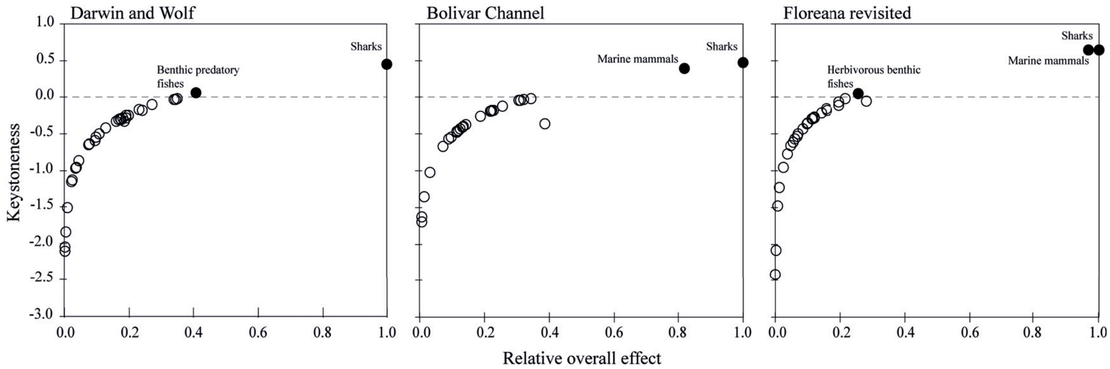

Fig. 6. Keystoneness for the functional compartments of the three major biogeographic regions. For each functional compartment, the keystoneness index (y axis) is reported against the overall effect (x axis). Overall effects are relative to the maximum effect measured in each trophic web, thus for x axis the scale is always between 0 and 1. In the graph functional compartments are ordered by decreasing keystoneness, therefore the keystone functional compartments are showed with black dots above the dashed line.

Comparing system attributes on a regional scale

While differences in the spatial delineation, system features and species compartmentalization may affect several of the summary statistics estimated by Ecopath, a comparison between Galápagos, and other Eastern Pacific systems in Peru (upwelling) and Costa Rica (non-upwelling) may still allow for some useful insights and is therefore attempted here. System size in terms of biomass (B) and total system throughput (T) differs greatly between oceanic and costal systems in the Eastern Pacific region (Fig. 7). However, very similar system responses to El Niño events of greatly reduced primary productivity have been described for two systems in the region, the Bolivar Channel in Galápagos (Wolff et al. 2012) and the Independence Bay in Peru (Taylor et al. 2008a). The comparison between these two systems suggests that the reduction of system size during El Niño event was even more pronounced in the Bolivar Channel due to the large reduction of primary production, 2 fold higher than in the Independence bay.

The energy flow reduction between the primary producers and the primary consumers results in secondary cascading effects through all trophic levels showing the profound effect of the El Niño on the whole food web, with most negative effects on predatory marine mammals in the Bolivar Channel.

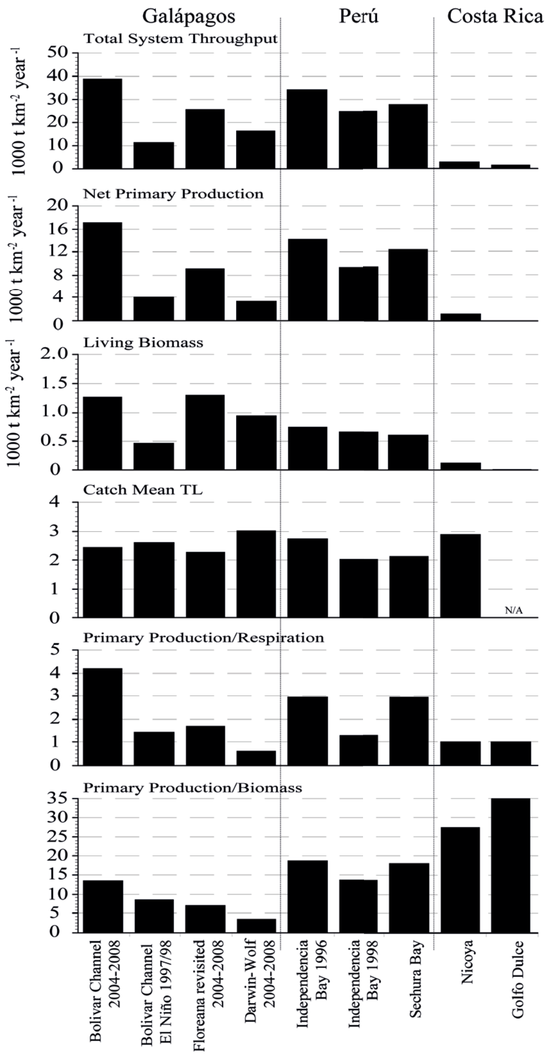

Fig. 7. Selected summary statistic compared between Ecopath models in the Eastern Pacific. Vertical lines separate models by country. N/A indicates values not provided by the author. (Bolivar Channel 2004-2008, Ruiz and Wolf 2011), (Bolivar Channel – El Niño 98, Wolff et al. 2012), (Floreana revisited, present study), (Darwin-Wolf 2004-2008, Ruiz et al. in prep.), (Independence Bay 1996 and 1998, Taylor et al. 2008a), (Sechura, Taylor et al. 2008b), (Nicoya, Wolff et al. 1998), (Golfo Dulce, Wolff et al. 1996).

Adaptive responses to the system stress during El Niño, such as changes in diet of several groups from small species prey to large predators, a slight increase in the TE, suggests an improvement in the efficiency in the system. The proliferation of two species during and after El Niño: the sea cucumber – *I. fuscus* in the Bolivar Channel, and the scallop – *Argopecten purpuratus* in the Independence bay clearly mediated the recycling of nutrients in the both systems through the utilization of the large amount of accumulated detritus during El Niño. In general it appears that the direct and indirect effects of El Niño strongly affect higher trophic level groups by the disruption of energy flows from intermediate to high trophic levels.

Conclusions

This study constitutes the first attempt to compare the trophic structure of three biogeographic regions in the Galápagos Marine Reserve (GMR), focusing on biomass distribution between trophic levels, ecosystem trophic structure and energy flows as important characteristics determining the function, stability, and maturity of the Galápagos marine ecosystems. In agreement with Worm and Duffy (2002), we also suggest that merging already existing biodiversity research in Galápagos with our standardized trophic modelling approach should result in a better understanding of fundamental aspects of the marine ecosystems studied.

Species composition and richness, biomass productivity, and trophic interactions can influence one another under a large range of environmental conditions, including fishing disturbances within the Galápagos Archipelago. Loss of functional dominant species, such as keystone species (Power et al. 1996), ecosystem engineers (Jones et al. 1994) or species with many trophic connections (Dunne et al. 2002), would have particularly strong effects and could induce rapid and violent changes in local biodiversity as has been observed during the last 50 years in the GMR.

Interaction through predation is not the only process that structures the trophic web of shallow rocky reef systems in Galápagos. The selectivity of target species by the local artisanal fishery plays an additional role in structuring the whole benthic community. However, direct effects are spatially variable, depending in part on the proximity to fishing ports around which most of the fishing activities are concentrated.

The three systems compared in this study were shown to differ in their oceanographic setting as well as in fishery impact. Conservation strategies need to take their functional and structural differences into account to avoid marine system disintegration through the development of a selective and unsustainable artisanal fishery.

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Synthesis

During the last five decades the Galápagos Archipelago has experienced changes in the marine community induced by: i) the high severity of two El Niño, 1982/83 and 1997/98, events, both with dramatic consequences for several marine species; ii) the exclusion of the industrial fishery through the establishment of the Galápagos Marine Reserve (GMR) in 1998; iii) the unsustainable local artisanal fishery, resulting mainly in the depletion of three benthic resources, the sea cucumber (*Isostichopus fuscus*), the spiny lobster (*Panulirus* spp.) and the Galápagos grouper (*Micteroperca olfax*); and iv) by the implementation of a spatial management scheme based on “no-take” zones to promote the recovery of resources and habitats.

The broad influence that natural and human impacts have on shallow rocky reef systems and the increase in evidences of the effects that fishing activities have in the GMR (Bustamante et al. 2008, Okey et al. 2004, Ruttenberg 2001, Sonnenholzner et al. 2009), pressures local decisions makers to minimize negative effects on Galápagos rocky reef ecosystems and to adequately implement management plans.

In order to allow a better understanding of the complexity of the Galápagos marine environment the goal of the present thesis was to identify differences in community structure and system functionality between the Galápagos biogeographic regions, focusing particularly on the potential mechanisms that could have generated or maintained the structural differences between contrasting biogeographic regions. The major findings of this thesis are summarized below.

Biogeographic differences in community structure

The ecosystem dynamics and species interactions in the shallow rocky reef systems of the Galápagos Archipelago are strongly influenced by the environmental variability. The three major biogeographic regions: the west cold upwelling system – Bolivar Channel, the south-central mixed system – Floreana and the far-north tropical system – Darwin and Wolf, show not only differences in species diversity and endemism as reported by Edgar et al (2004b) but also important differences in ecosystem structure, trophic interactions and energy flows (Chapter 4). Great differences in biomass have been identified between biogeographic regions and these clearly reflect the unique structure of each system, being the most contrasting finding between the Darwin and Wolf system and the Bolivar Channel system.

Clearly, system attributes are associated with specific biogeographic regions. For example, the Floreana system showed the highest total biomass among all the systems. Important differences were found in primary production, which is highest in the Bolivar Channel system attributed to the strong nutrient inputs due to upwelling processes. However, the net primary production is dominated by macroalgae in all three systems. The Darwin and Wolf system seems to be the most mature bio-region system due to high amounts of respiration and biomass in relation to its primary production. The differences observed between the three system attributes clearly confirm a gradient in the system development as influenced mainly by the contrasting oceanographic conditions.

Differences in trophic interactions and the role of keystone species can be observed between biogeographic systems. The influence of sharks is more relevant in the low productive system, where top-down control plays an important role restructuring the system, while marine mammals control the upwelling systems in the GMR (i.e. the Bolivar Channel system). The groups at the base of the food web such as phytoplankton, macroalgae and detritus control the bottom-up mechanism in all three ecosystems. Detritus is more important in the Floreana and the Darwin and Wolf system, whereas phytoplankton is crucial in the Bolivar Channel system.

The effect of El Niño in a highly primary productive system

The strongest changes in the Galápagos ecosystems have been associated with the occurrence of strong El Niño events. There, high mortalities largely reduced the overall biomass in the ecosystems.

Effects of El Niño due to the increase of water temperature were most evident in the cold-upwelling system, where an increment of ~7º C from the mean average temperature (14º – 18º C) during a long time period (> 6 months) disrupted the primary productivity in this system (Robinson and del Pino 1985). As it was discussed in Chapter 2, a reduction in primary producers (phytoplankton and macroalgae) decreases the energy flows between the system compartments, thus, showing a cascading effect through all trophic levels in the whole food web. However, smaller impacts were observed on several low trophic level compartments (e.g. sea cucumbers, sea stars, sea urchins, and predatory macroinvertebrates) that use the accumulated detritus as an important food source during El Niño events. Changes in trophic interactions emerge during El Niño by the reduction of prey biomass and the resulting increase in predatory pressure. The dynamic lead predators to consume opportunistically, thus, preying on unusual food items in an already stressed system.

The increase in respiration, the decrease in overall production and a larger fraction of the system's throughput recycled during El Niño suggest that the Bolivar Channel system became more efficient. Taylor et al. (2008) found a similar response in the Independence Bay system (Peru) during the El Niño event. Hence, the Bolivar Channel system compared to other tropical shallow water systems has more features of an upwelling system than that of a classical tropical system found in Galápagos and other coastal areas of the Eastern Tropical Pacific (ETP).

Dynamic simulations of El Niño effects in the Bolivar Channel system clearly confirm the negative trends for the majority of high trophic levels groups; except for sharks. Apparently, sharks successfully switch between preys, when the food spectrum changes during El Niño conditions. Contrastingly, seabirds (penguins and flightless cormorants) show great reductions in population numbers during the El Niño event (Valle-Castillo 1985, Vargas et al. 2006). Hence, the direction and magnitude of change in biomass induced by food shortage was the main reason for the increased species mortalities during El Niño.

Galápagos marine systems responses to artisanal fishing

Galápagos rocky reef systems have been increasingly influenced by both an increase in human population (Merlen 1995) and an intense artisanal fishery (Camhi 1995, Merlen 1995, MacFarland and Cifuentes 1996). Intensive removal of benthic predators near fishing ports (mainly in south-central areas) during the last three decades has gone down the food web through cascading effects (Ruttenbergs 2001, Sonnenholzner et al. 2009). In Chapter 3, the simulation of heavy fishing pressure resulted in strong biomass changes in the whole trophic web. The selectivity of the fishing gears used and the intensity of fishing perturbations have caused the dramatic change in the benthic community structure observed until today in some areas. The model simulation has clearly shown the mechanisms that lead to an emerging alternative system state dominated by sea urchin biomass, when the virtual trophic extinction of important benthic predatory fishes occurs in the system (e.g. McClanahan and Muthiga 1988, McClanahan et al. 1996, Ruttenbergs 2001).

The boom and bust dynamics of artisanal fisheries in Galápagos has created a complex mix of responses in the community structure and trophic interactions – proposed from anecdotal evidences – due to the depletion of marine biota, including key components of both high and low trophic levels (Bustamante et al. 2008). Differences in mean trophic level of the catch between modelled systems reflect the opportunistic and selective conduct of the artisanal fishery in the GMR. Target species and harvest intensity can also be related to differences in species distribution and abundances between biogeographic regions. The model outcomes also support the widely accepted paradigm that fishing by removing invertebrate-feeding fish allows to increase the biomass of sea urchins and as a consequence, the formation of overgrazed "barrens" of calcareous algae substrate in Galápagos (Ayling 1981, Breen and Mann 1976, Himmelman and Lavergne 1985).

The changes induced in community structure by target species depletion and replacement of fishing areas can result in dramatic effects in the Galápagos ecosystem through the reduction of the biogeographic differences in species composition.

Thus, understanding the different effects of fishing pressure on the natural bioregional variability of the food webs is essential to improve and develop new policies that protect the integrity of each biogeographic region within the actual spatial management plan of the Galápagos Marine Reserve.

Ecopath and Ecosism modelling approach and future prospects

Ecopath and Ecosim (EwE) modelling approach provides a relatively simple framework that is capable to consider the major components and trophic interactions of marine ecosystems. Thus, by tracking ecosystem changes over time, EwE provides a coherent link between ecological theories on ecosystem development properties.

The presented work is the first attempt to describe differences between biogeographic regions in terms of trophic interactions, system structure and biomass distribution induced by local environmental variability and fishing activities in the Galápagos rocky reef systems. Indirect estimations and assumptions of full exploitation of some resources were done during the models construction due to limitations of georeferenced fishery data. The large amount of data available from the long-term monitoring program established by the Charles Darwin Foundation during the last 15 years proved to be the strongest input for this EwE approach. However, it is important to complement ecological knowledge to elucidate, "who eats whom" in the system through experimental approaches or stomach content analysis.

Data collected by the fisheries monitoring program is crucial to construct models of fishing scenarios on rocky reef communities. However, additional georeferencial information needs to be included to allow a better quantification of the catches in specific areas of the Galápagos Marine Reserve in order to conduct a more robust analysis in the future.

Implications of modelling for conservation management

Food-web modelling approaches using EwE can hold a systematic framework for integrating ecosystem management processes. Thus, EwE provides a number of indicators that allows descriptions of the ecosystem features, which are linked to the functioning of marine ecosystems and the ecological services these systems provide (e.g. Pimm et al. 1991 – structure properties of the food webs; Pauly and Christensen 1995 – food web energy flows; Ulanowicz 1986 and Link 2002 – food web network analyses).

The potential of using EwE lies in the comparisons of systems, the identification of key species within the ecosystem, the definition of indicators for ecosystem shifts and the detection of structural changes induced by natural and anthropogenic perturbation, providing new insights into the understanding of how Galápagos rocky reef systems have evolved in this highly complex and dynamic environment, notably on system stability, maturity and food web organization. However, it is essential to understand that a model is just a simplification of the complexity observed in nature. Therefore, an EwE model, as any other model, has the capabilities and limitations of this approach (Christensen and Walters 2004).

EwE can obviously be considered as a tool for conservation as it accounts for the ecological complexity of marine ecosystems. The simplification of the ecosystem through a holistic approach helps to summarize the different sources of information and helps to identify gaps in knowledge. Therefore, it could direct future research lines on specific ecosystem (Christensen and Pauly 1993). In addition, it is important to recognize that although scientific knowledge about marine rocky reef systems in Galápagos is far from complete; this lack of information should not halt conservation strategies. At the very least, trophic models could provide a valuable baseline information that would allow to plan for future conservation strategies and the establishment of adequate fishing regimes.

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Appendix

Appendix 1. Input – output parameters of the steady-state model by the Bolivar Channel Ecosystem. P_i/B_i production rate, Q_i/B_i consumption rate, EE_i ecotrophic efficiency, GE_i gross efficiency, F_i fishing mortality, $M0_i$ non-predatory natural mortality, $M2_i$ predation mortality.

Functional groups / parameter	Trophic level	*Biomass* (t/km^2)	P_i/B_i (year)	Q_i/B_i (year)	EE_i	GE_i	Catch (year)	F_i	$M0_i$	$M2_i$
Sharks	3.89	2.600	0.240	4.900	0.721	0.049	0.180	0.069	0.067	0.104
Seabirds	3.50	0.030	5.400	60.000	0.000	0.090	-	-	5.400	0.000
Predatory marine mammals	3.49	1.000	0.080	27.000	0.796	0.003	-	-	0.016	0.064
Rays	3.43	9.500	0.250	3.500	0.949	0.071	0.980	0.103	0.013	0.134
Pelagic predatory fishes	3.40	3.900	0.420	2.200	0.828	0.191	0.490	0.126	0.072	0.222
Groupers	3.39	5.200	0.450	4.500	0.756	0.100	0.700	0.135	0.110	0.206
Barracudas	3.37	13.000	0.600	4.000	0.875	0.150	-	-	0.075	0.525
Benthic predatory fishes	3.36	20.000	0.750	4.000	0.854	0.188	4.720	0.236	0.109	0.405
Small benthic predatory fishes	3.19	8.000	1.200	6.000	0.783	0.200	-	-	0.261	0.939
Small predatory gastropods	2.96	17.000	0.900	4.000	0.963	0.225	-	-	0.033	0.867
Predatory macroinvertebrates	2.92	5.900	2.800	10.000	0.958	0.280	-	-	0.117	2.683
Predatory zooplankton	2.80	3.400	30.000	100.000	0.956	0.300	-	-	1.332	28.668
Lobsters	2.64	14.000	3.000	12.000	0.824	0.250	13.500	0.964	0.528	1.508
Small planktivorous reef fish	2.47	19.000	4.500	18.000	0.929	0.250	25.700	1.353	0.318	2.829
Planktivorous reef fish	2.34	10.000	1.500	15.000	0.852	0.100	-	-	0.221	1.279
Sea stars and sea urchins	2.28	77.000	2.000	8.200	0.932	0.244	-	-	0.137	1.863
Anemones and zoanthids	2.14	32.000	2.000	10.000	0.900	0.200	-	-	0.200	1.943
Benthic omnivorous fish	2.14	20.000	2.600	10.000	0.682	0.260	-	-	0.827	1.773
Mullets	2.11	23.000	2.800	11.000	0.871	0.255	19.300	0.839	0.362	1.598
Parrotfish	2.11	5.500	0.500	16.600	0.491	0.030	-	-	0.255	0.245
Gorgonians	2.07	29.000	2.400	15.000	0.936	0.160	-	-	0.154	2.246
Sponges and polychaetes	2.01	37.000	2.000	16.500	0.900	0.121	-	-	0.200	1.982
Small herbivorous gastropod	2.01	20.000	2.500	14.000	0.950	0.179	-	-	0.125	2.457
Sea turtles and marine iguanas	2.01	2.120	0.140	10.000	0.215	0.014	-	-	0.110	0.030
Herbivorous zooplankton	2.00	14.000	35.000	150.000	0.933	0.233	-	-	2.341	32.659
Sea cucumbers and other	2.00	12.000	0.120	3.360	0.920	0.036	0.660	0.055	0.010	0.055
Herbivorous benthic fish	2.00	25.000	1.000	16.000	0.695	0.063	-	-	0.305	0.695
Macroalgae	1.00	800.000	12.000	0.000	0.145	-	-	-	10.254	1.746
Phytoplankton	1.00	14.000	200.000	0.000	0.919	-	-	-	16.177	183.823
Detritus	1.00	500.000	-	-	0.166	-	-	-	-	-

Appendix 2. Predator-prey matrix by the steady-state model of Bolivar Channel Ecosystem.

Prey \ predator	1	2	3	4	5	6	7	8	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
1. Sharks			0.010																								
2. Seabirds																											
3. Predatory marine mammals	0.005																										
4. Rays	0.100																										
5. Pelagic predatory fishes	0.015		0.025																								
5. Groupers			0.010					0.010																			
7. Barracudas	0.250	0.050	0.040			0.020		0.025																			
8. Benthic predatory fishes	0.200		0.115			0.060	0.020																				
9. Small benthic predatory fishes	0.005		0.020			0.080	0.020	0.050																			
10. Small predatory gastropods						0.040		0.025			0.200																
11. Predatory macroinvertebrates			0.020	0.080												0.020											
12. Predatory zooplankton									0.050	0.005				0.150	0.050		0.050	0.050	0.010		0.010	0.005					
13. Lobsters			0.010	0.300		0.020		0.100	0.050																		
14. Small planktivorous reef fish	0.150	0.900	0.150		0.800	0.090	0.500	0.110	0.050																		
15. Planktivorous reef fish			0.150			0.040	0.150																				
16. Sea stars and sea urchins				0.600		0.070		0.340	0.100		0.100		0.500														
17. Anemones and zoanthids										0.340	0.020					0.060											
18. Benthic omnivorous fish	0.150		0.210			0.200	0.200	0.040	0.200																		
19. Mullets	0.020	0.050	0.100		0.200	0.260	0.060	0.075	0.350																		
20. Parrotfish			0.050																								
21. Gorgonians										0.100	0.160					0.063				0.100							
22. Sponges and polychaetes										0.150						0.100											
23. Small herbivorous gastropod						0.040		0.200		0.300	0.200																
24. Sea turtles and marine iguanas	0.005																										
25. Herbivorous zooplankton									0.050	0.005		0.800		0.200	0.250		0.050	0.050	0.090		0.050	0.005	0.010	0.010			
26. Sea cucumbers and other				0.020																							
27. Herbivorous benthic fish	0.100		0.090			0.080	0.050	0.025	0.150																		
28. Macroalgae											0.150		0.300			0.557		0.700		0.850			0.940	0.940	0.050		0.950
28. Phytoplankton												0.150		0.600	0.550		0.500	0.050	0.300		0.640	0.050			0.800		
30. Detritus										0.100	0.170	0.050	0.200	0.050	0.150	0.200	0.400	0.150	0.600	0.050	0.300	0.940	0.050	0.050	0.150	1.000	0.050
Sum	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Appendix 3. Input – output parameters of the reference state model (Ruiz and Wolff 2011) and of the El Niño (EN) 1997/98 model of the Bolivar Channel ecosystem. Biomasses values in bold correspond to groups that increased in biomass during EN 1997/98. B_i biomass, P_i/B_i production rate, Q_i/B_i consumption rate, EE_i ecotrophic efficiency, GE_i gross efficiency, F_i fishing mortality, $M0_i$ non-predatory natural mortality, $M2_i$ predation mortality.

Functional groups / parameter	TL	TL (EN)	B_i (t km^{-2})	B_i (t km^{-2}) (EN)	P_i/B_i (year^{-1})	P_i/B_i (year^{-1}) (EN)	Q_i/B_i (year^{-1})	Q_i/B_i (year^{-1}) (EN)	EE_i	EE_i (EN)	GE_i	GE_i (EN)	Catch (year^{-1})	Catch (year^{-1}) (EN)	F_i	F_i (EN)	$M0_i$	$M0_i$ (EN)	$M2_i$	$M2_i$ (EN)
Sharks	3.89	3.75	2.538	**4.201**	0.243	0.241	4.816	4.579	0.723	0.671	0.050	0.053	0.187	0.040	0.074	0.040	0.069	0.079	0.101	0.152
Seabirds	3.50	3.63	0.050	0.012	5.423	5.364	60.301	60.078	0.000	0.000	0.090	0.089	-	-	-	-	5.423	5.364	0.000	0.000
Predatory marine mammals	3.49	3.40	0.979	0.501	0.081	0.082	26.129	25.542	0.776	0.936	0.003	0.003	-	-	-	-	0.019	0.005	0.062	0.077
Rays	3.43	3.93	9.570	**11.650**	0.249	0.253	3.470	3.448	0.807	0.775	0.072	0.073	0.712	0.072	0.074	0.024	0.051	0.057	0.124	0.190
Pelagic predatory fishes	3.40	3.47	3.902	1.044	0.422	0.421	2.231	2.235	0.805	0.541	0.189	0.188	0.491	0.180	0.126	0.410	0.085	0.193	0.211	0.055
Groupers	3.39	3.32	5.197	**5.547**	0.450	0.459	4.448	4.328	0.736	0.890	0.101	0.106	0.702	0.700	0.135	0.275	0.121	0.051	0.194	0.282
Barracudas	3.37	3.49	13.058	**14.015**	0.611	0.636	3.931	3.613	0.825	0.866	0.155	0.176	-	-	-	-	0.109	0.085	0.502	0.551
Benthic predatory fishes	3.36	3.22	19.463	**25.158**	0.863	0.791	3.857	3.626	0.837	0.955	0.198	0.246	4.500	4.700	0.231	0.589	0.132	0.040	0.400	0.664
Small benthic predatory fishes	3.19	3.16	8.027	6.326	1.206	1.261	5.867	5.745	0.751	0.895	0.206	0.219	-	-	-	-	0.309	0.875	0.897	0.386
Small predators gastropods	2.96	2.88	17.564	10.321	0.919	0.921	4.006	4.009	0.865	0.933	0.229	0.230	-	-	-	-	0.130	0.062	0.789	0.859
Predatory macroinvertebrates	2.92	2.88	5.796	2.909	2.811	2.764	9.732	9.819	0.584	0.664	0.289	0.281	-	-	-	-	1.157	0.929	1.654	1.835
Predatory zooplankton	2.80	2.80	3.481	2.032	31.501	31.487	99.132	98.990	0.892	0.935	0.318	0.318	-	-	-	-	3.549	2.041	27.952	29.446
Lobsters	2.64	2.57	14.487	**19.557**	0.687	0.693	11.947	11.994	0.885	0.933	0.058	0.058	2.730	5.555	0.188	0.410	0.080	0.047	0.418	0.362
Small planktivorous reef fish	2.47	2.46	18.959	7.461	4.661	4.751	17.627	17.663	0.905	0.865	0.264	0.269	25.650	10.225	1.353	0.288	0.469	0.641	2.838	2.739
Planktivorous reef fish	2.34	2.34	10.126	8.580	1.522	1.548	15.045	14.993	0.811	0.813	0.101	0.103	-	-	-	-	0.302	0.290	1.220	1.258
Sea stars and sea urchins	2.28	2.17	81.744	**95.094**	2.238	2.272	7.851	7.795	0.832	0.864	0.285	0.291	-	-	-	-	0.393	0.309	1.845	1.963
Anemones and zoanthids	2.14	2.14	37.441	35.108	2.677	2.665	9.881	9.979	0.876	0.934	0.271	0.267	-	-	-	-	0.340	0.177	2.337	2.488
Benthic omnivorous fish	2.14	2.14	19.795	18.003	2.600	2.681	9.868	9.793	0.678	0.834	0.263	0.274	-	-	-	-	0.856	0.444	1.744	2.237
Mullets	2.11	2.11	22.682	7.960	2.869	2.966	10.942	10.954	0.850	0.842	0.262	0.271	18.900	1.448	0.833	0.061	0.458	0.469	1.578	2.315
Parrotfish	2.11	2.11	5.455	4.137	0.499	0.501	16.583	16.378	0.468	0.838	0.030	0.031	-	-	-	-	0.269	0.081	0.230	0.420
Gorgonians	2.07	2.07	29.209	17.076	2.395	2.402	14.926	14.958	0.643	0.933	0.160	0.161	-	-	-	-	0.914	0.161	1.481	2.241
Sponges and polychaetes	2.01	2.01	37.427	29.312	2.443	2.433	16.450	16.231	0.809	0.538	0.149	0.150	-	-	-	-	0.481	1.123	1.962	1.310
Small herbivorous gastropods	2.01	2.01	23.992	14.902	2.686	2.703	14.164	14.158	0.761	0.932	0.190	0.191	-	-	-	-	0.672	0.183	2.014	2.520
Sea turtles and marine iguanas	2.01	2.01	2.109	1.320	0.139	0.141	9.943	10.045	0.210	0.723	0.014	0.014	-	-	-	-	0.110	0.039	0.029	0.102
Herbivorous zooplankton	2.00	2.00	14.440	8.078	36.335	36.208	147.529	146.786	0.884	0.932	0.246	0.247	-	-	-	-	4.395	2.476	31.940	33.732
Sea cucumbers and other	2.00	2.00	11.975	**29.131**	0.123	0.123	3.382	3.341	0.752	0.523	0.036	0.037	0.432	1.071	0.036	0.299	0.031	0.059	0.055	0.028
Herbivorous benthic fish	2.00	2.00	25.172	14.293	0.999	1.367	15.914	15.999	0.684	0.848	0.063	0.085	-	-	-	-	0.325	0.208	0.674	1.159
Macroalgae	1.00	1.00	800.475	78.754	15.670	15.511	-	-	0.118	0.932	-	-	-	-	-	-	13.841	1.048	1.829	14.463
Phytoplankton	1.00	1.00	31.161	19.822	146.274	144.865	-	-	0.586	0.530	-	-	-	-	-	-	62.122	68.077	84.152	76.788
Detritus	1.00	1.00	500.000	500.000	-	-	-	-	0.166	0.545	-	-	-	-	-	-	-	-	-	-

Appendix 4. Predator-prey matrix of El Niño 1997/98 steady-state model of the Bolivar Channel Ecosystem. Values in parentheses represent the percentages of change in relation to values of the reference steady-state model. The diet matrix was constructed based on general knowledge from the literature, using FishBase (Froese and Pauly 2011) for fish groups and other published models (Opitz 1986, Ruiz and Wolff 2011, Taylor et al. 2008a, 2008b) for seabirds, marine mammals and invertebrate groups.

Prey\Predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1. Sharks			0.05 (80.0 %)																								
2. Seabirds																											
3. Predatory marine mammals	0.002 (-150.0 %)																										
4. Rays	0.115 (15.7 %)																										
5. Pelagic predatory fish	0.003 (-400.0 %)																										
6. Groupers	0.014 (100.0 %)		0.030 (66.7 %)					0.010																			
7. Barracudas	0.222 (-12.7 %)	0.246 (79.7 %)	0.04			0.02		0.025																			
8. Benthic predatory fishes	0.181 (-11.6 %)		0.074 (-54.1 %)		0.145 (100.0 %)	0.059	0.208 (90.4 %)																				
9. Small benthic predatory fishes	0.004 (-25.0 %)		0.001 (-1900.0 %)		0.070 (100.0 %)	0.030 (-166.7 %)	0.020	0.005 (-900.0 %)																			
10. Small predators gastropods						0.040		0.025			0.197 (0.5 %)																
11. Predatory macroinvertebrates			0.002 (-900.0 %)	0.040 (-100.0 %)												0.005 (-100 %)											
12. Predatory zooplankton									0.050	0.005				0.146	0.050		0.050	0.050	0.010		0.010	0.005					
13. Lobsters			0.010	0.050		0.020		0.049																			
14. Small planktivorous reef fish	0.004 (-3675.0 %)	0.442 (-103.8 %)	0.001		0.248 (-222.6 %)	0.089 (-1.1 %)	0.252 (-98.4 %)	0.030 (-270.0 %)	0.050																		
15. Planktivorous reef fish	0.005 (100.0 %)	0.050 (100.0 %)	0.149 (-0.7 %)		0.152 (100.0 %)	0.040	0.147 (-0.7 %)																				
16. Sea stars and sea urchins			0.002 (100.1 %)	0.640 (5.5 %)		0.260 (73.1 %)		0.492 (20.7 %)	0.355 (57.9 %)		0.100		0.401 (-25.4 %)														
17. Anemones and zoanthids										0.342 (-0.3 %)	0.020					0.098 (1.0 %)											
18. Benthic omnivorous fish	0.279 (46.2 %)	0.202 (100.0 %)	0.361 (41.9 %)		0.185 (100.0 %)	0.203 (1.5 %)	0.250 (20.0 %)	0.055 (27.3 %)	0.197 (-1.5 %)																		
19. Mullets	0.020	0.060 (18.3 %)	0.060 (-66.7 %)		0.200	0.069 (-278.7 %)	0.061	0.072 (-2.8 %)	0.150 (-132.0 %)																		
20. Parrotfish	0.044 (100.0 %)		0.030 (-63.3 %)				0.010 (100.0 %)																				
21. Gorgonians						0.050 (100.0 %)				0.100 (-1.0 %)	0.163 (0.6 %)					0.029 (3.4 %)				0.100							
22. Sponges and polychaetes								0.010 (100.0 %)		0.070 (-114.3 %)			0.100 (100.0 %)			0.015 (-553.3 %)											
23. Small herbivorous gastropods						0.040		0.202 (0.5 %)		0.301 (1.0 %)	0.200 (-0.5 %)																
24. Sea turtles and marine iguanas	0.007 (28.6 %)																										
25. Herbivorous zooplankton									0.050	0.005		0.796 (-0.1 %)		0.203 (2.0 %)	0.251 (0.4 %)		0.050	0.049	0.089 (-1.1 %)		0.050	0.005	0.010	0.010			
26. Sea cucumbers and other				0.02																							
27. Herbivorous benthic fish	0.100 (-1.0 %)		0.190 (52.1 %)			0.080	0.052 (1.9 %)	0.025	0.148 (-1.4 %)																		
28. Macroalgae											0.151		0.2 (-48.5 %)			0.566 (-0.2 %)		0.701 (-0.3 %)		0.849			0.939 (-0.1 %)	0.939	0.050		0.950 (0.1 %)
29. Phytoplankton												0.152		0.600 (-0.8 %)	0.549 (-0.4 %)		0.496 (-0.6 %)	0.050	0.296 (-0.3 %)		0.637 (-0.5 %)	0.049			0.799 (-0.3 %)		
30. Detritus				0.250 (2.0 %)						0.177 (44.6 %)	0.169 (-0.6 %)	0.052 (1.9 %)	0.299 (33.1 %)	0.051 (0.2 %)	0.150 (0.7 %)	0.287 (30.3 %)	0.404 (0.3 %)	0.150 (1.3 %)	0.605 (0.3 %)	0.051	0.303 (0.1 %)	0.941	0.051 (2.0 %)	0.051	0.151 (1.3 %)	1.00	0.050 (-2.0 %)
Sum	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Appendix 5. Input – output data Darwin and Wolf model. OI omnivory index, P_i/B_i production rate, Q_i/B_i consumption rate, EE_i ecotrophic efficiency, GE_i gross efficiency.

Group name	Trophic level	OI	Biomass (t km^{-2})	P_i/B_i (year)	Q_i/B_i (year)	EE_i	GE_i	Living Biomass (%)	Keystone index	Flow to detritus (t km^{-2} year)	Net efficiency	Fishing mort./ Total mort.
Sharks	4.01	1.613	55.860	0.241	4.712	0.800	0.051	5.952	0.124	55.330	0.064	0.013
Seabirds	3.92	0.942	0.100	5.453	60.619	0.821	0.090	0.011	-0.863	1.310	0.112	-
Hammerhead sharks	3.78	0.333	86.980	0.241	4.712	0.443	0.051	9.268	-0.593	93.644	0.064	-
Predatory marine mammals	3.63	0.828	0.195	0.041	16.127	0.856	0.003	0.021	-1.531	0.630	0.003	-
Dolphins	3.60	0.573	0.200	0.041	16.130	0.321	0.003	0.021	-1.845	0.651	0.003	-
Pelagic predatory fishes	3.42	0.827	75.050	0.418	2.228	0.946	0.188	7.997	-0.013	35.147	0.235	0.014
Groupers	3.34	0.351	1.230	0.462	2.773	0.913	0.167	0.131	-0.882	0.731	0.208	0.575
Planktivorous reef fish	3.01	0.128	36.440	1.994	12.533	0.948	0.159	3.883	-0.300	95.097	0.199	-
Rays	2.91	0.647	84.990	0.255	1.780	0.911	0.143	9.056	-0.025	32.188	0.179	-
Benthic predatory fishes	2.90	0.517	43.530	0.755	3.350	0.829	0.225	4.638	-0.188	34.774	0.282	0.076
Small planktivorous reef fish	2.79	0.277	0.690	4.513	16.328	0.865	0.276	0.074	-1.193	2.672	0.345	-
Small benthic predatory fishes	2.74	0.491	36.670	1.196	5.102	0.916	0.234	3.907	-0.249	41.121	0.293	<0.001
Lobsters	2.63	0.341	4.330	0.699	7.222	0.915	0.097	0.461	-0.971	6.512	0.121	0.089
Predatory macroinvertebrates	2.56	0.345	21.600	1.001	4.081	0.930	0.245	2.302	-0.178	19.144	0.307	-
Parrotfish	2.50	0.305	16.630	0.495	11.516	0.084	0.043	1.772	-0.499	45.845	0.054	-
Small predators gastropods	2.49	0.320	1.330	0.920	4.063	0.950	0.226	0.142	-0.303	1.142	0.283	-
Anemones and zoanthids	2.40	0.278	0.060	2.092	10.063	0.769	0.208	0.006	-2.050	0.150	0.260	-
Shrimps and small crabs	2.34	0.269	11.645	3.600	20.450	0.950	0.176	-	-0.544	49.724	0.220	-
Benthic omnivorous fish	2.29	0.250	12.590	2.607	9.825	0.763	0.265	1.342	-0.332	32.526	0.332	-
Predatory zooplankton	2.25	0.188	21.163	16.420	45.800	0.950	0.359	2.255	-0.294	211.225	0.448	-
Small herbivorous gastropods	2.00	-	17.803	2.536	13.910	0.928	0.182	1.897	-0.667	52.776	0.228	-
Sea turtles	2.19	0.199	0.870	0.139	10.131	0.435	0.014	0.093	-2.111	1.831	0.017	-
Sea stars and sea urchins	2.17	0.160	66.020	2.245	7.854	0.946	0.286	7.035	-0.176	111.660	0.357	-
Stony corals	2.11	0.110	138.782	1.488	15.093	0.950	0.099	14.788	-0.332	429.253	0.123	-
Mullets	2.00	0.009	5.110	2.811	10.644	0.620	0.264	0.545	-1.010	16.337	0.330	-
Herbivorous benthic fish	2.00	-	63.640	0.994	10.830	0.653	0.092	6.781	-0.285	159.790	0.115	0.001
Sponges and polychaetes	2.00	-	7.300	2.080	16.624	0.379	0.125	0.778	-1.145	33.708	0.156	-
Herbivorous zooplankton	2.00	0.081	26.785	23.684	61.200	0.950	0.387	2.854	-0.283	359.566	0.484	-
Sea cucumbers	2.00	-	14.970	0.122	3.342	0.912	0.037	1.595	-1.524	10.166	0.046	-
Macroalgae	1.00	-	76.480	15.687	-	0.927	-	8.149	-0.057	87.210	-	-
Phytoplankton	1.00	-	1.870	146.274	-	0.919	-	0.199	-0.421	22.170	-	-
Zooxanthella	1.00	-	6.912	280.000	-	0.950	-	0.737	-0.249	96.774	-	-
Detritus	1.00	0.424	100.000	-	-	0.850	-	-	-	-	-	-

Appendix 6. Predator-prey matrix of the Darwin and Wolf model.

Prey\ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
1. Sharks				0.021																									
2. Seabirds	0.002																												
3. Hammerhead sharks	0.035			0.024																									
4. Predatory marine mammals																													
5. Dolphins																													
6. Pelagic predatory fishes	0.111																												
7. Groupers	0.001																												
8. Planktivorous reef fishes	0.110	0.091	0.031	0.178	0.058	0.110	0.120			0.049																			
9. Rays	0.075																												
10. Benthic predatory fishes				0.070		0.050	0.063			0.109																			
11. Small planktivorous reef fish		0.063			0.002					0.015								0.001											
12. Small benthic predatory fishes			0.001	0.082		0.045	0.059			0.025		0.150																	
13. Lobsters	0.006						0.009			0.007																			
14. Predatory macroinvertebates									0.130					0.005															
15. Parrotfishes	0.001			0.004			0.030		0.001	0.002																			
16. Small predators gastropods														0.010		0.030		0.001											
17. Anemones and zoanthids																0.010													
18. Shrimps and small crabs										0.055		0.170																	
19. Benthic omnivorous fish	0.007		0.015	0.001		0.070	0.004		0.001	0.025									0.014										
20. Predatory zooplankton								0.450	0.040		0.350	0.030					0.120	0.100				0.150		0.040					
21. Small herbivorous gastropods										0.140		0.090		0.025		0.040			0.018										
22. Sea turtles																													
23. Sea stars and sea urchins							0.090		0.110	0.195		0.085	0.540	0.435				0.001	0.190										
24 Stony corals										0.028		0.010		0.005	0.450	0.350		0.100	0.001				0.150						
25. Mullets				0.040	0.040	0.050	0.085																						
26. Herbivorous benthic fish				0.196		0.175	0.310		0.068																				
27. Sponges and polychaetes												0.003	0.001	0.001					0.020				0.005						
28. Herbivorous zooplankton								0.450			0.350	0.002					0.250	0.100	0.007	0.250				0.060					
29. Sea cucumbers												0.005		0.001					0.005										
30. Macroalgae													0.100	0.100	0.150			0.098	0.380		0.750	0.850	0.485		0.100	0.800			
31. Phytoplankton								0.100			0.300						0.230			0.070				0.025				0.050	
32. Zooxanthella															0.300	0.150								0.850					
33. Detritus									0.150	0.350		0.455	0.359	0.418	0.100	0.420	0.400	0.600	0.365	0.680	0.250		0.360	0.025	0.800	0.200	1.000	0.050	1.000
Import	0.653	0.846	0.953	0.384	0.900	0.500	0.230		0.500																0.100			0.900	
Sum	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Appendix 7. Input data source for estimate values by the Darwin and Wolf model. *B* biomass, *P/B* production rate, *Q/B* consumption rate. *Nomenclature*: EM: Ecological monitoring, PM: Population monitoring, GE: guess estimate, EO: Ecopath output

Functional groups / Parameter	*B* (t km^{-2})	*P/B* (year^{-1})	*Q/B* (year^{-1})
Zooxanthella	EO	GE	-
Phytoplankton	ESA Globcolor database	Opitz (1996), Ruiz and Wolff (2011), Taylor et al. (2008)	-
Macroalgae	EO	Macchiavello et al. (1987), Ruiz and Wolff (2011)	-
Sea cucumbers	EM (2004 to 2008)	Brey (2001), Okey et al. (2004), Opitz (1996), Ruiz and Wolff (2011)	Okey et al. (2004), Opitz (1996), Ruiz and Wolff (2011)
Herbivorous zooplankton	EO	GE, Ruiz and Wolff (2011)	GE, Ruiz and Wolf (2011)
Sponges and polychaetes	EO	Okey et al. (2004), Opitz (1996), Ruiz and Wolff (2011)	Okey et al. (2004), Opitz (1996), Ruiz and Wolff (2011)
Herbivorous benthic fish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolf (2011)	Froese and Pauly (eds) (2011), Okey et al. 2004, Pauly et al. (2003), Ruiz and Wolff (2011)
Mullets	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Okey et al. (2004), Ruiz and Wolff (2011)
Stony corals	EO	Opitz (1996), Okey et al. (2004)	Opitz (1996), Okey et al. (2004)
Sea stars and sea urchins	EM (2004 to 2008)	Opitz (1996), Ortiz and Wolff (2002), Ruiz and Wolff (2011)	Opitz (1996), Ruiz and Wolff (2011)
Sea turtles	EM (2004 to 2008)	Okey et al. (2004), Ruiz and Wolff (2011)	Opitz (1996), Okey et al. (2004), Ruiz and Wolff (2011)
Small herbivorous gastropod	EO	Brey (2001), Opitz (1996), Ruiz and Wolff (2011)	Opitz (1996), Okey et al. (2004), Ruiz and Wolff (2011)
Predatory zooplankton	OM (2004 to 2006)	GE, Ruiz and Wolff (2011)	GE, Ruiz and Wolff (2011)
Benthic omnivorous fish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)
Shrimps and small crabs	EO		
Anemones and zoanthids	EO	Opitz (1996), Ruiz and Wolff (2011)	Opitz (1996), Ruiz and Wolff (2011)
Small predators gastropods	EM (2004 to 2008)	Brey (2001), Opitz (1996), Ruiz and Wolff (2011)	Opitz (1996), Okey et al. (2004), Ruiz and Wolff (2011)
Parrotfish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. 2003, Ruiz and Wolff (2011)
Predatory macroinvertebrates	EM (2004 to 2008)	Brey (2001), Ruiz and Wolff (2011)	Opitz (1996), Ruiz and Wolff (2011)
Lobsters	EM (2004 to 2008)	Brey (2001), Ruiz and Wolff (2011))	Opitz (1996), Ruiz and Wolff (2011)
Small benthic predatory fishes	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Ruiz and Wolff (2011)
Small planktivorous reef fish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Ruiz and Wolff (2011)
Benthic predatory fishes	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Ruiz and Wolff (2011)
Rays	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Ruiz and Wolff (2011)
Planktivorous reef fish	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Ruiz and Wolff (2011)
Groupers	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Ruiz and Wolff (2011)
Pelagic predatory fishes	EM (2004 to 2008)	Froese and Pauly (eds) (2011), Okey et al. (2004), Ruiz and Wolff (2011)	Froese and Pauly (eds) (2011), Pauly et al. (2003), Ruiz and Wolff (2011)
Dolphins	GE	Okey et al (2004)	Okey et al. (2004)
Predatory marine mammals	GE	Opitz (1996), Okey et al. (2004), Ruiz and Wolff (2011)	Opitz (1996), Okey et al. (2004), Ruiz and Wolff (2011)
Hammerhead sharks	EM (2004 to 2008)	Froese and Pauly (eds) (2011)	Froese and Pauly (eds) (2011)
Seabirds	GE	Opitz (1996), Moloney et al. (2005), Okey et al. (2004), Ruiz and Wolff (2011)	Opitz (1996), Okey et al. (2004), Ruiz and Wolff (2011)
Sharks	EM (2004 to 2008)	Opitz (1996), Okey et al. (2004), Ruiz and Wolff (2011)	Opitz (1996), Okey et al. (2004), Ruiz and Wolff (2011)

Appendix 8. Input – output data Floreana revisited model. OI omnivory index, P_i/B_i production rate, Q_i/B_i consumption rate, EE_i ecotrophic efficiency, GE_i gross efficiency.

Group name	Trophic level	OI	Biomass (t km^{-2})	P_i/B_i (year)	Q_i/B_i (year)	EE	GE	Living Biomass (%)	Keystone index	Flow to detritus (t km^{-2} year^{-1})	Net efficiency	Fishing mort./ Total mort.
Sharks	4.05	1.277	19.290	0.241	4.712	0.729	0.051	1.489	0.652	19.439	0.064	-
Barracudas	3.74	0.785	43.300	0.241	4.712	0.908	0.051	3.342	-0.026	41.768	0.064	-
Seabirds	3.73	0.373	0.010	5.453	60.619	0.500	0.090	0.001	-2.422	0.148	0.112	-
Predatory marine mammals	3.54	0.406	5.680	0.041	16.127	0.937	0.003	0.438	0.641	18.335	0.003	-
Groupers	3.43	0.139	6.440	0.462	2.773	0.933	0.167	0.497	-0.657	3.772	0.208	0.009
Pelagic predatory fishes	3.41	0.545	6.320	0.418	2.228	0.901	0.188	0.488	-0.607	3.077	0.235	0.007
Rays	3.08	0.526	52.300	0.255	1.780	0.941	0.143	4.037	-0.062	19.412	0.179	-
Planktivorous reef fish	2.91	0.220	36.230	1.994	12.533	0.929	0.159	2.797	-0.163	95.943	0.199	-
Benthic predatory fishes	2.81	0.420	43.730	0.755	3.350	0.906	0.225	3.375	-0.428	32.398	0.282	0.060
Small benthic predatory fishes	2.71	0.463	104.240	1.196	5.102	0.945	0.234	8.046	-0.175	113.199	0.293	-
Small planktivorous reef fish	2.68	0.313	14.470	4.513	16.328	0.910	0.276	1.117	-0.497	53.158	0.345	-
Lobsters	2.58	0.263	5.840	0.699	7.222	0.923	0.097	0.451	-0.950	8.749	0.121	0.023
Predatory macroinvertebrates	2.50	0.274	70.400	1.001	4.081	0.634	0.245	5.434	-0.210	83.282	0.307	-
Shrimps and small crabs	2.28	0.241	31.070	3.600	20.450	0.900	0.176	2.398	-0.369	138.261	0.220	-
Anemones and zoanthids	2.28	0.239	0.136	2.092	10.063	0.900	0.208	0.011	-2.087	0.303	0.260	-
Benthic omnivorous fish	2.26	0.200	8.250	2.607	9.825	0.734	0.265	0.637	-0.569	21.940	0.332	-
Predatory zooplankton	2.25	0.188	26.918	16.420	45.800	0.900	0.359	2.078	-0.318	290.770	0.448	-
Small predators gastropods	2.19	0.173	3.450	0.920	4.063	0.729	0.226	0.266	-0.346	3.663	0.283	-
Sea turtles	2.19	0.199	3.020	0.139	10.131	0.866	0.014	0.233	-1.491	6.175	0.017	-
Stony corals	2.11	0.110	57.335	1.488	15.093	0.950	0.099	4.426	-0.283	177.336	0.123	-
Parrotfish	2.09	0.091	8.680	0.495	11.516	0.880	0.043	0.670	-0.771	20.508	0.054	-
Sea stars and sea urchins	2.03	0.035	178.400	2.245	7.854	0.656	0.286	13.770	-0.110	418.108	0.357	-
Small herbivorous gastropods	2.00	-	34.855	2.536	13.910	0.900	0.182	2.690	-0.528	105.805	0.228	-
Herbivorous benthic fish	2.00	-	40.290	0.994	10.830	0.942	0.092	3.110	0.056	89.586	0.115	-
Sponges and polychaetes	2.00	-	5.340	2.080	16.624	0.950	0.125	0.412	-1.470	18.310	0.156	-
Herbivorous zooplankton	2.00	-	30.742	23.684	61.200	0.900	0.387	2.373	-0.320	449.092	0.484	-
Sea cucumbers	2.00	-	51.310	0.122	3.342	0.559	0.037	3.960	-1.215	37.057	0.046	0.013
Macroalgae	1.00	-	392.600	15.687	-	0.294	-	30.304	-0.047	4347.787	-	-
Phytoplankton	1.00	-	12.000	146.274	-	0.903	-	0.926	-0.271	170.059	-	-
Zooxanthella	1.00	-	2.898	280.000	-	0.950	-	0.224	-0.295	40.575	-	-
Detritus	1.00	-	300.000	-	-	0.423	-	-	-	-	-	-

Appendix 9. Predator-prey matrix of the Floreana revisited model.

Prey \ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1. Sharks				0.037																							
2. Barracudas	0.076			0.028																							
3. Seabirds																											
4. Predatory marine mammals	0.002																										
5. Groupers	0.015			0.015																							
6. Pelagic predatory fishes	0.026																										
7. Rays	0.138																										
8. Planktivorous reef fish	0.001	0.220	0.260	0.150	0.120	0.100			0.032																		
9. Benthic predatory fishes	0.097			0.053	0.110	0.110			0.074																		
10. Small benthic predatory fishes	0.078	0.095	0.140	0.082	0.180	0.045			0.001	0.150																	
11. Small planktivorous reef fish		0.220	0.285		0.108		0.052		0.050					0.001													
12. Lobsters	0.020			0.002	0.025				0.009																		
13. Predatory macroinvertebrates	0.020			0.111			0.320						0.010														
14. Shrimps and small crabs									0.070	0.170																	
15. Anemones and zoanthids																		0.010									
16. Benthic omnivorous fish	0.002	0.045	0.065	0.020	0.041	0.220			0.005																		
17. Predatory zooplankton							0.040	0.350		0.030	0.300			0.100	0.120				0.150	0.040							
18. Small predators gastropods													0.005	0.001				0.040									
19. Sea turtles	0.004																										
20. Stony corals									0.001	0.010			0.001	0.050		0.001		0.030			0.080	0.025					
21. Parrotfishes	0.001			0.038	0.006				0.001																		
22. Sea stars and sea urchins					0.120		0.110		0.268	0.085		0.560	0.439	0.001		0.190											
23. Small herbivorous gastropods									0.140	0.090			0.025			0.032		0.100									
24. Herbivorous benthic fish				0.300	0.290	0.175	0.028																				
25. Sponges and polychaetes										0.003		0.001	0.001			0.020						0.005					
26. Herbivorous zooplankton								0.350		0.002	0.300			0.100	0.130	0.007	0.250			0.060							
27. Sea cucumbers										0.005			0.001			0.005											
28. Macroalgae												0.100	0.100	0.198		0.380			0.850		0.420	0.600	0.750	0.800			
29. Phytoplankton								0.150			0.300				0.100		0.050			0.020						0.950	
30. Zooxanthella																		0.170		0.855	0.330						
31. Detritus							0.150	0.150	0.350	0.455	0.100	0.339	0.418	0.550	0.650	0.365	0.700	0.650		0.025	0.170	0.370	0.250	0.200	1.000	0.050	1.000
Import	0.520	0.420	0.250	0.165		0.350	0.300																				
Sum	1.000	1.000	1.000	1.001	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Appendix 10. Fish biomasses. The values are given in kg per m^2. Biomass values presented in this table were used as estimates for Ecopath models construction and comparisons rather than considered accurate absolute species or functional compartment biomasses estimates.

Functional Group Model / Species	Darwin and Wolf	Bolivar Channel	Floreana
Piscivores			
Barracudas			
Sphyraena idiastes		9.92	8.72
Benthic predatory fish			
Aulostomus chinensis	0.27	0.55	0.03
Fistularia commersonii	0.13	0.00	0.07
Paralabrax albomaculatus		0.80	
Canthidermis maculata	0.01		
Grouper			
Mycteroperca olfax	0.46	4.13	3.68
Pelagic predatory fish			
Caranx lugubris	0.47		
Caranx melampygus	2.38		
Caranx sexfasciatus	3.40		
Sarda orientalis			0.45
Seriola rivoliana	0.27	0.01	
Sharks			
Carcharhinus galapagensis	51.86		7.13
Sphyrna lewini	86.98		3.77
Triaenodon obesus	5.85	1.35	13.16
Small benthic predatory fish			
Dermatolepis dermatolepis	0.67	0.01	0.39
Haemulon scudderi		0.06	1.67
Haemulon sexfasciatum		0.81	0.49
Plagiotremus azaleus	0.02	0.03	0.01
Synodus lacertinus		0.01	0.01
Planktivores			
Planktivorous reef fish			
Xanthichthys mento	0.04		0.05
Chromis alta		0.33	0.00
Chromis atrilobata	0.35	0.30	0.29
Myripristis berndti	0.04	0.02	0.01
Myripristis leiognathos	0.04		0.03
Paranthias colonus	30.02	52.93	30.30
Taenioconger klausewitzi			0.02
Heteropriacanthus cruentatus	0.40		< 0.01
Hippocampus ingens		0.01	
Oxycirrhites typus		< 0.01	
Small planktivorous reef fish			
Apogon atradorsatus	0.15	5.82	0.71
Xenocys jessiae		33.40	< 0.01
Primary Consumer			
Mullets			
Mugil galapagensis	0.71	3.64	

Appendix 10. *Continuation…*

Functional Group Model / Species	Darwin and Wolf	Bolivar Channel	Floreana
Parrotfish			
Nicholsina denticulata		0.12	0.07
Scarus compressus	0.54	0.88	0.80
Scarus ghobban	0.37	2.35	1.69
Scarus perrico	0.48	0.16	1.28
Scarus rubroviolaceus	2.89	0.37	0.98
Herbivorous benthic fishes			
Acanthurus nigricans	2.36		0.11
Acanthurus xanthopterus	0.09		0.13
Girella freminvillei	0.02	20.22	7.98
Kyphosus analogus	0.43		
Kyphosus elegans	7.42	0.01	0.18
Prionurus laticlavius	15.56	5.89	28.54
Sectator ocyurus	0.46		
Secondary Consumer			
Benthic omnivorous fishes			
Aluterus scriptus			0.01
Arothron hispidus	< 0.01		0.01
Cantherhines dumerilii	0.10		
Canthigaster punctatissima	0.02	< 0.01	0.00
Chaetodon humeralis	0.00	0.02	0.02
Johnrandallia nigrirostris	0.90	0.12	0.48
Melichthys niger	3.71		0.12
Melichthys vidua	0.05		
Microspathodon bairdii	0.64		0.06
Microspathodon dorsalis	0.25	0.14	0.26
Nexilosus latifrons		0.01	
Ophioblennius steindachneri	0.76	0.20	0.33
Oplegnathus insignis		0.77	0.38
Stegastes arcifrons	1.48	0.10	0.11
Stegastes beebei	1.34	8.00	2.21
Zanclus cornutus	0.50		0.14
Benthic predatory fishes			
Balistes polylepis	0.56	0.02	0.22
Bodianus eclancheri		1.77	0.13
Bothus mancus			0.01
Caulolatilus princeps		0.66	0.05
Cephalopholis panamensis	0.33	0.03	0.38
Chilomycterus reticulatus	0.02	0.10	0.39
Cirrhitus rivulatus	1.09	0.18	0.49
Diodon holocanthus	0.09	0.14	0.24
Enchelycore lichenosa			0.01
Epinephelus analogus		0.06	0.01
Epinephelus labriformis	0.86	1.12	1.40

Appendix 10. *Continuation…*

Functional Group Model / Species	Darwin and Wolf	Bolivar Channel	Floreana
Benthic predatory fishes			
Gymnothorax dovii	8.52	< 0.01	0.01
Lutjanus aratus	0.01		0.06
Lutjanus argentiventris	0.30	0.05	7.17
Lutjanus novemfasciatus	0.14		
Lutjanus viridis	0.27	0.55	6.00
Muraena argus		0.01	
Muraena lentiginosa	1.23	< 0.01	< 0.01
Rypticus nigripinnis			< 0.01
Semicossyphus darwini		1.14	0.02
Sufflamen verres	1.39	0.14	0.89
Pelagic predatory fishes			
Caranx caballus	0.30		
Euthynnus lineatus	0.13		0.38
Rhinoptera steindachneri			0.09
Scomberomorus sierra	0.21	0.24	0.65
Rays			
Aetobatus narinari	1.53	0.40	2.49
Dasyatis brevis	1.89	1.99	1.67
Taeniura meyeni	1.11	2.76	2.75
Small benthic predatory fishes			
Abudefduf troschelii	0.02	6.19	1.80
Alphestes immaculatus	0.02	0.04	0.03
Anisotremus interruptus	0.07	0.19	12.17
Anisotremus scapularis		17.11	0.03
Arothron meleagris	0.56	0.06	0.28
Bodianus diplotaenia	3.12	3.17	3.56
Bothus leopardinus	0.14	< 0.01	
Calamus brachysomus		0.02	0.07
Calamus taurinus		0.25	0.37
Cirrhitichthys oxycephalus	0.18	0.01	0.01
Coryphopterus urospilus	< 0.01	< 0.01	< 0.01
Diodon hystrix	0.08	0.07	0.24
Eleotrica cableae		< 0.01	
Halichoeres dispilus	0.71	3.06	1.16
Halichoeres nicholsi	0.06	0.28	1.07
Heterodontus quoyi		0.68	
Holacanthus passer	1.71	1.49	2.00
Hoplopagrus guentherii	0.07		0.31
Labrisomus dendriticus	0.07	0.46	0.04
Lepidonectes corallicola	0.01	0.01	< 0.01
Malacoctenus tetranemus	< 0.01	< 0.01	< 0.01
Mulloidichthys dentatus	2.00		1.02
Myrichthys tigrinus		< 0.01	0.01

Appendix 10. *Continuation…*

Functional Group Model / Species	Darwin and Wolf	Bolivar Channel	Floreana
Small benthic predatory fishes			
Novaculichthys taeniourus	< 0.01		< 0.01
Odontoscion eurymesops		0.04	
Orthopristis forbesi	< 0.01	1.04	17.15
Ostracion meleagris	0.01	0.07	0.03
Pareques perissa		0.01	< 0.01
Prognathodes carlhubbsi		0.12	
Pseudobalistes naufragium		0.02	0.04
Rypticus bicolor	0.01		0.03
Sargocentron suborbitalis	0.01		0.03
Scorpaena plumieri mystes	0.12	0.09	0.08
Scorpaenodes xyris		0.00	
Serranus psittacinus	0.01	0.10	0.05
Sphoeroides angusticeps	< 0.01	0.01	
Sphoeroides annulatus	< 0.01	0.17	0.03
Sphoeroides lobatus		< 0.01	
Thalassoma grammaticum	0.19		0.02
Thalassoma lucasanum	5.58	0.20	0.46
Thalassoma purpureum	0.02		
Trachinotus stilbe	0.17		

Appendix 11. Macroinvertebrate biomasses. The values are given in kg per m^2. Biomass values presented in this table were used as estimates for Ecopath models construction and comparisons rather than considered accurate absolute species or functional compartment biomasses estimates.

Functional Group Model / Species	Darwin and Wolf	Bolivar Channel	Floreana
Predatory macroinvertebates			
Asteropsis carinifera	0.13	0.15	0.13
Astropecten armatus		0.45	
Heliaster cumingii		2.62	
Hexaplex princeps	7.16	4.31	16.26
Luidia bellonae		0.16	0.30
Neorapana grandis	1.17	1.17	
Pleuroploca princeps	152.58	45.77	47.68
Malea ringens		8.83	
Lobsters			
Panulirus gracilis	0.81		
Panulirus penicillatus	1.76		
Scyllarides astori	2.05	2.30	1.46
Sea cucumbers			
Holothuria arenicola		3.91	
Holothuria atra	12.89	2.76	21.18
Holothuria difficilis		0.99	
Holothuria fuscocinerea	3.68	2.76	3.80
Holothuria imitans			
Holothuria kefersteini	32.07	23.17	17.51
Isostichopus fuscus	5.03	10.97	4.22
Stichopus horrens		2.29	1.84
Sea stars and sea urchins			
Diadema mexicanum	55.41	16.33	2.20
Echinometra vanbrunti	0.28	34.85	0.42
Leiaster teres	0.24		
Linckia columbiae		0.27	0.04
Tripneustes depressus	1.75	7.15	
Caenocentrotus gibbosus		0.32	0.00
Centrostephanus coronatus	0.27	1.98	0.88
Eucidaris galapagensis	27.58	117.66	142.87
Lytechinus semituberculatus		20.39	0.27
Mithrodia bradleyi	0.24	0.24	0.30
Nidorellia armata	0.60	11.58	3.88
Pentaceraster cumingi	18.95	20.91	29.69
Pharia pyramidata	0.21	0.46	0.26
Phataria unifascialis	0.12	0.31	0.54
Toxopneustes roseus	1.91		
Small predators gastropods			
Thais planospira	0.18		